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Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically; Approved Standard—Ninth Edition

This document addresses reference methods for the determination of minimal inhibitory concentrations of aerobic bacteria by broth macrodilution, broth microdilution, and agar dilution.

A standard for global application developed through the Clinical and Laboratory Standards Institute consensus process.



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Abstract

Susceptibility testing is indicated for any organism that contributes to an infectious process warranting antimicrobial chemotherapy, if its susceptibility cannot be reliably predicted from knowledge of the organism's identity. Susceptibility tests are most often indicated when the causative organism is thought to belong to a species capable of exhibiting resistance to commonly used antimicrobial agents.

A variety of laboratory methods can be used to measure the *in vitro* susceptibility of bacteria to antimicrobial agents. This document describes standard broth dilution (macrodilution and microdilution [the microdilution method described in M07 is the same methodology outlined in ISO 20776-1])¹ and agar dilution techniques, and it includes a series of procedures to standardize the way the tests are performed. The performance, applications, and limitations of the current CLSI-recommended methods are also described.

The supplemental information (M100² tables) presented with this standard represents the most current information for drug selection, interpretation, and quality control using the procedures standardized in M07. These tables, as in previous years, have been updated and should replace tables published in earlier years. Changes in the tables since the previous edition (M100-S21³) appear in boldface type and are also summarized in the front of the document.

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It has been a distinct honor to serve as Chairholder of the Subcommittee on Antimicrobial Susceptibility Testing during the last three years. Many members of the subcommittee (which now numbers more than 180 volunteers including members, advisors, and observers) have been indispensable in the preparation of these documents. In addition, I would like to thank the working group chairholders of the Subcommittee on Antimicrobial Susceptibility Testing for their valuable contributions during the last three years. Those individuals who are not listed in the committee and working group lists in this document are as follows:

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Foreword

In this revision of CLSI document M07, several sections were added or revised as outlined below in the Summary of Changes. The latest version of the M100² tables, published as an annual volume, is made available with this document to ensure that users are aware of the latest subcommittee guidelines related to both methods and the tabular information normally presented in the annual tables.

Many other editorial and procedural changes in this edition of M07 resulted from meetings of the Subcommittee on Antimicrobial Susceptibility Testing since 2009. Specific changes for the M100² tables are summarized at the beginning of CLSI document M100.² The most important changes in M07 are summarized below.

Summary of Major Changes in This Document

CLSI Reference Methods vs Commercial Methods and CLSI vs US Food and Drug Administration Interpretive Criteria (Breakpoints) – Textbox

Added the recommendation for each laboratory to check with the manufacturer of its commercial susceptibility testing device for information on the breakpoints used in its system's software.

Section 4.1, Definitions

Clarified the definition for nonsusceptible.

Added a definition for breakpoint/interpretive criteria.

Section 6.2.1.2, β -Lactam/ β -Lactamase Inhibitor Combinations

Added information on the activity of β -lactamase inhibitors.

Section 6.2.1.3, Cephems (Including Cephalosporins)

Added information on a new subclass, cephalosporins with anti-methicillin-resistant *Staphylococcus aureus* (MRSA) activity.

Section 6.2.2.6, Nitroimidazoles

Added nitroimidazoles as a new section, which includes the antimicrobial agents metronidazole and tinidazole.

Section 6.2.2.7, Oxazolidinones

Added oxazolidinones as a new section, which includes the antimicrobial agent linezolid.

Section 6.2.2.9, Streptogramins

Added streptogramins as a new section, which includes the antimicrobial agents quinupristin-dalfopristin and linopristin-flopristin.

Section 6.2.2.10, Tetracyclines

Clarified that tigecycline is in a separate class, glycylcycline.

Section 6.2.2.11, Single-Drug Classes

Deleted streptogramins.

Added fusidic acid, macrocyclics, mupirocin, and spectinomycin.

Section 6.4, Suggested Guidelines for Routine and Selective Testing and Reporting

Added a recommendation for laboratories to develop a protocol to address isolates that are confirmed as resistant to all agents on their routine test panel.

Summary of Major Changes in This Document (Continued)

Section 11.4, *Streptococcus pneumoniae* and Other *Streptococcus* spp.

Clarified that studies using the agar dilution method have not been performed and reviewed by the subcommittee.

Section 12.1.1, Penicillin Resistance and β -lactamase

Added additional recommendations for the use of nitrocefin-based tests or penicillin zone-edge test for isolates of *Staphylococcus* with penicillin MICs ≤ 0.12 $\mu\text{g/mL}$ or zone diameters ≥ 29 mm for β -lactamase production before reporting as susceptible.

Section 12.1.3.2, Vancomycin Agar Screen

Added information on when variation in screen test results may occur.

Sections 12.3, Gram-Negative Bacilli and 12.3.1, Background

Clarified the active site for each of the molecular classes in the table.

Added OXY to Class A.

Added NDM to Class B.

Clarified the example for OXA in Class D.

Added information on β -lactamase enzymes and their resistance mechanisms.

Section 12.3.2, Extended-Spectrum β -Lactamases (ESBLs)

Provided additional information on ESBLs being inhibitor-susceptible enzymes.

Added information on how β -lactam interpretive breakpoints are set at MIC values to recognize ESBL activity.

Section 12.3.3, AmpC Enzymes

Expanded information on AmpC β -lactamases and their mode of action.

Section 12.3.4, Carbapenemases (Carbapenem-Resistant *Enterobacteriaceae*)

Combined previous sections on other β -lactamase-mediated resistance and metallo- β -lactamase, and provided more detailed information on carbapenemase activity in clinical isolates of *Enterobacteriaceae*, including addition of a table with examples.

Section 13, Inducible Clindamycin Resistance

Clarified the testing method to include specific disk placement for staphylococci.

Section 14.2, Selecting a β -Lactamase Test

Penicillin zone-edge test method added as an alternative method for detection of β -lactamase in staphylococci.

Section 16.5, Batch or Lot Quality Control

Added information regarding the use of Mueller-Hinton broth with or without cations.

Section 17.1, Application to Various Organism Groups

Added information for organisms not covered in M100² or other CLSI documents, along with recommendations for determining the need for susceptibility testing for these microorganisms.

Section 17.2, Misleading Results

Added cephalosporins with anti-MRSA activity to the second bullet.

Summary of Major Changes in This Document (Continued)

Appendix C, Conditions for Dilution Antimicrobial Susceptibility Tests

Clarified medium for *Staphylococcus* spp.

Added comment noting that studies using the agar dilution method have not been performed and reviewed by the subcommittee for *Streptococcus pneumoniae* and *Streptococcus* spp.

Added caution comment for testing *N. meningitidis* isolates.

Summary of CLSI Processes for Establishing Interpretive Criteria and Quality Control Ranges

The Clinical and Laboratory Standards Institute (CLSI) is an international, voluntary, nonprofit, interdisciplinary, standards-developing, and educational organization accredited by the American National Standards Institute, which develops and promotes the use of consensus-developed standards and guidelines within the health care community. These consensus standards and guidelines are developed to address critical areas of diagnostic testing and patient health care, and are developed in an open and consensus-seeking forum. CLSI is open to anyone, or any organization that has an interest in diagnostic testing and patient care. Information about CLSI is found at www.clsi.org.

The CLSI Subcommittee on Antimicrobial Susceptibility Testing reviews data from a variety of sources and studies (eg, *in vitro*, pharmacokinetics/pharmacodynamics, and clinical studies) to establish antimicrobial susceptibility test methods, interpretive criteria, and quality control (QC) parameters. The details of the data required to establish interpretive criteria, QC parameters, and how the data are to be presented for evaluation are described in CLSI document M23.⁴

Over time, a microorganism's susceptibility to an antimicrobial agent may decrease, resulting in a lack of clinical efficacy and/or safety. In addition, microbiological methods and QC parameters may be refined to ensure more accurate and better performance of susceptibility test methods. Because of this, CLSI continually monitors and updates information in its documents. Although CLSI standards and guidelines are developed using the most current information and thinking available at the time, the field of science and medicine is ever changing; therefore, standards and guidelines should be used in conjunction with clinical judgment, current knowledge, and clinically relevant laboratory test results to guide patient treatment.

Additional information, updates, and changes in this document are found in the meeting summary minutes of the Subcommittee on Antimicrobial Susceptibility Testing at www.clsi.org.

CLSI Reference Methods vs Commercial Methods and CLSI vs US Food and Drug Administration Interpretive Criteria (Breakpoints)

It is important for users of M02-A11 and M07-A9 to recognize that commercial susceptibility testing devices are not addressed in these standards. The methods described herein are generic reference procedures that can be used by clinical laboratories for routine susceptibility testing or to evaluate commercial devices for possible routine use. Results generated by the CLSI reference methods are used by the US Food and Drug Administration (FDA) to evaluate the performance of commercial systems before clearance is given for marketing in the United States. Clearance by the FDA indicates the agency has concluded that commercial devices provide susceptibility results that are substantially equivalent to results generated using the CLSI reference methods for the organisms and antimicrobial agents described in the manufacturer's approved package insert. Some laboratories could find that a commercial dilution, antibiotic gradient, colorimetric, turbidimetric, fluorometric, or other method is suitable for selective or routine use.

CLSI breakpoints may differ from those approved by various regulatory authorities for many reasons, including the following: different databases, differences in interpretation of data, variations in doses used in different parts of the world, and public health policies. Differences also exist because CLSI proactively evaluates the need for changing breakpoints. The reasons why breakpoints may change and the manner in which CLSI evaluates data and determines breakpoints are outlined in CLSI document M23.⁴

Following a decision by CLSI to change an existing breakpoint, regulatory authorities may also review data in order to determine how changing breakpoints may affect the safety and effectiveness of the antimicrobial agent for the approved indications. If the regulatory authority changes breakpoints, commercial device manufacturers may have to conduct a clinical laboratory trial, submit the data to the regulatory authority, and await review and approval. For these reasons, a delay of more than the suggested CLSI "tentative" period of one year may be required if an interpretive breakpoint change is to be implemented by a device manufacturer. In the United States, it is acceptable for laboratories that use FDA-cleared susceptibility testing devices to use existing FDA interpretive breakpoints. Either FDA or CLSI susceptibility interpretive breakpoints are acceptable to clinical laboratory accrediting bodies in the United States. Policies in other countries may vary. Each laboratory should check with the manufacturer of its antimicrobial susceptibility test system for additional information on the interpretive criteria used in its system's software.

Following discussions with appropriate stakeholders, such as infectious disease practitioners and the pharmacy department, as well as the Pharmacy and Therapeutics and Infection Control committees of the medical staff, newly approved or revised breakpoints may be implemented by clinical laboratories. CLSI broth dilution and agar dilution test breakpoints may be implemented as soon as they are published in M100.² If a device includes antimicrobial test concentrations sufficient to allow interpretation of susceptibility to an agent using the CLSI breakpoints, a laboratory could, after appropriate validation, choose to interpret and report results using CLSI breakpoints.

Subcommittee on Antimicrobial Susceptibility Testing Mission Statement

The Subcommittee on Antimicrobial Susceptibility Testing is composed of representatives from the professions, government, and industry, including microbiology laboratories, government agencies, health care providers and educators, and pharmaceutical and diagnostic microbiology industries. Using the CLSI voluntary consensus process, the subcommittee develops standards that promote accurate antimicrobial susceptibility testing and appropriate reporting.

The mission of the Subcommittee on Antimicrobial Susceptibility Testing is to:

- Develop standard reference methods for antimicrobial susceptibility tests.
- Provide quality control parameters for standard test methods.
- Establish interpretive criteria for the results of standard antimicrobial susceptibility tests.
- Provide suggestions for testing and reporting strategies that are clinically relevant and cost-effective.
- Continually refine standards and optimize detection of emerging resistance mechanisms through development of new or revised methods, interpretive criteria, and quality control parameters.
- Educate users through multimedia communication of standards and guidelines.
- Foster a dialogue with users of these methods and those who apply them.

The ultimate purpose of the subcommittee's mission is to provide useful information to enable laboratories to assist the clinician in the selection of appropriate antimicrobial therapy for patient care. The standards and guidelines are meant to be comprehensive and to include all antimicrobial agents for which the data meet established CLSI guidelines. The values that guide this mission are quality, accuracy, fairness, timeliness, teamwork, consensus, and trust.

Key Words

Agar dilution, antimicrobial susceptibility, broth dilution, macrodilution, microdilution, minimal inhibitory concentration

Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically; Approved Standard—Ninth Edition

1 Scope

This document describes the standard broth (macrodilution and microdilution) and agar dilution methods used to determine the *in vitro* susceptibility of bacteria that grow aerobically. It addresses preparation of broth and agar dilution tests, testing conditions (including inoculum preparation and standardization, incubation time, and incubation temperature), reporting of minimal inhibitory concentration (MIC) results, quality control (QC) procedures, and limitations of the dilution test methods. To assist the clinical laboratory, suggestions are provided on the selection of antimicrobial agents for routine testing and reporting. Standards for testing the *in vitro* susceptibility of bacteria that grow aerobically using the antimicrobial disk susceptibility testing method are found in CLSI document M02⁵; standards for testing the *in vitro* susceptibility of bacteria that grow anaerobically are found in CLSI document M11.⁶ Guidelines for standardized susceptibility testing of infrequently isolated or fastidious bacteria that are not included in CLSI documents M02,⁵ M07, or M11⁶ are available in CLSI document M45.⁷

2 Introduction

Either broth or agar dilution methods may be used to measure quantitatively the *in vitro* activity of an antimicrobial agent against a given bacterial isolate. To perform the tests, a series of tubes or plates is prepared with a broth or agar medium to which various concentrations of the antimicrobial agents are added. The tubes or plates are then inoculated with a standardized suspension of the test organism. After incubation at $35 \pm 2^\circ\text{C}$, the tests are examined and the MIC is determined. The final result is significantly influenced by methodology, which must be carefully controlled if reproducible results (intralaboratory and interlaboratory) are to be achieved.

This document describes reference standard broth dilution (macrodilution and microdilution) and agar dilution methods. The basics of these methods are derived, in large part, from information generated by the International Collaborative Study.⁸ Although these methods are standard reference methods, some are sufficiently practical for routine use in both clinical laboratories and research laboratories.

Commercial systems based primarily, or in part, on some of these methods are available and may provide results essentially equivalent to the CLSI methods described here. The US Food and Drug Administration (FDA) is responsible for the approval of commercial devices used in the United States. CLSI does not approve or endorse commercial products or devices.

The methods described in this document are intended primarily for testing commonly isolated aerobic or facultative bacteria that grow well after overnight incubation in unsupplemented Mueller-Hinton agar (MHA) or Mueller-Hinton broth (MHB). Alternative media and methods for some fastidious or uncommon organisms are described in Section 11 and M100² Tables 2E through 2I. Methods for testing anaerobic bacteria are outlined in CLSI document M11⁶ and in M100² Table 2J. Methods for testing infrequently isolated or fastidious bacteria not included in M02⁵ and M07 are found in CLSI document M45.⁷

This document, along with M100,² describes methods, QC, and interpretive criteria currently recommended for dilution susceptibility tests. When new problems are recognized or improvements in these criteria are developed, changes will be incorporated into future editions of this standard and also distributed in annual informational supplements (M100).²

3 Standard Precautions

Because it is often impossible to know what isolates or specimens might be infectious, all patient and laboratory specimens are treated as infectious and handled according to “standard precautions.” Standard precautions are guidelines that combine the major features of “universal precautions and body substance isolation” practices. Standard precautions cover the transmission of all known infectious agents and thus are more comprehensive than universal precautions, which are intended to apply only to transmission of blood-borne pathogens. Standard and universal precaution guidelines are available from the Centers for Disease Control and Prevention (CDC).⁹ For specific precautions for preventing the laboratory transmission of all known infectious agents from laboratory instruments and materials and for recommendations for the management of exposure to all known infectious diseases, refer to CLSI document M29.¹⁰

4 Terminology

4.1 Definitions

antimicrobial susceptibility test interpretive category – a classification based on an *in vitro* response of an organism to an antimicrobial agent at levels corresponding to blood or tissue levels attainable with usually prescribed doses of that agent.

- 1) **susceptible** – a category that implies that isolates are inhibited by the usually achievable concentrations of antimicrobial agent when the dosage recommended to treat the site of infection is used.
- 2) **intermediate** – a category that includes isolates with antimicrobial agent minimal inhibitory concentrations that approach usually attainable blood and tissue levels and for which response rates may be lower than for susceptible isolates; **NOTE:** The intermediate category implies clinical efficacy in body sites where the drugs are physiologically concentrated (eg, quinolones and β -lactams in urine) or when a higher than normal dosage of a drug can be used (eg, β -lactams). This category also includes a buffer zone, which should prevent small, uncontrolled, technical factors from causing major discrepancies in interpretations, especially for drugs with narrow pharmacotoxicity margins.
- 3) **resistant** – a category that implies that isolates are not inhibited by the usually achievable concentrations of the agent with normal dosage schedules and/or that demonstrate minimal inhibitory concentrations that fall in the range in which specific microbial resistance mechanisms (eg, β -lactamases) are likely, and clinical efficacy of the agent against the isolate has not been reliably shown in treatment studies.
- 4) **nonsusceptible** – a category used for isolates for which only a susceptible interpretive criterion has been designated because of the absence or rare occurrence of resistant strains. Isolates that have minimal inhibitory concentrations (MICs) above or zone diameters below the value indicated for the susceptible breakpoint should be reported as nonsusceptible; **NOTE 1:** An isolate that is interpreted as nonsusceptible does not necessarily mean that the isolate has a resistance mechanism. It is possible that isolates with MICs above the susceptible breakpoint that lack resistance mechanisms may be encountered within the wild-type distribution subsequent to the time the susceptible-only breakpoint is set; **NOTE 2:** For strains yielding results in the “nonsusceptible” category, organism identification and antimicrobial susceptibility test results should be confirmed. (See M100² Appendix A.)

breakpoint/interpretive criteria – minimal inhibitory concentration (MIC) or zone diameter value used to indicate susceptible, intermediate, and resistant as defined above.

For example, for antimicrobial X with interpretive criteria of:

	MIC (µg/mL)	Zone Diameter (mm)
Susceptible	≤ 4	≥ 20
Intermediate	8–16	15–19
Resistant	≥ 32	≤ 14

“Susceptible breakpoint” is 4 µg/mL or 20 mm.

“Resistant breakpoint” is 32 µg/mL or 14 mm.

D-zone test – a disk diffusion test using clindamycin and erythromycin disks placed in close proximity to detect the presence of inducible clindamycin resistance in staphylococci and streptococci.^{11,12}

minimal inhibitory concentration (MIC) – the lowest concentration of an antimicrobial agent that prevents visible growth of a microorganism in an agar or broth dilution susceptibility test.

quality assurance (QA) – a part of quality management focused on providing confidence that quality requirements will be fulfilled (ISO 9000)¹³; **NOTE:** The practice that encompasses all procedures and activities directed toward ensuring that a specified quality of product is achieved and maintained. In the testing environment, this includes monitoring all the raw materials, supplies, instruments, procedures, sample collection/transport/storage/processing, recordkeeping, calibrating and maintaining equipment, quality control, proficiency testing, training of personnel, and all else involved in the production of the data reported.

quality control (QC) – the operational techniques and activities that are used to fulfill requirements for quality (modified from ISO 9000)¹³; **NOTE:** A system for ensuring maintenance of proper standards by periodic inspection of the results and the operational techniques that are used to ensure accuracy and reproducibility.

saline – a solution of 0.85% to 0.9% NaCl (w/v).

4.2 Abbreviations and Acronyms

AST	antimicrobial susceptibility testing
ATCC	American Type Culture Collection
BHI	Brain Heart Infusion
BSC	biological safety cabinet
BSL	Biosafety Level (USA)
CAMHB	cation-adjusted Mueller-Hinton broth
CDC	Centers for Disease Control and Prevention
CFU	colony-forming units
CSF	cerebrospinal fluid
DNA	deoxyribonucleic acid
EDTA	ethylenediaminetetraacetic acid
ESBL	extended-spectrum β-lactamase
FDA	US Food and Drug Administration
HPLC	high-performance liquid chromatography
HTM	<i>Haemophilus</i> Test Medium
hVISA	heteroresistant vancomycin-intermediate <i>Staphylococcus aureus</i>

IU	International Units
KPC	<i>Klebsiella pneumoniae</i> carbapenemase
LHB	lysed horse blood
MHA	Mueller-Hinton agar
MHB	Mueller-Hinton broth
MHT	modified Hodge test
MIC	minimal inhibitory concentration
MRS	methicillin-resistant staphylococci
MRSA	methicillin-resistant <i>Staphylococcus aureus</i>
NAD	nicotinamide adenine dinucleotide
P-80	polysorbate 80
PBP	penicillin-binding protein
PBP 2a	penicillin-binding protein 2a
QA	quality assurance
QC	quality control
RNA	ribonucleic acid
TEM	Temoneira (first patient from whom a TEM β -lactamase-producing strain was reported)
US	United States
VRE	vancomycin-resistant enterococci

5 Indications for Performing Susceptibility Tests

Susceptibility testing is indicated for any organism that contributes to an infectious process warranting antimicrobial chemotherapy if its susceptibility cannot be reliably predicted from knowledge of the organism's identity. Susceptibility tests are most often indicated when the causative organism is thought to belong to a species capable of exhibiting resistance to commonly used antimicrobial agents. Mechanisms of resistance include production of drug-inactivating enzymes, alteration of drug targets, and altered drug uptake or efflux. Some organisms have predictable susceptibility to antimicrobial agents, and empiric therapy for these organisms is widely accepted. Susceptibility tests are seldom necessary when the infection is due to a microorganism recognized as susceptible to a highly effective drug (eg, the continued susceptibility of *Streptococcus pyogenes* to penicillin). For *S. pyogenes* isolates from penicillin-allergic patients, erythromycin or another macrolide may be tested to detect strains resistant to those agents. Susceptibility tests are also important in studies of the epidemiology of resistance and in studies of new antimicrobial agents.

Isolated colonies of each type of organism that may be pathogenic should be selected from primary agar plates and tested individually for susceptibility. Identification procedures are often performed at the same time. Mixtures of different types of microorganisms should not be tested on the same susceptibility test plate or panel.

The practice of conducting susceptibility tests directly with clinical material (eg, normally sterile body fluids and urine) should be avoided, except in clinical emergencies when the direct Gram stain suggests a single pathogen. When testing has been carried out directly with the clinical material, results should be reported as preliminary, and the susceptibility test must be repeated using the standardized methodology.

When the nature of the infection is not clear and the specimen contains mixed growth or normal flora (in which the organisms probably bear little relationship to the infectious process treated), susceptibility tests are often unnecessary, and the results may be misleading.

The MIC obtained using a dilution test may tell a physician the concentration of antimicrobial agent required at the site of infection to inhibit the infecting organism. However, the MIC does not represent an *absolute* value. The "true" MIC is somewhere between the lowest test concentration that inhibits the organism's growth (ie, the MIC reading) and the next lowest test concentration. If, for example, twofold

dilutions were used and the MIC is 16 µg/mL, the “true” MIC would be between 16 µg/mL and 8 µg/mL. Even under the best of controlled conditions, a dilution test may not yield the same end point each time it is performed. Generally, the acceptable reproducibility of the test is within one twofold dilution of the end point. To avoid greater variability, the dilution test must be standardized and carefully controlled as described herein.

MICs have been determined using concentrations derived traditionally from serial twofold dilutions indexed to the base 2 (eg, 1, 2, 4, 8, 16 µg/mL). Other dilution schemes have also been used, including use of as few as two widely separated or “breakpoint” concentrations, or concentrations between the usual values (eg, 4, 6, 8, 12, 16 µg/mL). The results from these alternative methods may be equally useful clinically; however, some are more difficult to control (see Section 16.3). When there is inhibition of growth at the lowest concentration tested, the true MIC value cannot be accurately determined and should be reported as equal to or less than the lowest concentration tested. To apply interpretive criteria when concentrations between the usual dilutions are tested, results falling between serial twofold dilutions should be rounded up to the next highest concentration (eg, an MIC of 6 µg/mL would become 8 µg/mL).

Whenever MIC results are reported to clinicians to direct therapy, an interpretive category (ie, susceptible, intermediate, or resistant) should accompany the MIC result based on the criteria outlined in M100² Tables 2A through 2J. When tests in which four or fewer consecutive concentrations are tested or when nonconsecutive concentrations are tested, an interpretive category result must be reported. The MIC result may also be reported, if desired.

6 Selection of Antimicrobial Agents for Routine Testing and Reporting

Selection of the most appropriate antimicrobial agents to test and to report is a decision best made by each clinical laboratory in consultation with the infectious disease practitioners and the pharmacy, as well as the pharmacy and therapeutics and infection control committees of the medical staff. The recommendations in M100² Tables 1A, 1B, and 1C for each organism group list agents of proven efficacy that show acceptable *in vitro* test performance. Considerations in the assignment of agents to specific test/report groups include clinical efficacy, prevalence of resistance, minimizing emergence of resistance, cost, FDA clinical indications for usage, and current consensus recommendations for first choice and alternative drugs. Tests of selected agents may be useful for infection control purposes.

6.1 Routine Reports

The agents in M100² Tables 1A, 1B, and 1C are recommendations that are considered appropriate for testing and reporting. To avoid misinterpretation, routine reports to physicians should include only those antimicrobial agents appropriate for therapeutic use, as suggested in M100² Tables 1A, 1B, and 1C. Agents may be added to or removed from these basic lists as conditions demand. Antimicrobial agents other than those appropriate for use in therapy may also be tested to provide taxonomic data and epidemiological information, but they should not be included on patient reports. However, such results should be available (in the laboratory) to the infection control practitioner and/or hospital epidemiologist.

6.2 Nonproprietary Names

To minimize confusion, all antimicrobial agents should be reported using official nonproprietary (ie, generic) names. To emphasize the relatedness of the many currently available antimicrobial agents, they may be grouped together by drug classes as follows (see M100² Glossary I).

6.2.1 β -Lactams (see M100² Glossary I, Part 1)

β -lactam antimicrobial agents all share the common, central, four-member β -lactam ring and inhibition of cell wall synthesis as the primary mode of action. Additional ring structures or substituent groups added to the β -lactam ring determine whether the agent is classified as a penicillin, cephem, carbapenem, or monobactam.

6.2.1.1 Penicillins

Penicillins are primarily active against non- β -lactamase-producing, aerobic, gram-positive, some fastidious, aerobic, gram-negative, and some anaerobic bacteria. Aminopenicillins (ampicillin and amoxicillin) are active against additional gram-negative species, including some members of the *Enterobacteriaceae*. Carboxypenicillins (carbenicillin and ticarcillin) and ureidopenicillins (mezlocillin and piperacillin) are active against a considerably expanded list of gram-negative bacteria, including many *Pseudomonas* and *Burkholderia* spp. Penicillinase-stable penicillins (cloxacillin, dicloxacillin, methicillin, nafcillin, and oxacillin) are active against predominantly gram-positive bacteria, including penicillinase-producing staphylococci.

6.2.1.2 β -Lactam/ β -Lactamase Inhibitor Combinations

These antimicrobial agents are combinations that include a β -lactam class antimicrobial agent and a second agent that has minimal antibacterial activity, but functions as an inhibitor of some β -lactamases. β -lactamase inhibitors generally do not have antimicrobial activity on their own, but will potentiate the activity of the β -lactam antimicrobial agent combined with it. Currently, three β -lactamase inhibitors are in use: clavulanic acid, sulbactam, and tazobactam. The results of tests of only the β -lactam portion of the combination against β -lactamase-producing organisms are often not predictive of susceptibility to the two-drug combination.

6.2.1.3 Cephems (Including Cephalosporins)

Different cephem antimicrobial agents exhibit somewhat different spectrums of activity against aerobic and anaerobic, gram-positive and gram-negative bacteria. The cephem antimicrobial class includes the classical cephalosporins; the agents in subclasses cephamycin, oxacephem, and carbacephems; as well as a new subclass, cephalosporins with anti-methicillin-resistant *Staphylococcus aureus* (MRSA) activity (see M100² Glossary I). Cephalosporins are often referred to as “first-,” “second-,” “third-,” or “fourth-generation” cephalosporins, based on the extent of their activity against the more antimicrobial agent-resistant, gram-negative aerobic bacteria. Not all representatives of a specific group or generation necessarily have the same spectrum of activity. Because of these differences in activities, representatives of each group may be selected for routine testing.

6.2.1.4 Penems

The penem antimicrobial class, which includes two subclasses—the carbapenems and penems—differs slightly in structure from the penicillin class; agents in this class are much more resistant to β -lactamase hydrolysis, which provides them with broad-spectrum activity against many gram-positive and gram-negative bacteria.

6.2.1.5 Monobactams

Monobactam antimicrobial agents are monocyclic β -lactams. Aztreonam, which has activity only against aerobic, gram-negative bacteria, is the only monobactam antimicrobial agent approved for use in the United States by the FDA.

6.2.2 Non- β -lactams (see M100² Glossary I, Part 2)

6.2.2.1 Aminoglycosides

Aminoglycosides are structurally related antimicrobial agents that inhibit bacterial protein synthesis at the ribosomal level. This class includes agents variously affected by aminoglycoside-inactivating enzymes, resulting in some differences in the spectrum of activity among the agents. Aminoglycosides are used primarily to treat aerobic, gram-negative rod infections or in synergistic combinations with cell wall-active antimicrobial agents (eg, penicillin, ampicillin, vancomycin) against some resistant, gram-positive bacteria, such as enterococci.

6.2.2.2 Folate Pathway Inhibitors

Sulfonamides and trimethoprim are chemotherapeutic agents with similar spectra of activity resulting from the inhibition of the bacterial folate pathway. Sulfisoxazole is among the most commonly used sulfonamides in the treatment of urinary tract infections; thus, it may be the appropriate selection for *in vitro* testing. Sulfamethoxazole is usually tested in combination with trimethoprim, because these two antimicrobial agents inhibit sequential steps in the folate pathway of some gram-positive and gram-negative bacteria.

6.2.2.3 Glycopeptides

Glycopeptide antimicrobial agents, which include vancomycin (in the glycopeptide subclass) and teicoplanin (in the lipoglycopeptide subclass), share a complex chemical structure and a principal mode of action of inhibition of cell wall synthesis at a different site than that of the β -lactams. The activity of this group is directed primarily at aerobic, gram-positive bacteria. Vancomycin is an accepted agent for treatment of a gram-positive bacterial infection in the penicillin-allergic patient, and it is useful for therapy of infections due to β -lactam-resistant, gram-positive bacterial strains (eg, MRSA and some enterococci).

6.2.2.4 Lipopeptides

Lipopeptides are a structurally related group of antimicrobial agents, for which the principal target is the cell membrane. The polymyxin subclass, which includes polymyxin B and colistin, has activity against gram-negative organisms. Daptomycin is a cyclic lipopeptide with activity against gram-positive organisms. Lipopeptide activity is strongly influenced by the presence of divalent cations in the medium used to test them. The presence of excess calcium cations inhibits the activity of the polymyxins, whereas the presence of physiological levels (50 mg/L) of calcium ions is essential for the proper activity of daptomycin.

6.2.2.5 Macrolides

Macrolides are structurally related antimicrobial agents that inhibit bacterial protein synthesis at the ribosomal level. Several members of this class currently in use may need to be considered for testing against fastidious, gram-negative bacterial isolates. For gram-positive organisms, only erythromycin may need to be tested routinely. The macrolide group of antimicrobial agents consists of several subgroups, including azithromycin, clarithromycin, dirithromycin, the ketolide telithromycin, and the fluoroketolide solithromycin.

6.2.2.6 Nitroimidazoles

Nitroimidazoles, which include metronidazole and tinidazole, are bactericidal agents that are converted intracellularly in susceptible organisms to metabolites that disrupt the host deoxyribonucleic acid (DNA); they are active only against strictly anaerobic bacteria.

6.2.2.7 Oxazolidinones

The oxazolidinone class is a class of antimicrobial agents with a unique mechanism of action that inhibits protein synthesis. The first agent approved in this class was linezolid which has activity against gram-positive organisms.

6.2.2.8 Quinolones

Quinolones (quinolones and fluoroquinolones) are structurally related antimicrobial agents that function primarily by inhibiting the DNA-gyrase or topoisomerase activity of many gram-positive and gram-negative bacteria. Some differences in spectrum may require separate testing of the individual agents.

6.2.2.9 Streptogramins

Streptogramins, which include quinupristin-dalfopristin and linopristin-flopristin, are combinations of two cyclic peptides produced by *Streptomyces* spp. They work synergistically to inhibit protein synthesis, mainly in gram-positive organisms, although they do have limited activity against some gram-negative and anaerobic organisms.

6.2.2.10 Tetracyclines

Tetracyclines are structurally related antimicrobial agents that inhibit protein synthesis at the ribosomal level of certain gram-positive and gram-negative bacteria. Agents in this group are closely related and, with few exceptions, only tetracycline may need to be tested routinely. Organisms that are susceptible to tetracycline are also considered susceptible to doxycycline and minocycline. However, some organisms that are intermediate or resistant to tetracycline may be susceptible to doxycycline, minocycline, or both. Tigecycline, a glycylcycline, is a derivative of minocycline that has activity against organisms that may be resistant to tetracyclines.

6.2.2.11 Single-Drug Classes

The following antimicrobial agents (antimicrobial classes) are currently the only members of their respective classes used in humans that are included in this document, and are appropriate for *in vitro* testing. These include chloramphenicol (phenicols), clindamycin (lincosamides), fusidic acid (steroidal), mupirocin (pseudomonic acid), spectinomycin (aminocyclitols), all of which inhibit protein synthesis; and rifampin (ansamycins) and fidaxomicin (macrocyclus), which are ribonucleic acid (RNA) synthesis inhibitors. Nitrofurantoin (nitrofurans), which is used only in the therapy of urinary tract infections, acts by inhibiting several protein synthesis-and-assembly steps at the ribosomal level. Fosfomycin (fosfomycins), approved by the FDA for urinary tract infections only, inhibits enzymes involved in cell wall synthesis.

6.3 Selection Guidelines

To make routine susceptibility testing relevant and practical, the number of agents tested should be limited. M100² Tables 1A, 1B, and 1C list those agents that fulfill the basic requirements for routine use in most clinical laboratories. The tables are divided into columns based on specific organisms or organism

groups, and then the various drugs are indicated in priority for testing to assist laboratories in the selection of their routine testing batteries.

The listing of drugs together in a single box designates clusters of agents for which interpretive results (susceptible, intermediate, or resistant) and clinical efficacy are similar. Within each box, an “or” between agents designates those agents for which cross-resistance and cross-susceptibility are nearly complete. This means combined major and very major errors are fewer than 3% and minor errors are fewer than 10%, based on a large collection of random clinical isolates tested. In addition, to qualify for an “or,” at least 100 strains with resistance to the agents in question must be tested, and a result of “resistant” must be obtained with all agents for at least 95% of the strains. “Or” is also used for comparable antimicrobial agents when tested against organisms for which “susceptible-only” interpretive criteria are provided (eg, cefotaxime or ceftriaxone with *Haemophilus influenzae*). Thus, results from one agent connected by an “or” could be used to predict results for the other agent. For example, *Enterobacteriaceae* susceptible to cefotaxime can be considered susceptible to ceftriaxone. The results obtained from testing cefotaxime would be reported and a comment could be included on the report that the isolate is also susceptible to ceftriaxone. When no “or” connects agents within a box, testing of one agent cannot be used to predict results for another, either owing to discrepancies or insufficient data.

6.4 Suggested Guidelines for Routine and Selective Testing and Reporting

As listed in M100² Tables 1A, 1B, and 1C, agents in Group A are considered appropriate for inclusion in a routine, primary testing panel as well as for routine reporting of results for the specified organism groups.

Group B includes antimicrobial agents that may warrant primary testing, but they may be reported only selectively, such as when the organism is resistant to agents of the same class, as in Group A. Other indications for reporting the result might include a selected specimen source (eg, a third-generation cephalosporin for enteric bacilli from cerebrospinal fluid [CSF] or trimethoprim-sulfamethoxazole for urinary tract isolates); a polymicrobial infection; infections involving multiple sites; cases of patient allergy, intolerance, or failure to respond to an agent in Group A; or for purposes of infection control.

Group C includes alternative or supplemental antimicrobial agents that may require testing in those institutions that harbor endemic or epidemic strains resistant to several of the primary drugs (especially in the same class [eg, β -lactams or aminoglycosides]); for treatment of patients allergic to primary drugs; for treatment of unusual organisms (eg, chloramphenicol for extraintestinal isolates of *Salmonella* spp.); or for reporting to infection control as an epidemiological aid.

Group U includes certain antimicrobial agents (eg, nitrofurantoin and certain quinolones) that are used only or primarily for treating urinary tract infections; these agents should not be routinely reported against pathogens recovered from other sites of infection. Other agents with broader indications may be included in Group U for specific urinary pathogens (eg, *Pseudomonas aeruginosa* and ofloxacin).

Group O (“other”) includes antimicrobial agents that have a clinical indication for the organism group, but are generally not candidates for routine testing and reporting in the United States.

Group Inv. (“investigational”) includes antimicrobial agents that are investigational for the organism group and have not yet been approved by the FDA for use in the United States.

Each laboratory should decide which agents in M100² Tables 1A, 1B, and 1C to report routinely (Group A) and which might be reported only selectively (Group B) in consultation with the infectious disease practitioners, the pharmacy, as well as the pharmacy and therapeutics and infection control committees of the health care institution. Selective reporting should improve the clinical relevance of test reports and help minimize the selection of multiresistant, health care-associated strains by overuse of broad-spectrum

agents. Results for Group B agents tested but not reported routinely should be available on request. Unexpected resistance, when confirmed, should be reported (eg, resistance to a secondary agent but susceptibility to a primary agent). In addition, each laboratory should develop a protocol to address isolates that are confirmed as resistant to all agents on their routine test panel. This protocol should include options for testing additional agents in-house or sending the isolate to a reference laboratory.

7 Antimicrobial Agents

7.1 Source

Obtain antimicrobial standards or reference powders directly from the drug manufacturer, from the United States Pharmacopoeia (www.usp.org), or from certain other commercial sources. Parenteral preparations should not be used for susceptibility testing. Acceptable powders bear a label that states the drug's generic name, lot number, potency (usually expressed in micrograms [μg] or International Units [IU] per milligram of powder), and expiration date. Store the powders as recommended by the manufacturer or at $\leq -20^\circ\text{C}$ in a desiccator (preferably in a vacuum). When the desiccator is removed from the refrigerator or freezer, allow the desiccator to warm to room temperature before opening to avoid condensation of water.

7.2 Weighing Antimicrobial Powders

All antimicrobial agents are assayed for standard units of activity. The assay units may differ widely from the actual weight of the powder and often may differ between drug production lots. Thus, a laboratory must standardize its antimicrobial solutions based on assays of the lots of antimicrobial powders that are used to make stock solutions.

The value for potency supplied by the manufacturer should include consideration of measures of purity (usually by high-performance liquid chromatography [HPLC] assay), water content (eg, by Karl Fischer analysis or by weight loss on drying), and the salt/counter-ion fraction (if the compound is supplied as a salt instead of free acid or base). The potency may be expressed as a percentage, or in units of $\mu\text{g}/\text{mg}$ (w/w).

In some cases, a certificate of analysis with values for each of these components may be provided with antimicrobial agent powders; in this case, an overall value for potency may not be provided, but can be calculated from HPLC purity, water content, and, when applicable, the active fraction for drugs supplied as a salt (eg, hydrochloride, mesylate). However, when applying these calculations, if any value is unknown or is not clearly determined from the certificate of analysis, it is advisable that the factors used in this calculation be confirmed with the supplier or the manufacturer. The following demonstrates an example calculation:

Example: meropenem trihydrate

Certificate of analysis data:

Assay purity (by HPLC): 99.8%

Measured water content (by Karl Fischer analysis): 12.1% (w/w)

Active fraction: 100% (supplied as the free acid and not a salt)

Potency calculation from above data:

Potency = (Assay purity) • (Active fraction) • (1 – Water Content)

Potency = (998) • (1.0) • (1 – 0.121)

Potency = 877 $\mu\text{g}/\text{mg}$ or 87.7%

Use either of the following formulas to determine the amount of powder (1) or diluent (2) needed for a standard solution:

$$\text{Weight (mg)} = \frac{\text{Volume (mL)} \cdot \text{Concentration (}\mu\text{g/mL)}}{\text{Potency (}\mu\text{g/mg)}} \quad (1)$$

or

$$\text{Volume (mL)} = \frac{\text{Weight (mg)} \cdot \text{Potency (}\mu\text{g/mg)}}{\text{Concentration (}\mu\text{g/mL)}} \quad (2)$$

Weigh the antimicrobial powder on an analytical balance that has been calibrated by approved reference weights from a national metrology organization. If possible, weigh more than 100 mg of powder. It is advisable to accurately weigh a portion of the antimicrobial agent in excess of that required and to calculate the volume of diluent needed to obtain the final concentration desired as in formula 2 above.

Example: To prepare 100 mL of a stock solution containing 1280 $\mu\text{g/mL}$ of antimicrobial agent with antimicrobial powder that has a potency of 750 $\mu\text{g/mg}$, 170 mg to 200 mg of the antimicrobial powder should be accurately weighed. If the actual weight is 182.6 mg, the volume of diluent needed is as follows:

$$\text{Volume (mL)} = \frac{182.6 \text{ mg} \cdot 750 \mu\text{g/mg}}{\frac{\text{Actual Weight (Potency)}}{\text{Desired Concentration}}} = 107.0 \text{ mL} \quad (3)$$

Therefore, dissolve 182.6 mg of antimicrobial powder in 107.0 mL of diluent.

7.3 Preparing Stock Solutions

Prepare antimicrobial agent stock solutions at concentrations of at least 1000 $\mu\text{g/mL}$ (eg, 1280 $\mu\text{g/mL}$) or 10 times the highest concentration to be tested, whichever is greater. Some antimicrobial agents, however, have limited solubility and may require preparation of lower stock concentrations. In all cases, consider directions provided by the drug's manufacturer as part of determining solubility.

Some drugs must be dissolved in solvents other than water. In such cases:

- Use a minimum amount of solvent to solubilize the antimicrobial powder.
- Finish diluting the final stock concentration with water or the appropriate diluent as indicated in M100² Table 5A.
- For potentially toxic solvents, consult the material safety data sheets available from the manufacturer (see M100² Table 5A).

Because microbial contamination is extremely rare, solutions that have been prepared aseptically but not filter sterilized are generally acceptable. If desired, however, solutions may be sterilized by membrane filtration. Paper, asbestos, or sintered glass filters, which may adsorb appreciable amounts of certain antimicrobial agents, must *not* be used. Whenever filtration is used, it is important to document the absence of adsorption by appropriate assay procedures.

Dispense small volumes of the sterile stock solutions into sterile glass, polypropylene, polystyrene, or polyethylene vials; carefully seal; and store (preferably at -60°C or below, but never at a temperature warmer than -20°C and never in a self-defrosting freezer). Vials may be thawed as needed and used the same day. Any unused stock solution should be discarded at the end of the day. Stock solutions of most

antimicrobial agents can be stored at -60°C or below for six months or more without significant loss of activity. In all cases, directions provided by the drug's manufacturer must be considered in addition to these general recommendations. Any significant deterioration of an antimicrobial agent should be reflected in the results of susceptibility testing using QC strains.

7.4 Number of Concentrations Tested

The concentrations tested for a particular antimicrobial agent should encompass the interpretive breakpoints provided in M100² Tables 2A through 2J, but the number of concentrations tested is the decision of the laboratory. However, it is advisable to choose a range that allows at least one QC organism to have on-scale values. Unusual concentrations may be tested for special purposes (eg, high concentrations of gentamicin and streptomycin may be tested to indicate a synergistic effect with a penicillin or glycopeptide against enterococci).

8 Inoculum Preparation for Dilution Tests

8.1 Turbidity Standard for Inoculum Preparation

To standardize the inoculum density for a susceptibility test, use a BaSO_4 turbidity standard equivalent to a 0.5 McFarland standard or its optical equivalent (eg, latex particle suspension). Prepare a BaSO_4 0.5 McFarland standard as described in Appendix B. Alternatively, a photometric device can be used.

8.2 Inoculum Preparation

8.2.1 Direct Colony Suspension Method

The direct colony suspension method is the most convenient method for inoculum preparation. This method can be used with most organisms; it is the recommended method for testing the fastidious organisms, *Haemophilus* spp., *Neisseria gonorrhoeae*, *Neisseria meningitidis*, and streptococci (see Section 11), and for testing staphylococci for potential methicillin or oxacillin resistance.

1. Prepare the inoculum by making a direct broth or saline suspension of isolated colonies selected from an 18- to 24-hour agar plate (use a nonselective medium, such as blood agar).
2. Adjust the suspension to achieve a turbidity equivalent to a 0.5 McFarland standard. This results in a suspension containing approximately 1 to 2×10^8 colony-forming units (CFU)/mL for *Escherichia coli* ATCC^{®a} 25922. To perform this step accurately, use either a photometric device or, if performed visually, use adequate light to visually compare the inoculum tube and the 0.5 McFarland standard against a card with a white background and contrasting black lines.

8.2.2 Growth Method

The growth method can be used alternatively and is sometimes preferable when colony growth is difficult to suspend directly and a smooth suspension cannot be made. It can also be used for nonfastidious organisms (except staphylococci) when fresh (24-hour) colonies, as required for the direct colony suspension method, are not available.

1. Select at least three to five well-isolated colonies of the same morphological type from an agar plate culture. Touch the top of each colony with a loop or sterile swab and transfer the growth into a tube containing 4 to 5 mL of a suitable broth medium, such as tryptic soy broth.

^a ATCC is a registered trademark of the American Type Culture Collection.

2. Incubate the broth culture at $35 \pm 2^{\circ}\text{C}$ until it achieves or exceeds the turbidity of the 0.5 McFarland standard (usually two to six hours).
3. Adjust the turbidity of the actively growing broth culture with sterile saline or broth to achieve a turbidity equivalent to that of a 0.5 McFarland standard. This results in a suspension containing approximately 1 to 2×10^8 CFU/mL for *E. coli* ATCC® 25922. To perform this step accurately, use either a photometric device or, if performed visually, use adequate light to visually compare the inoculum tube and the 0.5 McFarland standard against a card with a white background and contrasting black lines.

9 Agar Dilution Procedure

The agar dilution method for determining antimicrobial susceptibility is a well-established technique.^{8,14} The antimicrobial agent is incorporated into the agar medium, with each plate containing a different concentration of the agent. The inocula can be applied rapidly and simultaneously to the agar surfaces using an inoculum-replicating apparatus¹⁵ capable of transferring 32 to 36 inocula to each plate.

The general procedure for performing agar dilution is provided in this section. Variations for testing fastidious organisms are provided in Section 11.

9.1 Reagents and Materials

9.1.1 Mueller-Hinton Agar

Of the many media available, the subcommittee considers MHA the best for routine susceptibility testing of nonfastidious bacteria for the following reasons:

- It shows acceptable batch-to-batch reproducibility for susceptibility testing.
- It is low in inhibitors that affect sulfonamide, trimethoprim, and tetracycline susceptibility test results.
- It supports satisfactory growth of most pathogens.
- Large bodies of data and experience have been collected about susceptibility tests performed with this medium.

Although MHA is generally reliable for susceptibility testing, results obtained with some batches may, on occasion, vary significantly. Prepare MHA as described in Appendix B. Only MHA formulations that have been tested according to, and that meet the acceptance limits described in, CLSI document M06¹⁶ should be used.

1. Performance test new lots of medium before use as outlined in Section 16.5.
2. See Appendix B for instructions for preparation of MHA.
3. Check the pH of each batch of MHA as described in Appendix B.
4. Additions to MHA that must be made for testing specific antimicrobial agents are:
 - 2% NaCl when testing staphylococci and oxacillin, nafcillin, and methicillin
 - 25 µg/mL of glucose-6-phosphate when testing fosfomycin
5. Do not add calcium or magnesium cations to MHA.

9.1.2 Agar Media for Testing Fastidious Organisms

Media for testing fastidious organisms using methods described in this document are:

- MHA with 5% sheep blood
- MHA with 5% aged sheep blood (≥ 2 weeks old) (for testing *Helicobacter pylori*)
- *Haemophilus* Test Medium (HTM)
- GC agar base + 1% defined growth supplement

See Appendix B for instructions for preparation of these media.

9.1.3 Inoculum Replicators

Most currently available inoculum replicators transfer 32 to 36 inocula to each plate.¹⁴ Replicators with pins 3 mm in diameter deliver approximately 2 μL (range 1 to 3 μL) onto the agar surface. Those with smaller 1-mm pins deliver tenfold less, approximately 0.1 to 0.2 μL .¹⁷

9.2 Preparing Agar Dilution Plates

9.2.1 Dilution Scheme for Reference Method

Prepare intermediate ($10\times$) antimicrobial agent solutions by making successive 1:2, 1:4, and 1:8 dilutions using the dilution format described in M100² Table 6A, or by making serial twofold dilutions. Then, add one part of the $10\times$ antimicrobial solution to nine parts of molten agar.

9.2.2 Procedure

1. Add appropriate dilutions (see Section 9.2.1) of antimicrobial solution to molten test agars that have been allowed to equilibrate in a water bath to 45 to 50°C.
2. Mix the agar and antimicrobial solution thoroughly and pour into Petri dishes on a level surface to result in an agar depth of 3 to 4 mm.
3. Pour the plates quickly after mixing to prevent cooling and partial solidification in the mixing container. Avoid generating bubbles when mixing.
4. Allow the agar to solidify at room temperature, and either use the plates immediately or store them in sealed plastic bags at 2 to 8°C for up to five days for reference work, or longer for routine tests. In one study, agar plates containing cefaclor had to be prepared within 48 hours of use because of degradation of the drug, whereas cefamandole remained stable for greater than the recommended five days.¹⁸ In addition to cefaclor, other antimicrobial agents that are particularly labile are ampicillin, clavulanic acid, imipenem, and methicillin.

NOTE: Do not assume all antimicrobial agents will maintain their potency under these storage conditions. The user should evaluate the stability of plates from results obtained with QC strains and should develop applicable shelf-life criteria. This information is sometimes available from pharmaceutical manufacturers.

5. Allow plates stored at 2 to 8°C to equilibrate to room temperature before use. Make certain that the agar surface is dry before inoculating the plates. If necessary, place the plates in an incubator or laminar flow hood for approximately 30 minutes with their lids ajar to hasten drying of the agar surface.

9.2.3 Quality Control Frequency

Although weekly QC, as described in this standard, is acceptable for most drug/agar combinations, some may require more frequent testing (eg, the labile drugs described in Section 9.2.2). See Section 16 for further details.

9.2.4 Control Plates

Use drug-free plates prepared from the base medium (with or without supplements as indicated in Section 9.1) as growth controls.

9.3 Preparing the Inoculum

9.3.1 Inoculum Preparation

Prepare a standardized inoculum for the agar dilution method by either growing microorganisms to the turbidity of the 0.5 McFarland standard or suspending colonies directly to achieve the same density as described in Section 8. Preparation of the initial inoculum and final dilution may vary for certain fastidious organisms, such as *N. meningitidis* (see Section 11 and Appendix C).

9.3.2 Dilution of Inoculum Suspension

Cultures adjusted to the 0.5 McFarland standard contain approximately 1 to 2×10^8 CFU/mL with most species, and the final inoculum required is 10^4 CFU per spot of 5 to 8 mm in diameter. When using replicators with 3-mm pins that deliver 2 μ L, dilute the 0.5 McFarland suspension 1:10 in sterile broth or saline to obtain a concentration of 10^7 CFU/mL. The final inoculum on the agar will be approximately 10^4 CFU per spot. When using replicators with 1-mm pins that deliver 0.1 to 0.2 μ L, do not dilute the initial suspension. Optimally, use the adjusted suspensions for final inoculation within 15 minutes of preparation.

9.4 Inoculating Agar Dilution Plates

1. Arrange the tubes containing the adjusted and diluted bacterial suspensions (10^7 CFU/mL) in order in a rack. Place an aliquot of each well-mixed suspension into the corresponding well in the replicator inoculum block.
2. Mark the agar plates for orientation of the inoculum spots.
3. Apply an aliquot of each inoculum to the agar surface, either with an inocula-replicating device or with standardized loops or pipettes. Appropriate dilution of the inoculum suspension should be made depending on the volume of inoculum delivery so as to obtain a final concentration of 10^4 CFU/spot (see Section 9.3.2).
4. Inoculate a growth-control plate (no antimicrobial agent) first and then, starting with the lowest concentration, inoculate the plates containing the different antimicrobial concentrations. Inoculate a second growth control plate last to ensure there was no contamination or significant antimicrobial carryover during the inoculation.
5. Streak a sample of each inoculum on a suitable nonselective agar plate and incubate overnight to detect mixed cultures and to provide freshly isolated colonies in case retesting proves necessary.

9.5 Incubating Agar Dilution Plates

1. Allow the inoculated plates to stand at room temperature until the moisture in the inoculum spots has been absorbed into the agar, ie, until the spots are dry, but no more than 30 minutes. Invert the plates and incubate at $35 \pm 2^\circ\text{C}$ for 16 to 20 hours (see Sections 11 and 12, and Appendix C for exceptions).
2. Do not incubate the plates in an atmosphere with increased CO_2 when testing nonfastidious organisms, because the surface pH may be altered. However, incubate *N. gonorrhoeae*, *N. meningitidis*, and streptococci in an atmosphere containing 5% CO_2 . (See Section 11, Appendix C, and M100² Tables 2A through 2I.)

9.6 Determining Agar Dilution End Points

1. Place the plates on a dark, nonreflecting surface to determine the end points. Record the MIC as the lowest concentration of antimicrobial agent that completely inhibits growth, disregarding a single colony or a faint haze caused by the inoculum. With trimethoprim and the sulfonamides, antagonists in the medium may allow some slight growth; therefore, read the end point at the concentration in which there is 80% or greater reduction in growth as compared to the control.
2. If two or more colonies persist in concentrations of the agent beyond an obvious end point, or if there is no growth at lower concentrations but growth at higher concentrations, check the culture purity and repeat the test if required.

10 Broth Dilution Procedures (Macrodilution and Microdilution)

The general procedures for performing broth dilution are provided in this section. Variations for testing fastidious organisms are provided in Section 11.

10.1 Mueller-Hinton Broth

MHB is recommended as the medium of choice for susceptibility testing of commonly isolated, rapidly growing aerobic or facultative organisms⁸ because:

- It shows acceptable batch-to-batch reproducibility for susceptibility testing.
 - It is low in inhibitors that affect sulfonamide, trimethoprim, and tetracycline susceptibility test results.
 - It supports satisfactory growth of most pathogens.
 - A large body of data and experience has been gathered about tests performed with this medium.
1. The medium of choice for routine broth dilution testing is cation-adjusted MHB (CAMHB). Instructions for preparation are provided in Appendix B.
 2. Check the pH of each batch of MHB as described in Appendix B.
 3. Evaluate the MIC performance characteristics of each batch of broth using a standard set of QC organisms (see Section 16.3). If a new lot of MHB does not yield the expected MICs, the cation content should be investigated along with other variables and components of the test.
 4. To determine the suitability of the medium for sulfonamide and trimethoprim tests, perform MICs with *Enterococcus faecalis* ATCC[®] 29212. The end points should be easy to read (as 80% or greater reduction in growth as compared to the control). If the MIC for trimethoprim-sulfamethoxazole is $\leq 0.5/9.5 \mu\text{g/mL}$, the medium may be considered adequate.

5. Additions to CAMHB that must be made for testing specific antimicrobial agents are:

- 2% NaCl when testing oxacillin and staphylococci
- 0.002% polysorbate-80 (P-80) when testing dalbavancin and oritavancin
- Additional calcium to equal 50 mg/L when testing daptomycin

10.2 Broth Media for Fastidious Organisms

Media that can be used for testing fastidious organisms using methods described in this document are:

- CAMHB + 2.5% to 5% lysed horse blood (LHB)
- HTM broth

Instructions for preparation of these media are provided in Appendix B.

10.3 Macrodilution (Tube) Broth Method

10.3.1 Preparing and Storing Diluted Antimicrobial Agents

1. Use sterile 13- × 100-mm test tubes to conduct the test. If the tubes are to be saved for later use, be sure they can be frozen.
2. Close the tubes with loose screw caps, plastic or metal closure caps, or cotton plugs.
3. Use a growth control tube containing broth without antimicrobial agent for each organism tested.
4. Prepare the final twofold (or other) dilutions of antimicrobial agent volumetrically in the broth. The schedule in M100² Tables 7A and 7B provides a convenient and reliable procedure for preparing dilutions. A minimum final volume of 1 mL of each dilution is needed for the test. Use one pipette for measuring all diluents and then for adding the stock antimicrobial solution to the first tube. For each subsequent dilution step, use a new pipette. Because there is a 1:2 dilution of the drugs when an equal volume of inoculum is added, the antimicrobial dilutions are often prepared at double the desired final concentration.
5. Use the tubes on the day of preparation or immediately place them in a freezer at $\leq -20^{\circ}\text{C}$ (preferably at $\leq -60^{\circ}\text{C}$) until needed. Although the antimicrobial agents in frozen tubes usually remain stable for several months, certain agents (eg, clavulanic acid and imipenem) are more labile than others and should be stored at $\leq -60^{\circ}\text{C}$. Do not store the tubes in a self-defrosting freezer. Thawed antimicrobial solutions must not be refrozen; repeated freeze-thaw cycles accelerate the degradation of some antimicrobial agents, particularly β -lactams.

10.3.2 Inoculum Preparation and Inoculation

1. Prepare a standardized inoculum using either the direct colony suspension or growth method as described in Section 8.
2. Optimally within 15 minutes of preparation, dilute the adjusted inoculum suspension in broth so, after inoculation, each tube contains approximately 5×10^5 CFU/mL. This can be accomplished by diluting the 0.5 McFarland suspension 1:150, resulting in a tube containing approximately 1×10^6 CFU/mL. The subsequent 1:2 dilution in step 3 brings the final inoculum to 5×10^5 CFU/mL.

3. Within 15 minutes after the inoculum has been standardized as described above, add 1 mL of the adjusted inoculum to each tube containing 1 mL of antimicrobial agent in the dilution series (and a positive control tube containing only broth), and mix. This results in a 1:2 dilution of each antimicrobial concentration and a 1:2 dilution of the inoculum. It is advisable to perform a purity check of the inoculum suspension by subculturing an aliquot onto a nonselective agar plate for simultaneous incubation.

10.4 Broth Microdilution Method

This method is called “microdilution” because it involves the use of small volumes of broth dispensed in sterile, plastic microdilution trays that have round or conical bottom wells. Each well should contain 0.1 mL of broth. The broth microdilution method described in M07 is the same methodology outlined in ISO 20776-1.¹

10.4.1 Preparing and Storing Diluted Antimicrobial Agents

1. To prepare microdilution trays, make intermediate twofold (or other) dilutions of antimicrobial agent volumetrically in broth or sterile water. For the intermediate (10×) antimicrobial solutions, dilute the concentrated antimicrobial stock solution (see Section 7.3) as described in M100² Tables 7A and 7B or by making serial twofold dilutions. Use one pipette for measuring all diluents and then for adding the stock antimicrobial solution to the first tube. For each subsequent dilution step, use a new pipette. Dispense the antimicrobial/broth solutions into the plastic microdilution trays.
2. The dilution scheme outlined in M100² Table 7B must be used for diluting water-insoluble agents, such as dalbavancin.
3. The most convenient method of preparing microdilution trays is by use of a dispensing device and antimicrobial dilutions made in at least 10 mL of broth. The dispensing device then delivers 0.1 (±0.02) mL into each of the 96 wells of a standard tray.
4. If the inoculum is added by pipette as described in Section 10.4.2 (3), prepare the antimicrobial solutions at twice the desired final concentration, and fill the wells with 0.05 mL instead of 0.1 mL. Each tray should include a growth control well and a sterility (uninoculated) well.
5. Seal the filled trays in plastic bags and immediately place in a freezer at ≤ −20°C (preferably at ≤ −60°C) until needed. Although the antimicrobial agents in frozen trays usually remain stable for several months, certain agents (eg, clavulanic acid, imipenem) are more labile than others and should be stored at ≤ −60°C. Do not store the trays in a self-defrosting freezer. Thawed antimicrobial solutions must not be refrozen; repeated freeze-thaw cycles accelerate the degradation of some antimicrobial agents, particularly β-lactams.

10.4.2 Inoculum Preparation and Inoculation

1. Prepare a standardized inoculum using either the direct colony suspension or growth method as described in Section 8.
2. Optimally within 15 minutes of preparation, dilute the adjusted inoculum suspension in water, saline, or broth so, after inoculation, each well contains approximately 5×10^5 CFU/mL (range 2 to 8×10^5 CFU/mL). The dilution procedure to obtain this final inoculum varies according to the method of delivery of the inoculum to the individual wells and it must be calculated for each situation. For microdilution tests, the exact inoculum volume delivered to the wells must be known to make this calculation. For example, if the volume of broth in the well is 0.1 mL and the inoculum volume is 0.01 mL, then the 0.5 McFarland suspension (1×10^8 CFU/mL) should be diluted 1:20 to yield

5×10^6 CFU/mL. When 0.01 mL of this suspension is inoculated into the broth, the final test concentration of bacteria is approximately 5×10^5 CFU/mL (or 5×10^4 CFU/well in the microdilution method).

3. Within 15 minutes after the inoculum has been standardized as described above, inoculate each well of a microdilution tray using an inoculator device that delivers a volume that does not exceed 10% of the volume in the well (eg, ≤ 10 μ L of inoculum in 0.1 mL antimicrobial agent solution). Conversely, if a 0.05-mL pipette is used, it results in a 1:2 dilution of the contents of each well (containing 0.05 mL), as with the macrodilution procedure.
4. It is advisable to perform a purity check of the inoculum suspension by subculturing an aliquot onto a nonselective agar plate for simultaneous incubation.

10.5 Colony Counts of Inoculum Suspensions

Laboratories are encouraged to perform colony counts on inoculum suspensions at least quarterly to ensure that the final inoculum concentration routinely obtained closely approximates 5×10^5 CFU/mL for *E. coli* ATCC® 25922. Do this by removing a 0.01-mL aliquot from the growth control well or tube immediately after inoculation and diluting it in 10 mL of saline (1:1000 dilution). After mixing, remove a 0.1-mL aliquot and spread it over the surface of a suitable agar medium. After incubation, the presence of approximately 50 colonies indicates an inoculum density of 5×10^5 CFU/mL.

10.6 Incubation

1. Incubate the inoculated macrodilution tubes or microdilution trays at $35 \pm 2^\circ\text{C}$ for 16 to 20 hours in an ambient air incubator (for exceptions, see Sections 11 and 12, and Appendix C) within 15 minutes of adding the inoculum. To maintain the same incubation temperature for all cultures, do not stack microdilution trays more than four high.
2. To prevent drying, seal each tray in a plastic bag, with plastic tape, or with a tight-fitting plastic cover before incubating.
3. Incubation times may differ for fastidious organisms or some problem organisms with difficult-to-detect resistance; follow specific instructions in Appendix C, in Sections 11 to 13, or the appropriate table in M100.²

10.7 Determining Minimal Inhibitory Concentration End Points

The MIC is the lowest concentration of antimicrobial agent that completely inhibits growth of the organism in the tubes or microdilution wells as detected by the unaided eye. Viewing devices intended to facilitate reading microdilution tests and recording of results may be used as long as there is no compromise in the ability to discern growth in the wells.

1. Compare the amount of growth in the wells or tubes containing the antimicrobial agent with the amount of growth in the growth-control wells or tubes (no antimicrobial agent) used in each set of tests when determining the growth end points. For a test to be considered valid, acceptable growth (≥ 2 mm button or definite turbidity) must occur in the growth-control well.
2. With trimethoprim and the sulfonamides, antagonists in the medium may allow some slight growth; therefore, read the end point at the concentration in which there is $\geq 80\%$ reduction in growth as compared to the control.

3. When a single skipped well occurs in a microdilution test, read the highest MIC. Do not report results for drugs for which there is more than one skipped well.
4. Generally, microdilution MICs for gram-negative bacilli are the same or one twofold dilution lower than the comparable macrodilution MICs.¹⁹

11 Fastidious Organisms

Mueller-Hinton medium described previously for the rapidly growing aerobic pathogens is not adequate for susceptibility testing of fastidious organisms. If MIC tests are performed with fastidious organisms, the medium, QC procedures, and interpretive criteria must be modified to fit each organism. Dilution tests for the following organisms are described here:

- *H. influenzae* and *Haemophilus parainfluenzae*, using HTM (broth dilution only)
- *N. gonorrhoeae*, using GC agar base medium
- Streptococci, using LHB-supplemented CAMHB
- *N. meningitidis*, using LHB-supplemented CAMHB or MHA with sheep blood

Details for the conditions required for these tests are summarized in Appendix C and further explained below.

Certain fastidious bacteria other than those listed above may be tested by a dilution or disk diffusion method as described in CLSI document M45.⁷ Methods for testing anaerobic bacteria are provided in CLSI document M11.⁶

11.1 *Haemophilus influenzae* and *Haemophilus parainfluenzae*

When the designation *Haemophilus* spp. is used in Sections 11.1.1 and 11.1.2, it applies only to these two species. MIC testing of *H. influenzae* and *H. parainfluenzae* using HTM²⁰ has been developed only for broth dilution as described in Section 11.1.1. The agar dilution method using HTM has not been validated. Details of the medium preparation are described in Appendix B. If sulfonamides or trimethoprim are tested, add 0.2 IU thymidine phosphorylase to the medium aseptically. The pH should be 7.2 to 7.4.

H. influenzae ATCC® 10211 is recommended as a useful additional QC strain to verify the growth promotion properties of HTM. In particular, manufacturers of HTM are encouraged to use *H. influenzae* ATCC® 10211 as a QC test strain.

11.1.1 Test Procedure

Follow the general test procedure in Section 10 with the following exceptions:

1. Use the direct colony suspension procedure when testing *Haemophilus* spp. Using colonies taken directly from an overnight (preferably 20- to 24-hour) chocolate agar culture plate, prepare a suspension of the test organism in MHB or saline. Adjust the suspension with broth or saline using a photometric device to achieve a turbidity equivalent to a 0.5 McFarland standard. Suspensions prepared from a 20- to 24-hour agar plate contain approximately 1 to 2×10^8 CFU/mL, whereas those prepared from a 16- to 18-hour plate contain more cells (approximately 3 to 4×10^8 CFU/mL). Exercise care in preparing this suspension, considering the age of the plate from which the inoculum is prepared; inoculum concentrations higher than 5×10^5 CFU/mL may lead to false-resistant results with some β -lactam antimicrobial agents, particularly when β -lactamase-producing strains of *H.*

influenzae are tested. Use the suspension for tray inoculation within 15 minutes after adjusting the turbidity.

2. Incubate the trays or tubes at $35 \pm 2^\circ\text{C}$ in ambient air for 20 to 24 hours before reading the MICs.

11.1.2 Minimal Inhibitory Concentration Interpretive Criteria

The antimicrobial agents suggested for routine testing of *Haemophilus* spp. are listed in M100² Table 1B. Specific MIC interpretive criteria to be used when testing *Haemophilus* spp. are listed in M100² Table 2E.

11.2 *Neisseria gonorrhoeae*

MIC testing of *N. gonorrhoeae* has been developed only for agar dilution using GC agar base and 1% defined growth supplement as described in Appendix B and M100² Table 2F.

11.2.1 Test Procedure

Follow the general test procedure in Section 9 with the following exceptions:

1. Use the direct colony suspension procedure when testing *N. gonorrhoeae*. Using colonies taken directly from an overnight chocolate agar culture plate, prepare a suspension equivalent to that of a 0.5 McFarland standard in MHB or 0.9% phosphate buffered saline, pH 7.0. Use the suspension for plate inoculation within 15 minutes after adjusting the turbidity.
2. Incubate the plates at $36 \pm 1^\circ\text{C}$ (do not exceed 37°C) in an atmosphere of 5% CO_2 for 20 to 24 hours before reading the MICs.

11.2.2 Minimal Inhibitory Concentration Interpretive Criteria

The antimicrobial agents suggested for routine testing of *N. gonorrhoeae* are indicated in M100² Table 1B. Specific MIC interpretive criteria to be used when testing *N. gonorrhoeae* are listed in M100² Table 2F.

11.3 *Neisseria meningitidis*

Recommended precautions: Perform all antimicrobial susceptibility testing (AST) of *N. meningitidis* in a biological safety cabinet (BSC).²¹⁻²³ Manipulating *N. meningitidis* outside a BSC is associated with increased risk for contracting meningococcal disease. Laboratory-acquired meningococcal disease is associated with a case fatality rate of 50%. Exposure to droplets or aerosols of *N. meningitidis* is the most likely risk for laboratory-acquired infection. Rigorous protection from droplets or aerosols is mandated when microbiological procedures (including AST) are performed on *N. meningitidis* isolates.

If a BSC is unavailable, manipulation of these isolates should be minimized, limited to Gram staining or serogroup identification using phenolized saline solution, while wearing a laboratory coat and gloves and working behind a full face splash shield. Use BSL-3 practices, procedures, and containment equipment for activities with a high potential for droplet or aerosol production and for activities involving production quantities or high concentrations of infectious materials. If BSL-2 or BSL-3 facilities are not available, forward isolates to a reference or public health laboratory with a minimum of BSL-2 facilities.

Laboratorians who are exposed routinely to potential aerosols of *N. meningitidis* should consider vaccination according to the current recommendations of the CDC Advisory Committee on Immunization Practices (<http://www.cdc.gov/vaccines/recs/acip>). Vaccination decreases but does not eliminate the risk of infection, because it is less than 100% effective and does not provide protection against serogroup B, a frequent cause of laboratory-acquired cases.

Broth microdilution and agar dilution susceptibility testing of *N. meningitidis* have been validated for detection of possible emerging resistance. To date, resistance has mostly been found in older agents used for therapy (penicillin or ampicillin), or agents used for prophylaxis of case contacts. Because resistance to antimicrobial agents such as ceftriaxone or cefotaxime that are often employed for therapy of invasive diseases has not been detected, routine testing of isolates by clinical laboratories is not necessary.

Broth microdilution testing of *N. meningitidis* isolates is performed using CAMHB with 2% to 5% LHB; alternatively, agar dilution can be performed using MHA supplemented with 5% sheep blood. Details for preparation are provided in Appendix B.

11.3.1 Test Procedure

Follow the general test procedures as described in Sections 9 or 10 with the following exceptions:

1. Use the direct colony suspension procedure as described in Section 8.2.1 when testing *N. meningitidis*. Using colonies taken directly from an overnight (20- to 24-hour) enriched chocolate agar plate incubated at $35 \pm 2^\circ\text{C}$ in 5% CO_2 , prepare a suspension in saline to achieve a turbidity equivalent to a 0.5 McFarland standard. Use the suspension for tray or plate inoculation within 15 minutes after adjusting the turbidity.
2. Incubate trays or agar plates at $35 \pm 2^\circ\text{C}$ in 5% CO_2 for 20 to 24 hours before reading the MICs.

11.3.2 Minimal Inhibitory Concentration Interpretive Criteria

Specific MIC interpretive criteria and QC organisms used when testing *N. meningitidis* are listed in M100² Table 2I.

11.4 *Streptococcus pneumoniae* and Other *Streptococcus* spp.

MIC testing of *Streptococcus* spp. using CAMHB with 2.5% to 5% LHB has been developed only for broth dilution. Details of the medium preparation are described in Appendix B. Recent studies using the agar dilution method have not been performed and reviewed by the subcommittee.

11.4.1 Test Procedure

For broth dilution testing, follow the general test procedure in Section 10 with the following exceptions:

1. Use the direct colony suspension procedure as described in Section 8.2.1 when testing streptococci. Using colonies taken directly from an overnight (18- to 20-hour) sheep blood agar culture plate, prepare a suspension equivalent to that of a 0.5 McFarland standard in either MHB or saline. Use the suspension for inoculation within 15 minutes after adjusting the turbidity.
2. Incubate the trays in ambient air at $35 \pm 2^\circ\text{C}$ for 20 to 24 hours before reading the MICs.
3. Inducible clindamycin resistance can be identified in β -hemolytic streptococci using the methods described in Section 13.

Recent studies using the agar dilution method have not been performed and reviewed by the subcommittee. If agar dilution is performed, follow the general test procedure in Section 9 with the exceptions noted above and incubate in 5% CO_2 if necessary for growth.

11.4.2 Minimal Inhibitory Concentration Interpretive Criteria

The antimicrobial agents suggested for routine testing of pneumococci and other streptococci are indicated in M100² Table 1B. Specific MIC interpretive criteria used when testing *Streptococcus pneumoniae* and other streptococci are listed in M100² Tables 2G, and 2H-1 and 2H-2, respectively.

12 Organisms Requiring Special Consideration

This section discusses organism groups or particular resistance mechanisms for which there are significant testing issues. Testing issues regarding both dilution and disk diffusion testing are discussed here.

12.1 Staphylococci

12.1.1 Penicillin Resistance and β -lactamase

Most staphylococci are resistant to penicillin, and penicillin is rarely an option for treatment of staphylococcal infections. Penicillin-resistant strains of staphylococci produce β -lactamase, and testing penicillin instead of ampicillin is preferred. Penicillin should be used to test the susceptibility of all staphylococci to all penicillinase-labile penicillins, such as amoxicillin, ampicillin, azlocillin, carbenicillin, mezlocillin, piperacillin, and ticarcillin.

Some β -lactamase-producing staphylococcal isolates test susceptible to penicillin. Because staphylococcal β -lactamase is readily inducible, there is a risk of this occurring if penicillin were used to treat such strains. For this reason it is recommended that isolates of *Staphylococcus* with penicillin MICs ≤ 0.12 $\mu\text{g/mL}$ or zone diameters ≥ 29 mm be tested for β -lactamase production before reporting the isolate as penicillin susceptible. Several tests for β -lactamase production have been described. These include nitrocefin-based tests or evaluating the zone edge of a penicillin disk diffusion test (ie, a fuzzy zone edge indicates no β -lactamase production, whereas a sharp edge indicates β -lactamase production).²⁴ The penicillin disk diffusion zone-edge test was more sensitive than nitrocefin-based tests for detection of β -lactamase production in *S. aureus*. The penicillin zone-edge test is recommended if only one test is used for β -lactamase detection in *S. aureus*. However, some laboratories may choose to perform a nitrocefin-based test first, and if this test is positive report the results as positive for β -lactamase (or penicillin resistant). If the nitrocefin test is negative, it is recommended that the penicillin disk diffusion zone-edge test be performed before reporting penicillin susceptibility results in cases in which penicillin may be used for therapy for *S. aureus*. For coagulase-negative staphylococci, including *Staphylococcus lugdunensis*, only nitrocefin-based tests are recommended. For the most current recommendations to detect β -lactamases in staphylococcal species, see M100² Table 2C.

12.1.2 Methicillin/Oxacillin Resistance

12.1.2.1 Background

Historically, resistance to the antistaphylococcal, penicillinase-stable penicillins (eg, methicillin, nafcillin, and oxacillin) has been referred to as “methicillin resistance,” and the acronyms “MRSA” (for methicillin-resistant *S. aureus*) or “MRS” (for methicillin-resistant staphylococci) are still commonly used, even though methicillin is no longer the agent of choice for testing or treatment. In this document, resistance to these agents may be referred to using several terms (eg, “MRS,” “methicillin resistance,” or “oxacillin resistance”). Most resistance to oxacillin in staphylococci is mediated by the *mecA* gene, which directs the production of a supplemental penicillin-binding protein (PBP), PBP 2a, and is expressed either homogeneously or heterogeneously. Homogeneous resistance is easily detected with standard testing methods, whereas heterogeneous expression may be more difficult to detect with some methods because only a fraction of the population (eg, 1 in 100 000 cells) expresses the resistance phenotype. In the past,

the presence of resistance to other classes of agents suggested an isolate was oxacillin resistant. However, some MRSA, such as those found in community-associated infections, may not be multidrug resistant.

12.1.2.2 Organism Groups

S. lugdunensis are now grouped with *S. aureus* when determining methicillin/oxacillin resistance. Most *S. lugdunensis* are β -lactamase negative and nearly all are oxacillin susceptible. Oxacillin-susceptible, *mecA*-negative strains exhibit oxacillin MICs in the range of 0.25 $\mu\text{g/mL}$ to 1 $\mu\text{g/mL}$, whereas *mecA*-positive strains usually exhibit MICs $\geq 4 \mu\text{g/mL}$, characteristics more like *S. aureus* than other coagulase-negative staphylococci. Therefore, the presence of *mecA*-mediated resistance in *S. lugdunensis* is detected more accurately using the *S. aureus* interpretive criteria than the criteria for coagulase-negative staphylococci. *S. lugdunensis* should be assumed to be included with *S. aureus* in this section and in M100² Table 2C. Oxacillin and cefoxitin testing methods for coagulase-negative staphylococci exclude *S. lugdunensis*.

12.1.2.3 Methods for Detection of Oxacillin Resistance

Either oxacillin- or cefoxitin-based methods can be used for detection of *mecA*-mediated resistance in staphylococci. Oxacillin disk diffusion methods should not be used for *S. lu gdunensis* and other coagulase-negative staphylococci. Cefoxitin-based methods predict the presence of *mecA*-mediated resistance only; their use is preferred to tests using oxacillin because they are better predictors of the presence of *mecA* than are oxacillin-based methods, including the oxacillin-salt agar screening plate. Because of the rare occurrence of oxacillin resistance mechanisms other than *mecA* in *S. aureus*, some *S. aureus* may be encountered that are oxacillin resistant but *mecA* negative; these generally test as cefoxitin susceptible.

- All methods require the use of the direct colony suspension method for the preparation of inoculum (see Section 8.2).
- Incubate tests to detect MRS for a full 24 hours at $35 \pm 2^\circ\text{C}$ when using oxacillin (testing at temperatures above 35°C may not detect MRS, especially when using oxacillin) before reporting as susceptible. Incubate tests using cefoxitin for 16 to 20 hours for *S. aureus* and *S. lugdunensis* and 24 hours for coagulase-negative staphylococci.
- For the most current recommendations regarding testing and reporting, refer to M100² Table 2C.

12.1.2.4 Oxacillin-Based Methods

- Of the penicillinase-stable penicillins, oxacillin is preferred for *in vitro* testing. Oxacillin is more resistant to degradation in storage and is more likely to detect heteroresistant staphylococcal strains. Cloxacillin should not be used because it may not detect oxacillin-resistant *S. aureus*. Oxacillin susceptibility test results can be applied to the other penicillinase-stable penicillins (eg, cloxacillin, dicloxacillin, flucloxacillin, methicillin, nafcillin).
- The addition of NaCl (2% w/v; 0.34 mol/L) is required for both agar and broth dilution testing of oxacillin to improve the detection of heteroresistant MRSA. For disk diffusion testing, MHA should not be supplemented.
- For disk diffusion testing, if an oxacillin disk is used, examine the zone of inhibition around the oxacillin disk for light growth using transmitted light (plate held up to light); any discernable growth within the zone of inhibition of oxacillin is indicative of oxacillin resistance.

- If oxacillin-intermediate results (disk diffusion testing) are obtained for *S. aureus*, perform testing for *mecA* or PBP 2a, the cefoxitin MIC or cefoxitin disk test, an oxacillin MIC test, or the oxacillin-salt agar screening test. Report the result of the alternative test rather than the oxacillin-intermediate result (see Section 12.1.2.5 for reporting oxacillin when using cefoxitin as a surrogate test).

12.1.2.5 Cefoxitin-Based Methods

- The results of tests using cefoxitin (either broth microdilution or disk diffusion tests using a 30- μ g cefoxitin disk) and alternative breakpoints (see M100² Table 2C) can be used to predict *mecA*-mediated oxacillin resistance in *S. aureus*. Cefoxitin tests are equivalent to oxacillin MIC tests in sensitivity and specificity for *S. aureus*.
- For coagulase-negative staphylococci, currently only the cefoxitin disk diffusion test has been validated for prediction of *mecA*-mediated resistance. The cefoxitin disk diffusion test has equivalent sensitivity to oxacillin MIC tests but greater specificity (ie, the cefoxitin disk test more accurately identifies oxacillin-susceptible strains than the oxacillin MIC test). There are no oxacillin disk diffusion recommendations for coagulase-negative staphylococci.
- For both *S. aureus* and coagulase-negative staphylococci, the cefoxitin disk test is easier to read than the oxacillin disk test; thus, cefoxitin is the preferred disk when performing disk diffusion.
- For disk diffusion testing of *S. lugdunensis*, only the cefoxitin disk test should be used.
- For all staphylococci, read the zone of inhibition around the cefoxitin disk using reflected light.
- Cefoxitin is used as a surrogate for detecting oxacillin resistance. Based on the cefoxitin result, report oxacillin as susceptible or resistant.

12.1.2.6 Molecular Detection Methods

Tests for the *mecA* gene or the protein produced by *mecA*, the PBP 2a (also called PBP2'), are the most accurate methods for prediction of resistance to oxacillin.

12.1.2.7 Reporting

- Resistance may be reported any time growth is observed after a minimum of 16 hours of incubation.
- If a cefoxitin-based test is used, cefoxitin is used as a surrogate for detecting oxacillin resistance. Based on the cefoxitin result, report oxacillin as susceptible or resistant.
- Report isolates of staphylococci that carry *mecA*, or that produce PBP 2a, the *mecA* gene product, as oxacillin resistant. Report isolates that do not carry *mecA* or do not produce PBP 2a as oxacillin susceptible if oxacillin MICs are ≤ 2 μ g/mL. Because of the rare occurrence of resistance mechanisms other than *mecA*, report isolates that are negative for the *mecA* gene or do not produce PBP 2a, but for which oxacillin MICs are ≥ 4 μ g/mL as oxacillin resistant.
- Oxacillin MIC interpretive criteria for coagulase-negative staphylococci correlate with the presence or absence of the gene encoding *mecA* for *Staphylococcus epidermidis*. These interpretive criteria may overcall resistance for other coagulase-negative staphylococci (eg, *Staphylococcus saprophyticus*). For coagulase-negative staphylococci other than *S. epidermidis*, testing for *mecA* or PBP 2a may be appropriate for strains for which the oxacillin MICs are 0.5 to 2.0 μ g/mL. For coagulase negative staphylococci, the cefoxitin disk test is preferred (see Section 12.1.2.5).

- Report oxacillin-resistant staphylococci as resistant to all other penicillins, carbapenems, cepheims (with the exception of cephalosporins with anti-MRSA activity), and β -lactam/ β -lactamase inhibitors, regardless of *in vitro* test results for those agents. This recommendation is based on the fact that most cases of documented MRS infections have responded poorly to β -lactam therapy, or because convincing clinical data have yet to be presented that document clinical efficacy for those agents in MRS infections.
- For oxacillin-susceptible strains, report results for cepheims, β -lactam/ β -lactamase inhibitor combinations, and carbapenems, if tested, according to the results generated using routine interpretive criteria.

12.1.3 Vancomycin Resistance in *Staphylococcus aureus*

In 2006 (M100-S16), the interpretive criteria for vancomycin and *S. aureus* were lowered to ≤ 2 $\mu\text{g/mL}$ for susceptible, 4 to 8 $\mu\text{g/mL}$ for intermediate, and ≥ 16 $\mu\text{g/mL}$ for resistant.²⁵ For coagulase-negative staphylococci, they remain at ≤ 4 $\mu\text{g/mL}$ for susceptible, 8 to 16 $\mu\text{g/mL}$ for intermediate, and ≥ 32 $\mu\text{g/mL}$ for resistant.

The first occurrence of a strain of *S. aureus* with reduced susceptibility to vancomycin (MICs 4 to 16 $\mu\text{g/mL}$) was reported from Japan in 1997,²⁶ followed by reports from the United States and France.²⁷ The exact mechanisms of resistance that result in elevated MICs are unknown, although they likely involve alterations in the cell wall and changes in several metabolic pathways. To date, most vancomycin-intermediate *S. aureus* strains appear to have developed from MRSA.

Since 2002, *S. aureus* strains for which the vancomycin MICs ranged from 32 to 1024 $\mu\text{g/mL}$ have been reported in the United States. All of these strains contained a *vanA* gene similar to that found in enterococci.^{28,29} These strains are reliably detected by the broth microdilution reference method, the disk diffusion method, and the vancomycin agar screen test (see Section 12.1.3.2) when the tests are incubated for a full 24 hours at $35 \pm 2^\circ\text{C}$.

12.1.3.1 Methods for Detection of Reduced Susceptibility to Vancomycin

S. aureus with vancomycin MICs ≥ 32 $\mu\text{g/mL}$ can be detected by either MIC, disk diffusion, or the vancomycin agar screen test. In order to recognize strains of staphylococci for which the vancomycin MICs are 4 to 16 $\mu\text{g/mL}$, MIC testing must be performed and the tests incubated for a full 24 hours at $35 \pm 2^\circ\text{C}$. Strains with vancomycin MICs < 32 $\mu\text{g/mL}$ are not detected by disk diffusion, even with 24-hour incubation. The vancomycin agar screen test may be used to detect isolates of *S. aureus* with vancomycin MICs ≥ 8 $\mu\text{g/mL}$; however, this medium does not consistently detect *S. aureus* with vancomycin MICs of 4 $\mu\text{g/mL}$.

Ability of Various Methods to Detect Levels of Vancomycin Susceptibility in *S. aureus*

Vancomycin MIC ($\mu\text{g/mL}$)	MIC Method	Disk Diffusion Method*	Vancomycin Agar Screen
≤ 2 (S)	Yes	No	Yes
4 (I)	Yes	No	Variable
8 (I)	Yes	No	Yes
16 (R)	Yes	No	Yes
≥ 32 (R)	Yes	Yes	Yes

*Strains of *S. aureus* for which the vancomycin zone diameter is ≥ 7 mm may have MICs from ≤ 2 to 16 $\mu\text{g/mL}$. If disk diffusion testing is performed, the identification of isolates showing no zone of inhibition should be confirmed. Isolates of *S. aureus* producing vancomycin zones of ≥ 7 mm should not be reported as susceptible without performing a vancomycin MIC test.

Until further data on the prevalence or clinical significance of isolates with reduced susceptibility to vancomycin are known, laboratories may choose to examine MRSA strains more carefully for elevated MICs to vancomycin.

12.1.3.2 Vancomycin Agar Screen

Perform the test using the following procedure by inoculating an isolate of *S. aureus* onto Brain Heart Infusion (BHI) agar that has been supplemented with 6 µg/mL of vancomycin.

1. Prepare a direct colony suspension equivalent to a 0.5 McFarland standard as is done for MIC or disk diffusion testing (see Section 8.2.1).
2. Use a micropipette to deliver a 10-µL drop to the agar surface. (Alternatively, use a swab from which the excess liquid has been expressed as for the disk diffusion test and spot an area at least 10 to 15 mm in diameter.)
3. Incubate the plate at $35 \pm 2^\circ\text{C}$ in ambient air for a full 24 hours.
4. Examine the plate carefully, using transmitted light, for evidence of small colonies (> 1 colony) or a film of growth. Greater than 1 colony or a film of growth suggests reduced susceptibility to vancomycin.
5. Confirm results for *S. aureus* that grow on the BHI vancomycin agar screen by repeating identification tests and performing vancomycin MIC tests using a CLSI reference dilution method or other validated MIC method.
6. For QC, use:
 - *E. faecalis* ATCC® 29212 or *S. aureus* ATCC® 29213 (vancomycin susceptible) – negative control. (Do not use *S. aureus* ATCC® 25923 as a negative control, because it may give false-positive results.)
 - *E. faecalis* ATCC® 51299 (vancomycin resistant) – positive control.
7. Do not reuse plates after incubation.

Many *S. aureus* isolates with vancomycin MICs of 4 µg/mL do not grow on this vancomycin agar screen media (see Section 12.1.3.1). Also, there are insufficient data to recommend using this agar screen test for coagulase-negative staphylococci.

12.1.3.3 Heteroresistant Vancomycin-Intermediate *Staphylococcus aureus*³⁰

When first described in 1997, heteroresistant vancomycin-intermediate *S. aureus* (hVISA) isolates were those *S. aureus* that contained subpopulations of cells (typically 1 in every 100 000 to 1 000 000 cells) for which the vancomycin MICs were 8 to 16 µg/mL, ie, in the intermediate range. Because a standard broth microdilution test uses an inoculum of 5×10^5 CFU/mL, these resistant subpopulations go undetected and the vancomycin MICs determined for such isolates are in the susceptible range (formerly between 1 and 4 µg/mL). Many physicians and microbiologists initially were skeptical that heteroresistance would result in clinical treatment failures with vancomycin, because such strains were susceptible to vancomycin by the standard CLSI broth microdilution reference method. However, after reviewing both clinical and laboratory data, CLSI lowered the intermediate breakpoint for vancomycin (for *S. aureus* isolates only, not coagulase-negative staphylococci) from 8 to 4 µg/mL, and the resistant breakpoint from ≥ 32 to ≥ 16 µg/mL to make the breakpoints more predictive of clinical outcome. Thus, the vancomycin-susceptible

breakpoint for *S. aureus* is now ≤ 2 $\mu\text{g/mL}$, the intermediate range is 4 to 8 $\mu\text{g/mL}$, and the resistant range is ≥ 16 $\mu\text{g/mL}$. This captures many of the hVISA strains for which the vancomycin MICs are 4 $\mu\text{g/mL}$, which previously would have been designated as susceptible. Yet, some *S. aureus* strains for which the vancomycin MICs are 1 to 2 $\mu\text{g/mL}$ may still be hVISAs.

Determining the population analysis profiles of *S. aureus* isolates (ie, plating a range of dilutions of a standard inoculum of *S. aureus* [10^1 to 10^8 CFU] on a series of agar plates containing a range of vancomycin concentrations, plotting the population curve, dividing the bacterial counts by the area under the curve, and comparing the ratio derived to *S. aureus* control strains Mu3 and Mu50^{31,32}) has become the *de facto* best available method for investigating the clinical relevance of hVISA strains in several large surveillance studies. However, this technique is labor intensive and not suitable for routine clinical laboratories. Unfortunately, there is no standardized technique at this time that is convenient and reliable for detecting hVISA strains. The inability of both automated and standard reference susceptibility testing methods to detect the hVISA phenotype makes it difficult to identify those infections that may not respond to vancomycin therapy. Thus, confirming the presence of a heterogeneously resistant strain of *S. aureus* remains a difficult challenge.

12.1.3.4 Reporting

Vancomycin-susceptible staphylococci should be reported following the laboratory's routine reporting protocols. For strains determined to be vancomycin nonsusceptible (ie, those with MICs ≥ 4 $\mu\text{g/mL}$ and/or growth on BHI vancomycin screen agar), preliminary results should be reported following routine reporting protocols; final results should be reported after confirmation by a reference laboratory. See M100² Table 2C for the most recent recommendations for testing and reporting.

12.1.4 Inducible Clindamycin Resistance

Inducible clindamycin resistance can be identified using the method described in Section 13, and in Section 12 of CLSI document M02.⁵

12.1.5 Mupirocin Resistance

Rates of high-level mupirocin resistance may increase (ie, MICs ≥ 512 $\mu\text{g/mL}$) in *S. aureus* and are associated with the presence of the plasmid-mediated *mupA* gene.³³⁻³⁵ High-level mupirocin resistance can be detected using either routine disk diffusion or broth microdilution tests.³⁶ For disk diffusion using a 200- μg mupirocin disk, incubate the test a full 24 hours and read carefully for any haze or growth using transmitted light. No zone of inhibition = the presence of high-level mupirocin resistance; any zone of inhibition = the absence of high-level resistance. In a recent study, the majority of *mupA*-negative isolates demonstrated mupirocin 200- μg zone diameters > 18 mm. For broth microdilution testing, an MIC of ≥ 512 $\mu\text{g/mL}$ = high-level mupirocin resistance; MICs ≤ 256 $\mu\text{g/mL}$ = the absence of high-level resistance. For dilution testing, a single well containing 256 $\mu\text{g/mL}$ of mupirocin may be tested. For the one-concentration test, growth = high-level mupirocin resistance; no growth = the absence of high-level resistance.

12.2 Enterococci

12.2.1 Penicillin/Ampicillin Resistance

Enterococci may be resistant to penicillin and ampicillin because of production of low-affinity, PBPs or, rarely, because of the production of β -lactamase. Either the agar or broth dilution test accurately detects isolates with altered PBPs, but does not reliably detect isolates that produce β -lactamase. The rare β -lactamase-producing strains of enterococci are detected best by using a direct, nitrocefin-based, β -lactamase test (see Section 14.2). A positive β -lactamase test predicts resistance to penicillin, and

amino-, carboxy-, and ureidopenicillins. Strains of enterococci with ampicillin and penicillin MICs ≥ 16 $\mu\text{g/mL}$ are categorized as resistant. However, enterococci with low levels of penicillin (MICs ≤ 64 $\mu\text{g/mL}$) or ampicillin (MICs ≤ 32 $\mu\text{g/mL}$) resistance may be susceptible to synergistic killing by these penicillins in combination with gentamicin or streptomycin (in the absence of high-level resistance to gentamicin or streptomycin) if high doses of the penicillin are used. Enterococci possessing higher levels of penicillin (MICs ≥ 128 $\mu\text{g/mL}$) or ampicillin (MICs ≥ 64 $\mu\text{g/mL}$) resistance may not be susceptible to the synergistic effect.^{37,38} Physicians' requests to determine the actual MIC of penicillin or ampicillin for blood and CSF isolates of enterococci should be considered.

12.2.2 Vancomycin Resistance

Accurate detection of vancomycin-resistant enterococci (VRE) by the agar or broth dilution test requires incubation for a full 24 hours (rather than 16 to 20 hours) before reporting as susceptible and careful examination of the plates, tubes, or wells for evidence of faint growth. A vancomycin agar screen test may also be used, as described in Section 12.2.3 and in M100² Table 2D Supplemental Table 1.

12.2.3 Vancomycin Agar Screen

The vancomycin agar screening-plate procedure can be used in addition to the dilution methods described in Section 12.2.2 for the detection of VRE. Perform the test using the following procedure by inoculating an enterococcal isolate onto BHI agar that has been supplemented with 6 μg of vancomycin/mL.³⁹

1. Prepare a direct colony suspension equivalent to a 0.5 McFarland standard as is done for MIC or disk diffusion testing.
2. Inoculate the plate using either a 1- to 10- μL loop or a swab.
 - (a) Using a loop, spread the inoculum in an area 10 to 15 mm in diameter.
 - (b) Using a swab, express as for the disk diffusion test and then spot an area at least 10 to 15 mm in diameter.
3. Incubate the plate at $35 \pm 2^\circ\text{C}$ in ambient air for a full 24 hours and examine carefully, using transmitted light, for evidence of growth, including small colonies (> 1 colony) or a film of growth, indicating vancomycin resistance (also see M100² Table 2D Supplemental Table 1).
4. For QC, use:
 - *E. faecalis* ATCC[®] 29212 (vancomycin susceptible) – negative control
 - *E. faecalis* ATCC[®] 51299 (vancomycin resistant) – positive control
5. Do not reuse plates after incubation.

12.2.4 High-Level Aminoglycoside Resistance

High-level resistance to gentamicin and/or streptomycin indicates that an enterococcal isolate will not be killed by the synergistic action of a penicillin or glycopeptide combined with that aminoglycoside.³⁷ Agar or broth high-concentration gentamicin (500 $\mu\text{g/mL}$) and streptomycin (1000 $\mu\text{g/mL}$ with broth microdilution; 2000 $\mu\text{g/mL}$ with agar) tests can be used to screen for this type of resistance (see M100² Table 2D Supplemental Table 1). QC of these tests is also explained in M100² Table 2D Supplemental Table 1. Other aminoglycosides need not be tested, because their activities against enterococci are not superior to gentamicin or streptomycin.

12.3 Gram-Negative Bacilli

12.3.1 Background

The major mechanism of resistance to β -lactam antimicrobial agents in gram-negative bacilli is production of β -lactamase enzymes. Many different types of enzymes have been reported. β -lactamases may be named after the primary substrates that they hydrolyze, the biochemical properties of the β -lactamases, strains of bacteria from which the β -lactamase was detected, a patient from whom a β -lactamase-producing strain was isolated, etc.⁴⁰ For example, TEM is an abbreviation for Temoneira, the first patient from whom a TEM β -lactamase-producing strain was reported. β -lactamases may be classified as molecular Class A, B, C, or D enzymes.⁴¹

Class	Active Site	Examples
A	Inhibitor-susceptible (rare exceptions)	TEM-1, SHV-1, KPCs, OXY, and most ESBLs (including CTX-M)
B	Metallo- β -lactamases	Metalloenzymes; VIM, IMP, SPM, NDM
C	Inhibitor-resistant β -lactamases	AmpC
D	Oxacillin-active β -lactamases that may be inhibitor susceptible	OXA (including rare ESBL phenotypes)

β -lactamase enzymes in all four classes inactivate β -lactam antimicrobial agents at different rates. The genes encoding β -lactamases may be located on chromosomes and expressed with or without induction or carried on plasmids in single or multiple copies. An isolate may produce β -lactamases and possess other resistance mechanisms such as porin mutations that restrict antimicrobial access to their active binding sites in the bacterial cell. The variety of β -lactam resistance mechanisms encountered in gram-negative bacteria gives rise to a continuum of antimicrobial activities expressed as a range of MIC values. One would expect the interpretive breakpoint to be the MIC or zone diameter value that differentiates β -lactamase/other resistance mechanism-negative strains (susceptible) from β -lactamase/other resistance mechanism-positive strains (resistant). However, weak β -lactamase activity or low-level β -lactamase expression may not necessarily mean that the isolate will be refractory to β -lactam therapy. In practice, some isolates that are interpreted as susceptible will produce β -lactamases that have clinically inconsequential enzyme activity. These may be ESBL, AmpC, or carbapenemase-type enzymes as described in Sections 12.3.2, 12.3.3, and 12.3.4.

Identification of a specific β -lactamase resistance mechanism (eg, ESBL, *Klebsiella pneumoniae* carbapenemase [KPC], NDM) is not required or necessary for the determination of a susceptible or resistant interpretation. However, the identification of a specific enzyme may be useful for infection control procedures or epidemiological investigations. The M100² Supplemental Tables with Table 2A describe tests that can be used to screen for and confirm the presence of ESBLs in *E. coli*, *K. pneumoniae*, *Klebsiella oxytoca*, and *Proteus mirabilis*, and carbapenemase production in *Enterobacteriaceae*.

12.3.2 Extended-Spectrum β -Lactamases

ESBLs are inhibitor-susceptible enzymes in Classes A and D that arise by mutations in genes for common plasmid-encoded β -lactamases, such as TEM-1, SHV-1, and OXA-10; or may be only distantly related to a native enzyme, as in the case of the CTX-M β -lactamases. ESBLs may confer resistance to penicillins, cephalosporins, and aztreonam in clinical isolates of *K. pneumoniae*, *K. oxytoca*, *E. coli*, *P. mirabilis*,⁴² and other genera of the family *Enterobacteriaceae*.⁴¹ β -lactam interpretive breakpoints are set at MIC values that recognize ESBL activity, in combination with other resistance mechanisms, likely to predict clinical success or failure. While susceptible MIC values are found, generally, in ESBL-negative gram-negative bacilli, some susceptible strains will contain ESBL genes producing low amounts of enzyme or enzyme that has poor hydrolytic activity. These strains are categorized correctly as susceptible when using CLSI breakpoints changes first published in M100-S20.⁴³

A similar native enzyme, OXY (formerly KOXY or K1), in *K. oxytoca* acts as an extended-spectrum penicillinase, inactivating amino- and carboxypenicillins. When OXY enzymes are overproduced as a result of promoter mutations, resistance to ceftriaxone and aztreonam (but not ceftazidime), as well as resistance to all combinations of β -lactams and β -lactamase inhibitors, results. Although strains producing OXY enzymes may result in a positive ESBL confirmatory test, OXY enzymes are generally not considered ESBLs. MIC and zone diameter values correctly predict susceptible and resistant interpretations.

12.3.3 AmpC Enzymes

The AmpC β -lactamases are chromosomal or plasmid-encoded enzymes.⁴⁴ Isolates that produce AmpC enzymes have a similar antimicrobial susceptibility profile to those that produce ESBLs in that they show reduced susceptibility to penicillins, cephalosporins, and aztreonam. However, in contrast to ESBLs, AmpC β -lactamases also inactivate cephamycins, ie, bacteria expressing AmpC enzyme test as resistant to cefoxitin and cefotetan. In addition, AmpC-producing strains resistant to the current β -lactamase inhibitor combination agents, and strains producing AmpC enzyme may test resistant to carbapenems if accompanied by a porin mutation or in combination with overexpression of specific efflux pumps.

Chromosomal AmpC β -lactamases are found in *Enterobacter*, *Citrobacter*, *Serratia*, and some other gram-negative species, and are usually expressed in low amounts but can be induced to produce higher amounts by penicillins, carbapenems, and some cepheims such as cefoxitin. The expanded-spectrum cephalosporins (cephalosporin subclasses III and IV) do not induce AmpC enzymes but can be hydrolyzed by them. Use of cephalosporins also may select for stably derepressed chromosomal mutants, which can emerge during therapy.⁴⁵

AmpC enzymes can be carried on plasmids that are transmissible among bacteria. Although plasmid-mediated AmpC enzymes evolved from native chromosomal enzymes among a diverse group of bacteria, they are found primarily in clinical isolates of *K. pneumoniae* and *E. coli*.

There are no CLSI-validated phenotypic tests to confirm the presence of plasmid-encoded AmpC β -lactamases in clinical isolates. Strains carrying both ESBLs and plasmid-encoded AmpC β -lactamases are common in some geographical regions. The current susceptibility breakpoints for drugs affected by these combinations of enzymes, first published in M100-S20,⁴³ are the best approach for providing guidance for treatment of these strains.

12.3.4 Carbapenemases (Carbapenem-Resistant *Enterobacteriaceae*)

Carbapenemase activity in clinical isolates of *Enterobacteriaceae* occurs as a result of β -lactamase enzymes in Classes A, B, and D. KPC-type enzymes within Class A, NDM-type enzymes within Class B,

and OXA-type enzymes within Class D represent major families of clinical importance (see the table below). The presence of KPC-type enzymes can be confirmed using the modified Hodge test as described in M100² Supplemental Tables with Table 2A. NDM-type and other metallo- β -lactamase enzymes require zinc for activity and are inhibited by substances such as ethylenediaminetetraacetic acid (EDTA), which binds zinc. *Stenotrophomonas maltophilia*, *Bacillus anthracis*, and some strains of *Bacteroides fragilis* produce a chromosomal metallo- β -lactamase. Other metalloenzymes may be carried on plasmids and can occur in *Acinetobacter* spp., *P. aeruginosa*, *Serratia marcescens*, *K. pneumoniae*, and, increasingly, other *Enterobacteriaceae*. There is no CLSI-validated phenotypic test to confirm the presence of metallo- β -lactamases in clinical isolates. Current susceptibility breakpoints for drugs affected by these carbapenemases, first published in M100-S20-U,⁴⁶ are the recommended approach for providing guidance for treatment of infection by *Enterobacteriaceae* containing KPC- and NDM-type enzymes.

β -Lactamases With Carbapenemase Activity

β -Lactamase Class*	Found In	Examples
A	<i>K. pneumoniae</i> and other <i>Enterobacteriaceae</i>	KPC, SME
B	<i>Enterobacteriaceae</i> <i>P. aeruginosa</i> <i>Acinetobacter baumannii</i>	Metallo- β -lactamases inhibited by EDTA (IMP, VIM, NDM)
D	<i>A. baumannii</i> <i>Enterobacteriaceae</i>	OXA

* Carbapenemases have not yet been found in Class C.

12.4 *Streptococcus pneumoniae*

12.4.1 Penicillin and Third-Generation Cephalosporin Resistance

Penicillin, and cefotaxime, ceftriaxone, or meropenem, should be tested by a reliable MIC method and reported routinely with CSF isolates of *S. pneumoniae*. Such isolates should also be tested against vancomycin using the MIC or disk method. Consult M100² Table 2G for reporting of penicillins and third-generation cephalosporins, because there are specific interpretive criteria that must be used depending on the site of infection and the penicillin formulation used for therapy. In M100² Table 2G, breakpoints are listed for intravenous penicillin therapy for meningitis and nonmeningitis infections. Separate breakpoints are included for therapy of less severe infections with oral penicillin.

Amoxicillin, ampicillin, cefepime, cefotaxime, ceftriaxone, cefuroxime, ertapenem, imipenem, and meropenem may be used to treat pneumococcal infections; however, reliable disk diffusion susceptibility tests with these agents do not yet exist. Their *in vitro* activity is best determined using an MIC method.

13 Inducible Clindamycin Resistance

Macrolide-resistant isolates of *S. aureus*, coagulase-negative *Staphylococcus* spp., and β -hemolytic streptococci may express constitutive or inducible resistance to clindamycin (methylation of the 23S ribosomal RNA encoded by the *erm* gene, also referred to as MLS_B [macrolide, lincosamide, and type B streptogramin] resistance) or may be resistant only to macrolides (efflux mechanism encoded by the *msrA* gene in staphylococci or a *mef* gene in streptococci). Infections caused by staphylococci with inducible clindamycin resistance may fail to respond to clindamycin therapy.⁴⁷ However, the clinical significance of inducible resistance in streptococci is unclear. Inducible clindamycin resistance can be detected for all staphylococci and β -hemolytic streptococci using a disk diffusion test with clindamycin and erythromycin disks placed in proximity^{11,12} or as a single well test using broth microdilution.

Using disk diffusion, the test is done by placing a 2- μ g clindamycin disk either 15 to 26 mm away (for staphylococci) or 12 mm away (for streptococci) from the edge of a 15- μ g erythromycin disk on a standard blood agar plate used for the inoculum purity check or by using the standard disk diffusion procedure with MHA (see CLSI document M02,⁵ Section 12). Flattening of the clindamycin zone adjacent to the erythromycin disk (referred to as a D-zone) indicates inducible clindamycin resistance. Following incubation, organisms that do not show flattening of the clindamycin zone adjacent to the erythromycin disk in an erythromycin-resistant isolate should be reported as tested (ie, susceptible or intermediate to clindamycin).

For staphylococci and β -hemolytic streptococci, inducible clindamycin resistance can also be detected with the broth microdilution test using the combination of erythromycin and clindamycin. For staphylococci, erythromycin 4 μ g/mL and clindamycin 0.5 μ g/mL are used together in a single well of a broth microdilution panel; for streptococci, the combination of erythromycin 1 μ g/mL and clindamycin 0.5 μ g/mL is used. The broth microdilution test applies only to isolates that are erythromycin resistant and clindamycin susceptible or intermediate; for these isolates, growth in the well indicates the presence of inducible clindamycin resistance. Following incubation, organisms that do not grow in the broth microdilution well should be reported as tested (ie, susceptible or intermediate to clindamycin).

Organisms that grow in the microdilution well or that show flattening of the clindamycin zone adjacent to the erythromycin disk (referred to as a D-zone) have inducible clindamycin resistance. Recommendations for how these strains should be reported, recommendations for QC, and the most recent updates for testing are provided in M100² Tables 2C, 2H-1, 4A, and 4B.

14 β -Lactamase Tests

14.1 Purpose

A rapid β -lactamase test may yield clinically relevant results earlier than an MIC test with *Haemophilus* spp., *N. gonorrhoeae*, and *Moraxella catarrhalis*; a β -lactamase test is the only reliable test for detecting β -lactamase-producing *Enterococcus* spp. β -lactamase testing may also clarify the susceptibility test results of staphylococci to penicillin determined by broth microdilution, especially in strains with borderline MICs (0.06 to 0.25 μ g/mL).

A positive β -lactamase test result predicts the following:

- Resistance to penicillin, ampicillin, and amoxicillin among *Haemophilus* spp., *M. catarrhalis*, and *N. gonorrhoeae*
- Resistance to penicillin, and amino-, carboxy-, and ureidopenicillins among staphylococci and enterococci

A negative β -lactamase test result does not rule out β -lactam resistance due to other mechanisms. Do not use β -lactamase tests for members of the *Enterobacteriaceae*, *Pseudomonas* spp., and other aerobic, gram-negative bacilli, because the results may not be predictive of susceptibility to the β -lactams most often used for therapy.

14.2 Selecting a β -Lactamase Test

Nitrocefin-based tests are the preferred method for testing enterococci, *Haemophilus* spp., *M. catarrhalis*, and *N. gonorrhoeae*.⁴⁸ Acidimetric β -lactamase tests have generally produced acceptable results with *Haemophilus* spp., *N. gonorrhoeae*, and staphylococci. Iodometric tests may be used for testing *N. gonorrhoeae*, but only nitrocefin-based tests should be used to test *M. catarrhalis*.⁴⁹ Accurate detection of

β -lactamase in staphylococci may require alternative methods, such as induction of the enzyme or evaluation of the penicillin disk diffusion zone-edge for *S. aureus* (see Section 12.1.1 and M100² Table 2C). Induction can be easily accomplished by testing the growth from the zone margin surrounding an oxacillin or cefoxitin disk test. Care must be exercised when using these assays to ensure accurate results, including testing of known positive and negative control strains at the time clinical isolates are examined (see the manufacturer's recommendations for commercial tests).

15 Reporting of Minimal Inhibitory Concentration Results

The MIC results determined as described in this document may be reported directly to clinicians for patient-care purposes. However, it is essential for an understanding of the data by all clinicians that an interpretive category result also is provided routinely. Recommended interpretive categories for various MIC values are included in tables for each organism group and are based on evaluation data as described in CLSI document M23.⁴ For most agents, these categories are developed by determining MIC population distributions of a large number of isolates, including those with known mechanisms of resistance relevant to the particular class of drug. Second, the MICs are analyzed in relation to the pharmacokinetics of the drug from normal dosing regimens. Finally, when feasible, the tentative *in vitro* interpretive criteria are analyzed in relation to studies of clinical efficacy and microbiological eradication in the treatment of specific pathogens as outlined in CLSI document M23.⁴

See Section 4.1 for definitions of the interpretive categories susceptible, intermediate, resistant, and nonsusceptible.

16 Quality Control and Quality Assurance Procedures

16.1 Purpose

In AST, QC includes the procedures to monitor the performance of a test system to ensure reliable results. This is achieved by, but not limited to, the testing of QC strains with known susceptibility to the antimicrobial agents tested. The goals of a QC program are to monitor the following:

- Precision (repeatability) and accuracy of susceptibility test procedures
- Performance of reagents used in the tests
- Performance of persons who carry out the tests and read the results

A comprehensive quality assurance (QA) program helps to ensure that testing materials and processes consistently provide quality results. QA includes, but is not limited to, monitoring, evaluating, taking corrective actions (if necessary), recordkeeping, calibration and maintenance of equipment, proficiency testing, training, and QC.

16.2 Quality Control Responsibilities

Modern laboratories rely heavily on pharmaceutical and diagnostic product manufacturers for provision of reagents, media, or test systems for the performance of antimicrobial susceptibility tests. Although this section is intended to apply only to the standard reference methods, it may be applicable to certain commercially available test systems that are based primarily, or in part, on these methods.

Manufacturers and users of antimicrobial susceptibility tests have a shared responsibility for quality. The primary purpose of QC testing performed by manufacturers (in-house reference methods or commercial methods) is to ensure that the test has been appropriately manufactured. The primary purpose of QC testing performed by laboratories (users) is to ensure that the tests are maintained and performed appropriately.

A logical division of responsibility and accountability may be described as follows:

- Manufacturers (in-house or commercial products):
 - Antimicrobial stability
 - Antimicrobial labeling
 - Potency of antimicrobial stock solutions
 - Compliance with good manufacturing practices (eg, quality management system standards)
 - Integrity of product
 - Accountability and traceability to consignee
- Laboratories (users):
 - Storage under the environmental conditions recommended by the manufacturer (to prevent drug deterioration)
 - Proficiency of personnel performing tests
 - Use of current CLSI standards (or manufacturer's instructions for use) and adherence to the established procedure (eg, inoculum preparation, incubation conditions, interpretation of end points)

Manufacturers should design and recommend a QC program that allows users to evaluate those variables (eg, inoculum density, storage/shipping conditions) that most likely could cause user performance problems and to determine that the test is performing correctly when used according to established protocols.

16.3 Selection of Strains for Quality Control and Quality Assurance

Use of carefully selected QC strains allows the microbiologist to have confidence that the test is performing within acceptable standards, and thus that the test results are likely to be reliable.

Each QC strain should be obtained from a recognized source (eg, ATCC®). All CLSI-recommended QC strains appropriate for the antimicrobial agent and reference method should be evaluated and expected results established according to the procedures described in CLSI document M23.⁴ Users of commercial systems should follow the QC recommendations in their instructions for use.

- Ideal strains for QC of dilution tests have MICs that fall near the middle of the concentration range tested for all antimicrobial agents (eg, an ideal QC strain would be inhibited at the fourth dilution of a seven-dilution series, but strains with MICs at either the third or fifth dilution would also be acceptable). In certain instances with newer, more potent antimicrobial agents, it may be necessary to test additional QC strains not normally tested in order to provide on-scale values.
- When three or fewer adjacent doubling dilutions of an antimicrobial agent are tested by dilution methods, QC procedures must be modified.
 - One possible alternative is to use one control organism with a modal MIC that is equal to, or no less than, one doubling dilution of the lower concentration and a second control organism with a modal MIC that is equal to, or no greater than, one doubling dilution of the higher concentration. The combination of results derived from testing these two strains should provide for at least one on-scale end point.
 - Another strategy is to selectively test the most labile agents included in the panels (eg, clavulanic acid combinations, imipenem, and cefaclor).

QC strains and their characteristics are described in Appendix D. Some of these are listed as “QC strains” and others as “supplemental QC strains.” These can be defined as follows:

QC strains are tested regularly (eg, daily, weekly) to ensure the test system performs as expected and produces results that fall within specified limits listed in M100.² The QC strains recommended in this document should be included if a laboratory performs CLSI reference MIC testing as described herein. For commercial test systems, manufacturers' recommendations should be followed for all QC procedures.

Supplemental QC strains are used to assess a particular characteristic of a test or test system in select situations or may represent alternative QC strains. For example, *H. influenzae* ATCC[®] 10211 is more fastidious than *H. influenzae* ATCC[®] 49247 or *H. influenzae* ATCC[®] 49766, and is used to ensure HTM can adequately support the growth of clinical isolates of *H. influenzae* and *H. parainfluenzae*. Supplemental QC strains may possess susceptibility or resistance characteristics specific for one or more special tests listed in CLSI documents M02⁵ and M07. They can be used to assess a new test, for training new personnel, for competency assessment, etc. It is not necessary to include supplemental QC strains in routine daily or weekly AST QC programs.

Expected QC results for individual antimicrobial agents and tests are listed in M100² Tables 4A, 4B, and 4C.

16.4 Storing and Testing Quality Control Strains

- Test the QC strains by standard dilution test procedures described herein using the same materials and methods that are used to test clinical isolates.
- Proper organism storage and maintenance is required to ensure acceptable performance of QC strains (also refer to Appendix E).
 - For prolonged storage, maintain stock cultures at –20°C or below (preferably at ≤ –60°C or in liquid nitrogen) in a suitable stabilizer (eg, 50% fetal calf serum in broth, 10% to 15% glycerol in tryptic soy broth, defibrinated sheep blood, or skim milk) or in a freeze-dried state without significant risk of altering their antimicrobial susceptibility.
 - Subculture frozen or freeze-dried stock cultures onto appropriate media (eg, tryptic soy or blood agar for nonfastidious strains, or enriched chocolate or blood agar for fastidious strains) and incubate under the appropriate conditions for the organism (primary subculture). Subculture frozen or lyophilized cultures twice before use in testing. The second subculture is referred to as Day 1 working culture.
 - Store subcultures at 2 to 8°C or as appropriate for the organism type.
 - Prepare working cultures by subculturing the QC strains onto agar plates to obtain isolated colonies for testing. Prepare a new working culture each day.
 - Prepare a new subculture each week to create working cultures (eg, prepare working cultures from the same subculture for up to seven days; then prepare a new subculture on day 8).
 - Prepare new primary subcultures at least monthly from frozen, freeze-dried, or commercial cultures (eg, subculture each week for no more than three successive weeks). For best results, some strains may require preparation of new subcultures more frequently (eg, every two weeks).
- If an unexplained result suggests a change in the organism's inherent susceptibility, prepare a new primary subculture or working culture, or obtain a fresh stock culture of the QC strain. See Section 16.9 for additional guidance.

16.5 Batch or Lot Quality Control

1. Test each new batch or lot of macrodilution tubes, microdilution trays, or agar dilution plates with the appropriate QC strains to determine if MICs obtained with the batch or lot fall within the expected range (see M100² Tables 4A, 4B, and 4C); if they do not, the batch or lot must be rejected.
2. Incubate at least one uninoculated tube, microdilution tray, or agar plate from each batch overnight to verify sterility of the medium.
3. MHB used to prepare macrodilution tubes or microdilution trays can be obtained either without added cations or already supplemented with them (see Appendix B3.1). For lots of MHB without added cations, the levels present in the medium must be obtained from the manufacturer before preparing the medium as CAMHB. For lots received as CAMHB, further supplementation is not necessary. Some antimicrobial agents are affected by differences in cation content (eg, aminoglycosides and daptomycin, see M100² Table 4G) and, if QC problems are encountered with them, new lots of panels may need to be tested to verify acceptable cation content. For MHB, determine the MIC of gentamicin for *P. aeruginosa* ATCC® 27853 and compare it with the expected range (see M100² Table 4A). If the MIC is low, the broth may need to be supplemented with cations as directed in Section 10.1 and Appendix B3. When daptomycin is tested, levels of calcium cations in broth should be 50 mg/L.
4. Records should be kept of the lot numbers of all materials and reagents used in performing susceptibility tests.
5. *H. influenzae* ATCC® 10211 is recommended as a useful additional QC strain to verify the growth promotion properties of HTM. In particular, manufacturers of HTM are encouraged to use *H. influenzae* ATCC® 10211 as a QC test strain.

16.6 Minimal Inhibitory Concentration Quality Control Limits

Acceptable MIC QC limits for a single QC test (single-drug/single-organism combination) are listed in M100² Tables 4A, 4B, and 4C.

16.7 Frequency of Quality Control Testing (also refer to Appendix A and M100² Table 4F)

Monitor the overall performance of the test system using the QC limits by testing the appropriate QC strains each day the test is performed or, if satisfactory performance is documented (see Section 16.7.2.1), test the QC strains weekly. The weekly QC testing option outlined in Section 16.7.2 is not applicable when MIC tests are performed less than once a week. QC testing should be performed each test day for MIC tests performed less than once a week.

16.7.1 Daily Testing

Performance is satisfactory for daily QC testing when no more than three out of 30 results obtained on consecutive test days for each antimicrobial agent/organism combination are outside the acceptable limit stated in M100² Tables 4A, 4B, and 4C. Corrective action by the laboratory is required when this frequency is exceeded (see Section 16.9).

16.7.2 Weekly Testing

16.7.2.1 Demonstrating Satisfactory Performance for Conversion From Daily to Weekly Quality Control Testing

- Test all applicable QC strains for 20 or 30 consecutive test days, and document results.

- To convert from daily to weekly QC testing, no more than one out of 20 or three out of 30 MICs for each antimicrobial agent/organism combination may be outside the acceptable MIC limits stated in M100² Tables 4A, 4B, and 4C.

16.7.2.2 Implementing Weekly Quality Control Testing

- Weekly QC testing may be performed once satisfactory performance has been documented (see Section 16.7.2.1).
- Perform QC testing once per week and whenever any reagent component of the test (eg, a new lot of broth from the same manufacturer) is changed.
- If any of the weekly QC results are out of the acceptable range, corrective action is required (see Section 16.9).
- Refer to M100² Table 4F for guidance on QC frequency with new materials or test modifications.
- These guidelines can also be used for testing systems in which an MIC is determined using three or fewer adjacent doubling dilutions of an antimicrobial agent.
- For some drugs, QC records may indicate the need for testing to be done more frequently than once a week, because of the relatively rapid degradation of the drug (see Sections 9.2.2 and 10.4.1).

16.8 Frequency of Quality Control Testing for Screening Tests

The frequency of QC screening tests is as follows:

- QC of vancomycin and high-level aminoglycoside screen plates may be performed weekly if these tests are routinely used (ie, at least once a week) in the laboratory and criteria for converting from daily to weekly testing have been met.
- QC of screening tests may be needed each day of testing if the test is not performed routinely (at least once a week) or if the antimicrobial agent is labile (eg, oxacillin agar screen for *S. aureus*).

16.9 Corrective Action

16.9.1 Out-of-Control Result Due to Identifiable Error

If the cause of out-of-control results can be identified, correct the issue, document the reason, and retest the strain on the day the error is observed. If the repeated result is within range, no further corrective action is required. If a problem with the QC strain is suspected or identified, obtain a new working culture or subculture and retest as soon as possible.

The Troubleshooting Guide in M100² Table 4G provides guidance for troubleshooting and corrective action for out-of-range QC. Causes for the out-of-control results may include, but are not limited to:

- QC strain
 - Use of the wrong QC strain
 - Improper storage
 - Inadequate maintenance (eg, use of the same working culture for > 1 month)
 - Contamination
 - Nonviability
 - Changes in the organism (eg, mutation, loss of plasmid)

- Testing supplies
 - Improper storage or shipping conditions
 - Contamination
 - Inadequate volume of broth in tubes or wells
 - Use of damaged (eg, cracked, leaking) panels, plates, cards, tubes
 - Use of expired materials
- Testing process
 - Use of the wrong incubation temperature or conditions
 - Inoculum suspensions incorrectly prepared or adjusted
 - Inoculum prepared from a plate incubated for the incorrect length of time
 - Inoculum prepared from differential or selective media containing anti-infective agents or other growth-inhibiting compounds
 - Use of wrong reagents, ancillary supplies
 - Incorrect reading or interpretation of test results
 - Transcription error
- Equipment
 - Not functioning properly or out of calibration (eg, pipettes)

16.9.2 Out-of-Control Result With No Error Identified

16.9.2.1 Immediate Corrective Action

If the reason for the out-of-control result cannot be identified, corrective action is required as follows.

- Test the out-of-control antimicrobial agent/organism combination on the day the error is observed and/or as soon as a new working culture or subculture is available. Monitor for five consecutive test days. Document all results.
 - If all five MICs for the antimicrobial agent/organism combination are within the acceptable ranges, as defined in M100² Tables 4A, 4B, and 4C, no additional corrective action is necessary.
 - If any of the five MICs are still outside the acceptable range, additional corrective action is required (see Section 16.9.2.2 and M100² Tables 4A, 4B, and 4C).
- Daily control tests must be continued until final resolution of the problem is achieved.

16.9.2.2 Additional Corrective Action

When immediate corrective action does not resolve the problem, the problem is likely due to a system error rather than a random error. Additional investigation and corrective action is required. Refer to Section 16.9.1 and M100² Table 4G, Troubleshooting Guide for assistance.

If necessary, obtain a new QC strain (either from freezer storage or a reliable source) and new lots of materials (including new turbidity standards), possibly from different manufacturers. If the problem appears to be related to a manufacturer, contact and provide the manufacturer with the test results. It is also helpful to exchange QC strains and materials with another laboratory using the same method in order to determine the root cause of unexplained system problems. Until the problem is resolved, it may be necessary to use an alternative test method.

If a problem is identified and corrected, documentation of satisfactory performance for another five days is required to return to weekly QC testing. If a problem is not identified, but results go back into control

without any specific corrective action, documentation of satisfactory performance for another 20 or 30 consecutive days is required in order to return to weekly QC testing (see Section 16.7.2.1).

16.10 Reporting Patient Results When Out-of-Control Tests Occur

Whenever an out-of-control result occurs or corrective action is necessary, careful assessment of whether to report patient test results should be made on an individual patient basis, taking into account if the source of the error, when known, is likely to have affected relevant patient test results. Considerations may include, but are not limited to:

- Size and direction of error (eg, slightly or significantly increased MIC, slightly or significantly decreased MIC).
- Is the patient result close to the interpretive breakpoint?
- Results with other QC organisms.
- Results with other antimicrobial agents.
- Is the QC strain/antimicrobial agent an indicator for a procedural or storage issue (eg, inoculum dependent, heat labile)? Refer to M100² Table 4G, Troubleshooting Guide.

Options to consider include suppressing the results for an individual antimicrobial agent; retrospectively reviewing individual patient or cumulative data for unusual patterns; and using an alternative test method or a reference laboratory until the problem is resolved.

16.11 Verification of Patient Test Results

Multiple test parameters are monitored by following the QC recommendations described in this standard. However, acceptable results derived from testing QC strains do not guarantee accurate results when testing patient isolates. It is important to review all of the results obtained from all drugs tested on a patient's isolate before reporting the results. This should include, but not be limited to, ensuring that:

- The antimicrobial susceptibility results are consistent with the identification of the isolate
- The results from individual agents within a specific drug class follow the established hierarchy of activity rules (eg, third-generation cephalosporins are more active than first- or second-generation cephalosporins against *Enterobacteriaceae*)
- The isolate is susceptible to those agents for which resistance has not been documented (eg, vancomycin and *Streptococcus* spp.) and for which only "susceptible" interpretive criteria exist in M100²

Unusual or inconsistent results should be verified by checking for the following:

- Previous results on the patient (eg, Did the patient have the same isolate with an unusual antibiogram previously?)
- Previous QC performance (eg, Is there a similar trend or observation with recent QC testing?)
- Problems with the testing supplies, process, or equipment (see Section 16.9.1 and M100² Table 4G, Troubleshooting Guide)

If a reason for the unusual or inconsistent result cannot be ascertained, a repeat of the susceptibility test or the identification, or both, is in order. Sometimes, it is helpful to use an alternative test method for the repeat test. A suggested list of results that may require verification is included in M100² Appendix A. Each laboratory must develop its own policy for verification of unusual or inconsistent antimicrobial susceptibility test results. This policy should emphasize those results that may significantly impact patient care.

16.12 Other Control Procedures

16.12.1 Growth Control

Each microdilution broth tray, macrodilution broth series, and agar dilution plate series should include a growth control of basal medium without antimicrobial agent to assess viability of the test organisms. With the broth tests, the growth control also serves as a turbidity control for reading end points.

16.12.2 Purity Control

Following inoculation of agar or broth dilution tests, streak a sample from each inoculum on a suitable agar plate and incubate overnight to detect mixed cultures and to provide freshly isolated colonies in case retesting proves necessary. This step is particularly important for broth dilutions in which mixed cultures are likely to go unrecognized.

16.12.3 Inoculum Control

Perform QC plate counts at least quarterly with representative inocula to ensure that the 0.5 McFarland standard and the procedures for standardizing and diluting inocula remain under control. To do this, remove samples for plate counts immediately after inoculation from the growth-control well of microdilution trays, the growth-control tube of a macrodilution series, or in agar dilution, a random reservoir well of the replicator seed block (see Section 10.5).

16.12.4 End-point Interpretation Control

Monitor end-point interpretation periodically to minimize variation in the interpretation of MIC end points among observers. All laboratory personnel who perform these tests should independently read a selected set of dilution tests. Record the results and compare to the results obtained by an experienced reader. All readers should agree within ± 1 twofold concentration increment of one another.⁵⁰

17 Limitations of Dilution Test Methods

17.1 Application to Various Organism Groups

The dilution methods described in this document are standardized for testing rapidly growing pathogens, which include *Staphylococcus* spp., *Enterococcus* spp., the *Enterobacteriaceae*, *P. aeruginosa*, *Pseudomonas* spp., *Acinetobacter* spp., *Burkholderia cepacia*, and *S. maltophilia*, and they have been modified for testing some fastidious organisms such as *H. influenzae* and *H. parainfluenzae* (M100² Table 2E), *N. gonorrhoeae* (M100² Table 2F), *N. meningitidis* (M100² Table 2I), and streptococci (M100² Tables 2G, 2H-1, and 2H-2). For organisms excluded from M100² Tables 2A through 2J and not covered in other CLSI guidelines or standards, such as CLSI document M45,⁷ studies are not yet adequate to develop reproducible, definitive standards to interpret results. These organisms may require different media, require different atmospheres of incubation, or show marked strain-to-strain variation in growth rate. For these microorganisms, consultation with an infectious disease specialist is recommended for guidance in determining the need for susceptibility testing and in the interpretation of results. Published reports in the medical literature and current consensus recommendations for therapy of uncommon microorganisms

may obviate the need for testing of such organisms. If testing is necessary, a dilution method usually is the most appropriate testing method, and this may require submitting the organism to a reference laboratory.

17.2 Misleading Results

Dangerously misleading results can occur when certain antimicrobial agents are tested and reported as susceptible against specific organisms. These combinations include, but may not be limited to, the following:

- First- and second-generation cephalosporins, cephamycins, and aminoglycosides against *Salmonella* and *Shigella* spp.
- Penicillins, β -lactam/ β -lactamase inhibitor combinations, cephems (except for cephalosporins with anti-MRSA activity), and carbapenems against oxacillin-resistant *Staphylococcus* spp.
- Aminoglycosides (except high concentrations), cephalosporins, clindamycin, and trimethoprim-sulfamethoxazole against *Enterococcus* spp.

17.3 Development of Resistance and Testing of Repeat Isolates

Isolates that are initially susceptible may become intermediate or resistant after initiation of therapy. Therefore, subsequent isolates of the same species from a similar body site should be tested in order to detect resistance that may have developed. This can occur within as little as three to four days and has been noted most frequently in *Enterobacter*, *Citrobacter*, and *Serratia* spp. with third-generation cephalosporins; in *P. aeruginosa* with all antimicrobial agents; and in staphylococci with quinolones. For *S. aureus*, vancomycin-susceptible isolates may become vancomycin intermediate during the course of prolonged therapy.

In certain circumstances, testing of subsequent isolates to detect resistance that may have developed might be warranted earlier than within three to four days. The decision to do so requires knowledge of the specific situation and the severity of the patient's condition (eg, an isolate of *Enterobacter cloacae* from a blood culture on a premature infant). Laboratory guidelines on when to perform susceptibility testing on repeat isolates should be determined after consultation with the medical staff.

18 Screening Tests

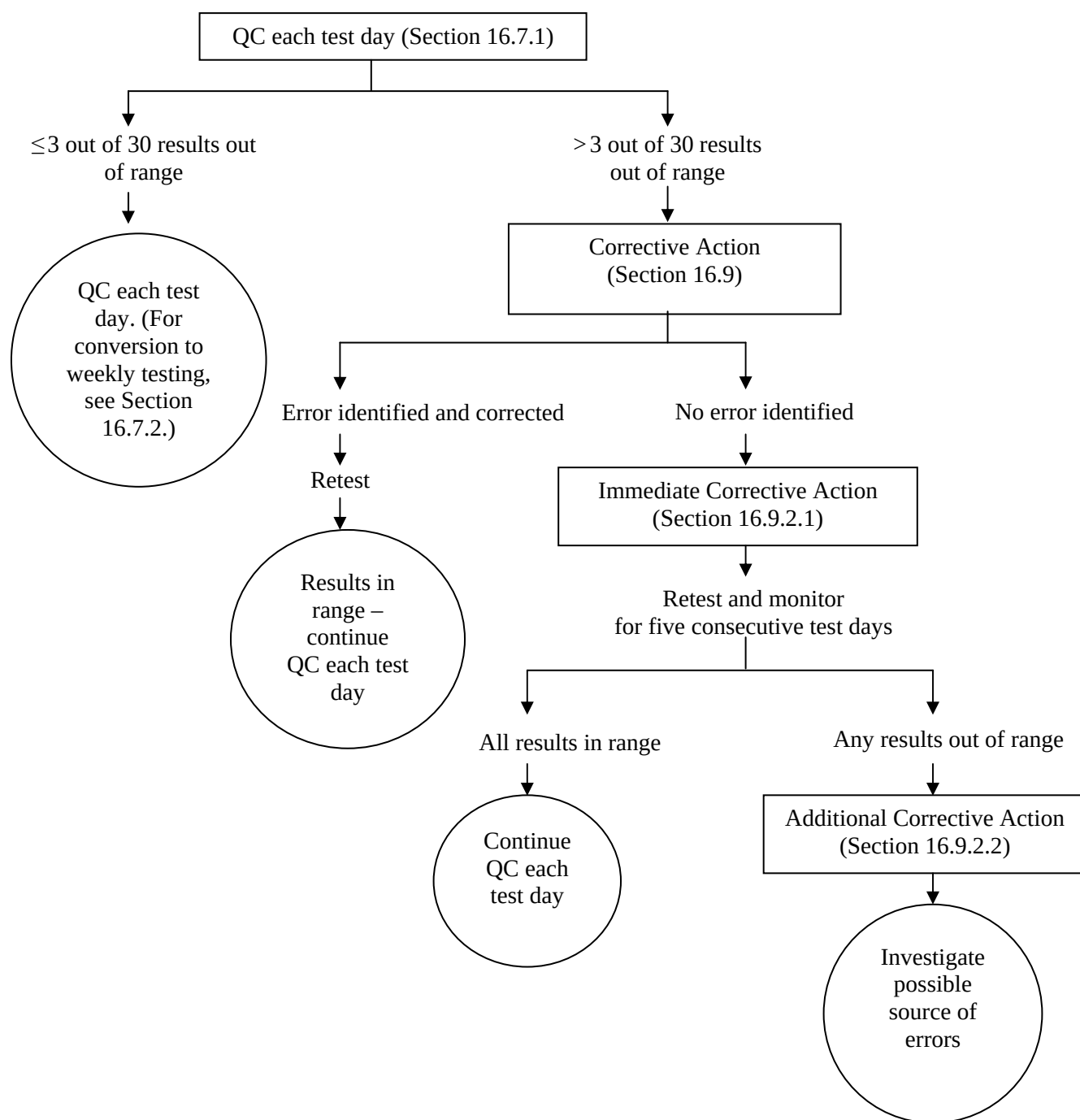
Screening tests, as described in this document and in M100,² characterize an isolate as susceptible or resistant to one or more antimicrobial agents based on a specific resistance mechanism or phenotype. Some screening tests have sufficient sensitivity and specificity such that results of the screen can be reported without additional testing. Others require further testing to confirm the presumptive results obtained with the screen test. The details for each screening test, including test specifications, limitations, and additional tests needed for confirmation, are provided in M100² Instructions for Use of Tables 1 and 2, Section VII and in the M100² Table 2 locations specified there.

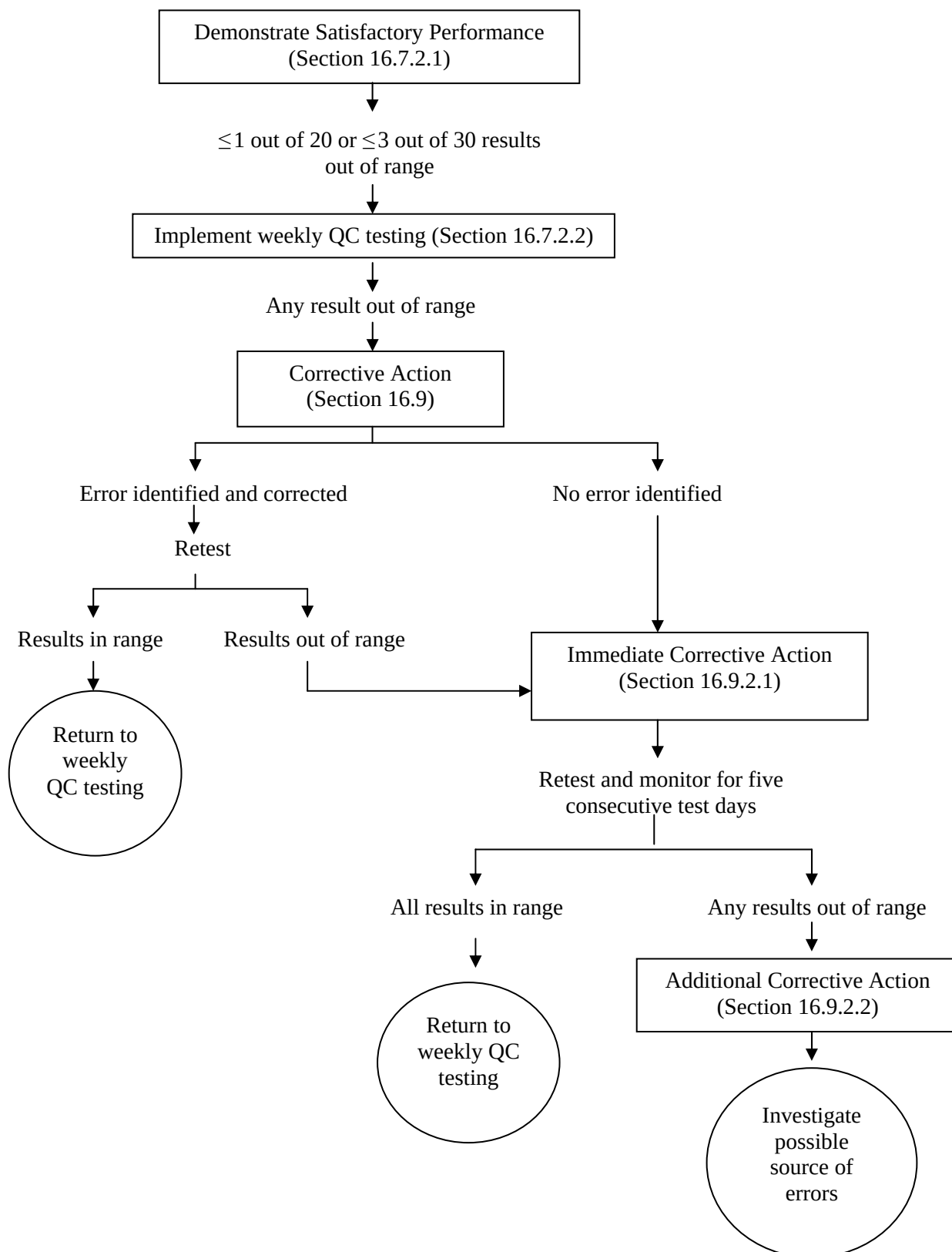
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Appendix A. Quality Control Protocol Flow Charts**Quality Control (QC) Protocol: Daily Testing**

Appendix A. (Continued)**QC Protocol: Weekly Testing**

Appendix B. Preparation of Supplements, Media, and Reagents

B1 Supplements

B1.1 Cation Stock Solutions

1. To prepare a magnesium stock solution, dissolve 8.36 g of $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ in 100 mL of deionized water. This solution contains 10 mg of Mg^{++}/mL .
2. To prepare a calcium stock solution, dissolve 3.68 g of $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$ in 100 mL of deionized water. This solution contains 10 mg of Ca^{++}/mL .
3. Sterilize by membrane filtration and store at 2 to 8°C.

B1.2 Lysed Horse Blood (50%)

1. Aseptically mix equal volumes of defibrinated horse blood and sterile, deionized water (now 50% lysed horse blood [LHB]).
2. Freeze and thaw the 50% LHB until the cells are thoroughly lysed (about five to seven freeze-thaw cycles).
3. For use in broth dilution tests, the combination of broth and LHB must be clear, and this can be accomplished by centrifuging the 50% LHB at $12\,000 \times g$ for 20 minutes. Decant the supernatant and recentrifuge if necessary.
4. Aseptically add appropriate amounts of the 50% LHB to the cation-adjusted Mueller-Hinton broth (CAMHB) to yield a final concentration of 2.5% to 5% LHB.
5. Check the pH after aseptic addition of the blood to the autoclaved and cooled broth.
6. The blood may be added to microdilution trays either when they are first dispensed or after they are thawed just before inoculation. If added after plate preparation, the blood must be added along with the inoculum so further dilution of the antimicrobial agent in the tray does not occur (see Section 10.4.2), and the final concentration of LHB in the well is 2.5% to 5%. Alternatively, add 5 μL of 50% LHB to each 100- μL well before tray inoculation. This is only acceptable if total volume of liquids added to the well does not exceed 10 μL .
7. The 50% LHB can be stored frozen at $\leq -20^\circ\text{C}$ indefinitely.

NOTE: LHB prepared for use in broth microdilution is available from commercial sources.

B2 Agar Media

B2.1 Mueller-Hinton Agar

Use Mueller-Hinton agar (MHA) from a commercially available dehydrated base. Only MHA formulations that have been tested according to, and that meet the acceptance limits described in, CLSI document M06¹ should be used.

1. Prepare according to the manufacturer's directions.

Appendix B. (Continued)

2. For use in preparation of agar dilution plates, immediately after autoclaving, allow the agar to cool to 45 to 50°C in a water bath before aseptically adding antimicrobial solutions and heat-labile supplements, and pouring the plates (see Section 9.2.2).
3. Check the pH of each batch of MHA when the medium is prepared. The exact method used depends largely on the type of equipment available in the laboratory. The agar medium should have a pH between 7.2 and 7.4 at room temperature, and must therefore be checked after gelling. If the pH is less than 7.2, certain drugs will appear to lose potency (eg, aminoglycosides, macrolides), whereas other agents may appear to have excessive activity (eg, tetracyclines). If the pH is greater than 7.4, the opposite effects can be expected. Check the pH by one of the following means:
 - Macerate enough agar to submerge the tip of a pH electrode.
 - Allow a small amount of agar to solidify around the tip of a pH electrode in a beaker or cup.
 - Use a surface electrode.
4. Do not add supplemental calcium or magnesium cations to MHA.

B2.2 Mueller-Hinton Agar + 2% NaCl

1. Follow instructions in Section B2.1 except that for oxacillin, methicillin, or nafcillin testing of staphylococci, add 20 g NaCl to 1 L of MHA (2% w/v) before autoclaving.

B2.3 Mueller-Hinton Agar With 5% Sheep Blood or 2.5% to 5% Lysed Horse Blood

1. Prepare MHA as described above through Section B2.1 (2). When it has cooled to 45 to 50°C, add 50 mL of defibrinated sheep blood or 25 to 50 mL of LHB to 1 L of MHA.
2. Check the pH after aseptic addition of the blood to the autoclaved and cooled medium. The final pH should be the same as unsupplemented MHA, pH 7.2 to 7.4.

NOTE: For testing *Helicobacter pylori*, the sheep blood should be ≥ 2 weeks old before using to prepare the agar medium.

B2.4 GC Agar + 1% Defined Growth Supplement

1. Use a 1% defined growth supplement that contains the following ingredients per liter:
 - 1.1 g L-cystine
 - 0.03 g guanine HCl
 - 0.003 g thiamine HCl
 - 0.013 g *p*-aminobenzoic acid
 - 0.01 g vitamin B₁₂
 - 0.1 g thiamine pyrophosphate (cocarboxylase)
 - 0.25 g nicotinamide adenine dinucleotide (NAD)
 - 1 g adenine
 - 10 g L-glutamine
 - 100 g glucose
 - 0.02 g ferric nitrate
 - 25.9 g L-cysteine HCl

Appendix B. (Continued)

2. Prepare 1 L of single strength GC agar base from a commercially available dehydrated base according to the manufacturer's instructions.
3. After autoclaving, cool to 45 to 50°C in a 45 to 50°C water bath.
4. Add 10 mL of 1% defined growth supplement.

B3 Broth Media

B3.1 Cation-Adjusted Mueller-Hinton Broth

Obtain dehydrated Mueller-Hinton broth (MHB) base from a commercially available source that is accessible in either of the following two forms:

- Containing the desired concentration of cations (20 to 25 mg of Ca^{++} /L and 10 to 12.5 mg of Mg^{++} /L)
- Without supplemental cation, in which case, the cation content is usually inadequate

The first form does not require further cation adjustment (except for testing daptomycin when the broth must be supplemented to 50 mg/L of Ca^{++}), if prepared according to the manufacturer's recommendations.

For the second form, the amount of Ca^{++} and Mg^{++} cations present in the medium must be determined before it is prepared. This information is usually available in the certificate of analysis for the lot of medium obtained; if not, contact the manufacturer for that information. The concentration of cations already present in the broth must be accounted for when calculating the amount of Ca^{++} or Mg^{++} , if any, that needs to be added.

The instructions for cation adjustments below should be followed only when the initial MHB has been certified by the manufacturer or measured by the user to contain no or inadequate amounts of Ca^{++} and Mg^{++} when assayed for total divalent cation content by atomic absorption spectrophotometry. Inadequate cation content may lead to erroneous results. For testing daptomycin, MHB must be supplemented to 50 mg/L of Ca^{++} .

1. Prepare MHB according to the manufacturer's recommendations, autoclave, and before adding supplemental cations, chill overnight at 2 to 8°C (or in an ice bath if broth is to be used the same day).
2. With stirring, add 0.1 mL of chilled Ca^{++} or Mg^{++} stock solution per liter of broth for each desired increment of 1 mg/L in the final concentration in the adjusted MHB. It is preferable to add the Mg^{++} before adding the Ca^{++} . This medium is called CAMHB.

Example:

For MHB that contains:

- 4.3 mg/L Ca^{++}
- 3.8 mg/L Mg^{++}

Appendix B. (Continued)

(a) For Ca^{++} , the final amount needed is 20 to 25 mg/L.

$$25 \text{ mg/L} - 4.3 \text{ mg/L} = 20.7 \text{ mg/L}$$

[Final amount needed – Amount in medium = Amount to be added]

The medium needs 20.7 mg/L added.

$$20.7 \text{ mg/L} \cdot 0.1 \text{ mL} = 2.07 \text{ mL} \text{ [0.1 mL for each 1 mg/L needed]}$$

Therefore, add 2.07 mL of Ca^{++} stock per liter.

NOTE: For testing daptomycin, supplement the broth to 50 mg/L of Ca^{++} .

(b) For Mg^{++} , the final amount needed is 10 to 12.5 mg/L.

$$12.5 \text{ mg/L} - 3.8 \text{ mg/L} = 8.7 \text{ mg/L}$$

[Final amount needed – Amount in medium = Amount to be added]

The medium needs 8.7 mg/L added.

$$8.7 \text{ mg/L} \cdot 0.1 \text{ mL} = 0.87 \text{ mL} \text{ [0.1 mL for each 1 mg/L needed]}$$

Therefore, add 0.87 mL of Mg^{++} stock per liter.

3. Check the pH after additions of the cations. The final pH should be 7.2 to 7.4.

B3.2 Cation-Adjusted Mueller-Hinton Broth + 2% NaCl

1. Follow instructions in Section B3.1 except that for oxacillin, methicillin, or nafcillin testing of staphylococci, add 20 g NaCl to 1 L of MHB (2% w/v) before autoclaving.

B3.3 Cation-Adjusted Mueller-Hinton Broth + 2.5% to 5% Lysed Horse Blood

1. Prepare CAMHB as described in Section B3.1, except for each liter of broth desired, prepare with 50 to 100 mL less deionized water to account for the addition of the blood.
2. After autoclaving and cooling, add 50 to 100 mL of 50% LHB to 1 L of broth (see Section B1.2 for preparation of LHB).
3. Check the pH after aseptic addition of the blood to the medium. The final pH should be the same as CAMHB, pH 7.2 to 7.4.

B3.4 Cation-Adjusted Mueller-Hinton Broth + 0.002% Polysorbate 80 (P-80)

1. Prepare a 2% solution of P-80 by diluting full-strength P-80 50-fold (ie, 1 mL P-80 + 49 mL of deionized water).
2. Add 1 mL of the 2% P-80 solution per liter of broth before autoclaving.
3. The final pH should be the same as CAMHB, pH 7.2 to 7.4.

Appendix B. (Continued)

B3.5 *Haemophilus* Test Medium Broth

In its broth form, *Haemophilus* Test Medium (HTM) consists of the following ingredients:

- MHB
 - 15 µg/mL β-NAD
 - 15 µg/mL bovine or porcine hematin
 - 5 g/L yeast extract
 - 0.2 International Units (IU)/mL thymidine phosphorylase (if sulfonamides or trimethoprim are to be tested)
1. Prepare a fresh hematin stock solution by dissolving 50 mg of hematin powder in 100 mL of 0.01 mol/L NaOH with heat, and stirring until the powder is thoroughly dissolved.
 2. Prepare an NAD stock solution by dissolving 50 mg of NAD in 10 mL of distilled water; filter sterilize.
 3. Prepare MHB from a commercially available dehydrated base according to the manufacturer's directions, adding 5 g of yeast extract and 30 mL of hematin stock solution to 1 L of MHB.
 4. After autoclaving, cool to 45 to 50°C; aseptically add cations, if needed, as in CAMHB.
 5. If sulfonamides or trimethoprim are to be tested, add 0.2 IU thymidine phosphorylase to the medium aseptically.
 6. Aseptically add 3 mL of the NAD stock solution.
 7. The pH should be 7.2 to 7.4.

NOTE: *Haemophilus influenzae* (ATCC^{®a} 10211) is recommended as a useful additional quality control (QC) strain to verify the growth promotion properties of HTM. In particular, manufacturers of HTM are encouraged to use *H. influenzae* ATCC[®] 10211 as a supplemental QC test strain.

B4 Reagents

B4.1 0.5 McFarland Turbidity Standard

1. Prepare a 0.048 mol/L BaCl₂ (1.175% w/v BaCl₂•2H₂O) stock solution.
2. Prepare a 0.18 mol/L (0.36 N) H₂SO₄ (1% v/v) stock solution.
3. With constant stirring to maintain a suspension, add 0.5 mL of the BaCl₂ solution to 99.5 mL of the H₂SO₄ stock solution.
4. Verify the correct density of the turbidity standard by measuring absorbance using a spectrophotometer with a 1-cm light path and matched cuvettes. The absorbance at 625 nm should be 0.08 to 0.13 for the 0.5 McFarland standard.
5. Transfer the barium sulfate suspension in 4- to 6-mL aliquots into screw-cap tubes of the same size as those used for standardizing the bacterial inoculum.

^a ATCC[®] is a registered trademark of the American Type Culture Collection.

Appendix B. (Continued)

6. Tightly seal the tubes and store in the dark at room temperature.
7. Vigorously agitate the barium sulfate turbidity standard on a vortex mixer before each use and inspect for a uniformly turbid appearance. Replace the standard if large particles appear. Mix latex particle suspensions by inverting gently, not on a vortex mixer.
8. The barium sulfate standards should be replaced or their densities verified monthly.

NOTE: McFarland standards made from latex particle suspension are commercially available. When used, they should be mixed by inverting gently (not on a vortex mixer) immediately before use.

Reference for Appendix B

- ¹ CLSI. *Protocols for Evaluating Dehydrated Mueller-Hinton Agar; Approved Standard—Second Edition*. CLSI document M06-A2. Wayne, PA: Clinical and Laboratory Standards Institute; 2006.

Appendix C. Conditions for Dilution Antimicrobial Susceptibility Tests

C1. Conditions for Dilution Antimicrobial Susceptibility Tests for Nonfastidious Organisms

Organism/ Organism Group	M100 ¹ Table	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Minimal Quality Control	Comments/Modifications
<i>Enterobacteriaceae</i>	2A	CAMHB; MHA	Direct colony suspension in MHB or saline, or growth method	35 ± 2°C; ambient air	16 to 20 hours	<i>Escherichia coli</i> ATCC ^{®a} 25922 <i>E. coli</i> ATCC [®] 35218 (for β-lactam/β-lactamase inhibitor combinations)	
<i>Pseudomonas aeruginosa</i>	2B-1	CAMHB; MHA	Direct colony suspension in MHB or saline, or growth method	35 ± 2°C; ambient air	16 to 20 hours	<i>E. coli</i> ATCC [®] 25922 <i>P. aeruginosa</i> ATCC [®] 27853 <i>E. coli</i> ATCC [®] 35218 (for β-lactam/β-lactamase inhibitor combinations)	
<i>Acinetobacter</i> spp.	2B-2	CAMHB; MHA	Direct colony suspension in MHB or saline, or growth method	35 ± 2°C; ambient air	20 to 24 hours	<i>E. coli</i> ATCC [®] 25922 <i>P. aeruginosa</i> ATCC [®] 27853 <i>E. coli</i> ATCC [®] 35218 (for β-lactam/β-lactamase inhibitor combinations)	
<i>Burkholderia cepacia</i>	2B-3	CAMHB; MHA	Direct colony suspension in MHB or saline, or growth method	35 ± 2°C; ambient air	20 to 24 hours	<i>E. coli</i> ATCC [®] 25922 <i>P. aeruginosa</i> ATCC [®] 27853 <i>E. coli</i> ATCC [®] 35218 (for β-lactam/β-lactamase inhibitor combinations)	
<i>Stenotrophomonas maltophilia</i>	2B-4	CAMHB; MHA	Direct colony suspension in MHB or saline, or growth method	35 ± 2°C; ambient air	20 to 24 hours	<i>E. coli</i> ATCC [®] 25922 <i>P. aeruginosa</i> ATCC [®] 27853 <i>E. coli</i> ATCC [®] 35218 (for β-lactam/β-lactamase inhibitor combinations)	
Other Non- <i>Enterobacteriaceae</i>	2B-5	CAMHB; MHA	Direct colony suspension in MHB or saline, or growth method	35 ± 2°C; ambient air	16 to 20 hours	<i>E. coli</i> ATCC [®] 25922 <i>P. aeruginosa</i> ATCC [®] 27853 <i>E. coli</i> ATCC [®] 35218 (for β-lactam/β-lactamase inhibitor combinations)	

Appendix C. (Continued)

C1. (Continued)

Organism/ Organism Group	M100 ¹ Table	Medium	0.5 McFarland Inoculum	Incubation	Incubation Time	Minimal Quality Control ^b	Comments/Modifications
<i>Staphylococcus</i> spp.	2C	CAMHB; CAMHB with 2% NaCl for oxacillin, methicillin, and nafcillin; CAMHB supplemented to 50 µg/mL calcium (daptomycin); MHA; MHA with 2% NaCl for oxacillin, methicillin, and nafcillin	Direct colony suspension in MHB	35 ± 2°C; ambient air	16 to 20 hours; 24 hours for oxacillin, methicillin, nafcillin, and vancomycin	<i>Staphylococcus aureus</i> ATCC [®] 29213 <i>E. coli</i> ATCC [®] 35218 (for β-lactam/β-lactamase inhibitor combinations)	Agar dilution has not been validated for daptomycin. Testing at temperatures above 35°C may not detect MRS.
<i>Enterococcus</i> spp.	2D	CAMHB; CAMHB supplemented to 50 µg/mL calcium (daptomycin); MHA	Direct colony suspension in MHB or saline, or growth method	35 ± 2°C; ambient air	16 to 20 hours; 24 hours for vancomycin	<i>Enterococcus faecalis</i> ATCC [®] 29212	Agar dilution has not been validated for daptomycin.

Appendix C. (Continued)

C2. Conditions for Dilution Antimicrobial Susceptibility Tests for Fastidious Organisms

Organism/ Organism Group	M100 ¹ Table	Medium	0.5 McFarland Inoculum ^b	Incubation	Incubation Time	Minimal Quality Control ^b	Comments/Modifications
<i>Haemophilus influenzae</i> <i>Haemophilus parainfluenzae</i>	2E	HTM broth	Direct colony suspension in MHB or saline prepared from an overnight (preferably 20- to 24-hour) chocolate agar plate ^c	35 ± 2°C; ambient air	20 to 24 hours	<i>H. influenzae</i> ATCC [®] 49247 <i>H. influenzae</i> ATCC [®] 49766 <i>E. coli</i> ATCC [®] 35218 (for amoxicillin-clavulanate acid)	
<i>Neisseria gonorrhoeae</i>	2F	GC agar base with 1% defined growth supplement ^d	Direct colony suspension in MHB or 0.9% phosphate-buffered saline, pH 7.0, prepared from an overnight chocolate agar plate incubated in 5% CO ₂	36 ± 1°C (do not exceed 37°C); 5% CO ₂	20 to 24 hours	<i>N. gonorrhoeae</i> ATCC [®] 49226	The use of a cysteine-free supplement is required for agar dilution tests with carbapenems and clavulanic acid. Cysteine-containing defined growth supplements <i>do not</i> significantly alter dilution test results with other drugs.
<i>Streptococcus pneumoniae</i>	2G	CAMHB with 2.5% to 5% LHB broth; CAMHB with 2.5% to 5% LHB supplemented to 50 µg/mL calcium for daptomycin; MHA + 5% sheep blood	Direct colony suspension in MHB or saline using colonies from an overnight (18- to 20-hour) sheep blood agar plate	35 ± 2°C; ambient air (CO ₂ if necessary for growth with agar dilution)	20 to 24 hours	<i>S. pneumoniae</i> ATCC [®] 49619	Recent studies using the agar dilution method have not been performed or reviewed by the subcommittee. Agar dilution has not been validated for daptomycin.
<i>Streptococcus</i> spp.	2H-1 and 2H-2	CAMHB with 2.5% to 5% LHB; CAMHB with 2.5% to 5% LHB supplemented to 50 µg/mL calcium for daptomycin; MHA + 5% sheep blood	Direct colony suspension in MHB or saline	35 ± 2°C; ambient air (CO ₂ if necessary for growth with agar dilution)	20 to 24 hours	<i>S. pneumoniae</i> ATCC [®] 49619	Recent studies using the agar dilution method have not been performed or reviewed by the subcommittee. Agar dilution has not been validated for daptomycin.
<i>N. meningitidis</i>	2I	CAMHB with 2.5% to 5% LHB; MHA with 5% sheep blood	Direct colony suspension in MHB or saline prepared from an overnight chocolate agar plate incubated in 5% CO ₂ ^d	35 ± 2°C; 5% CO ₂	20 to 24 hours	<i>S. pneumoniae</i> ATCC [®] 49619 (ambient air or 5% CO ₂ ; azithromycin QC test must be incubated in ambient air) <i>E. coli</i> ATCC [®] 25922 (ambient air or 5% CO ₂ ; for ciprofloxacin, nalidixic acid, minocycline, and sulfisoxazole)	Caution: Perform all tests in a BSC.

Appendix C. (Continued)

C2. (Continued)

Organism/ Organism Group	M100 ¹ Table	Medium	0.5 McFarland Inoculum ^b	Incubation	Incubation Time	Minimal Quality Control ^b	Comments/Modifications
Anaerobes	2J	Brucella broth with hemin (5 µg/mL), vitamin K ₁ (1 µg/mL), and 5% LHB; Brucella agar with hemin (5 µg/mL), vitamin K ₁ (1 µg/mL), and 5% laked sheep blood (see CLSI document M11 ² for method of preparation for these media)	Direct colony suspension in Brucella broth or growth method (see CLSI document M11 ² for exact methods for inoculum preparation)	36 ± 1°C; anaerobically	Broth, 46 to 48 hours; agar, 42 to 48 hours	<i>Bacteroides fragilis</i> ATCC [®] 25285 <i>Bacteroides thetaiotaomicron</i> ATCC [®] 29741 <i>Clostridium difficile</i> ATCC [®] 700057 <i>Eubacterium lentum</i> ATCC [®] 43055 Test any two for agar dilution; test one for a single broth dilution test	Anaerobes

Abbreviations: BSC, biological safety cabinet; CAMHB, cation-adjusted Mueller-Hinton broth; HTM, *Haemophilus* Test Medium; LHB, lysed horse blood; MHA, Mueller-Hinton agar; MHB, Mueller-Hinton broth; MRS, methicillin-resistant staphylococci; QC, quality control.

Footnotes

^a ATCC[®] is a registered trademark of the American Type Culture Collection.

^b See specific M100¹ Supplemental Tables for additional QC recommendations for screening and confirmatory tests.

^c This suspension will contain approximately 1 to 4 × 10⁸ colony-forming units (CFU)/mL. Exercise care in preparing this suspension, because higher inoculum concentrations may lead to false-resistant results with some β-lactam antimicrobial agents, particularly when β-lactamase-producing strains of *H. influenzae* are tested.

^d Colonies grown on sheep blood agar may be used for inoculum preparation. However, the 0.5 McFarland suspension obtained from sheep blood agar will contain fewer CFU/mL. This must be considered when preparing the final dilution before panel inoculation, as guided by colony counts.

References for Appendix C

¹ CLSI. *Performance Standards for Antimicrobial Susceptibility Testing; Twenty-Second Informational Supplement*. CLSI document M100-S22. Wayne, PA: Clinical and Laboratory Standards Institute; 2012.

² CLSI. *Methods for Antimicrobial Susceptibility Testing of Anaerobic Bacteria; Approved Standard—Seventh Edition*. CLSI document M11-A7. Wayne, PA: Clinical and Laboratory Standards Institute; 2007.

Appendix D. Quality Control Strains for Antimicrobial Susceptibility Tests (refer to current edition of M100¹ for the most updated version of this table)

Quality Control Strain	Organism Characteristics	Disk Diffusion Tests	MIC Tests	Screening Tests	Other
<i>Enterococcus faecalis</i> ATCC ^{®a} 29212			•Nonfastidious gram-positives	•Vancomycin agar •High-level aminoglycoside resistance	•Assess suitability of medium for sulfonamide or trimethoprim MIC tests. ^e
<i>E. faecalis</i> ATCC [®] 51299	•Resistant to vancomycin (<i>VanB</i>) and high-level aminoglycosides			•Vancomycin agar •High-level aminoglycoside resistance	•Assess suitability of cation content in each batch/lot of Mueller-Hinton for daptomycin broth microdilution
<i>Escherichia coli</i> ATCC [®] 25922	•β-Lactamase negative	•Nonfastidious gram-negatives • <i>Neisseria meningitidis</i>	•Nonfastidious gram-negatives • <i>N. meningitidis</i>		
<i>E. coli</i> ATCC [®] 35218	•Contains plasmid-encoded TEM-1 β-lactamase (non-ESBL) ^{b,c,f,g}	•β-Lactam/β-lactamase inhibitor combinations	•β-Lactam/β-lactamase inhibitor combinations		
<i>Haemophilus influenzae</i> ATCC [®] 49247	•BLNAR	• <i>Haemophilus</i> spp.	• <i>Haemophilus</i> spp.		
<i>H. influenzae</i> ATCC [®] 49766	•Ampicillin susceptible	• <i>Haemophilus</i> spp. (more reproducible with selected β-lactams)	• <i>Haemophilus</i> spp. (more reproducible with selected β-lactams)		
<i>Klebsiella pneumoniae</i> ATCC [®] 700603	•Contains SHV-18 ESBL ^{c,f,g}	•ESBL screen and confirmatory tests	•ESBL screen and confirmatory tests		
<i>Neisseria gonorrhoeae</i> ATCC [®] 49226	• CMRNG	• <i>N. gonorrhoeae</i>	• <i>N. gonorrhoeae</i>		
<i>Pseudomonas aeruginosa</i> ATCC [®] 27853 ^d	• Contains inducible AmpC β-lactamase	•Nonfastidious gram-negatives	•Nonfastidious gram-negatives		•Assess suitability of cation content in each batch/lot of Mueller-Hinton for gentamicin MIC and disk diffusion.
<i>Staphylococcus aureus</i> ATCC [®] 25923	•β-Lactamase negative • <i>mecA</i> negative •Little value in MIC testing due to extreme susceptibility to most drugs	•Nonfastidious gram-positives			

Appendix D. (Continued)

Quality Control Strain	Organism Characteristics	Disk Diffusion Tests	MIC Tests	Screening Tests	Other
<i>S. aureus</i> ATCC® 29213	- Weak β-lactamase–producing strain <i>mecA</i> negative		Nonfastidious gram-positives	Oxacillin agar	
<i>S. aureus</i> ATCC® 43300	Oxacillin-resistant, <i>mecA</i> positive	Cefoxitin disk diffusion testing	Cefoxitin MIC testing	Oxacillin agar	
<i>S. aureus</i> ATCC® BAA-1708	High-level mupirocin resistance mediated by the <i>mupA</i> gene	Screening test for high-level mupirocin resistance	Screening test for high-level mupirocin resistance		
<i>Streptococcus pneumoniae</i> ATCC® 49619	Penicillin intermediate by altered penicillin binding protein	<i>S. pneumoniae</i> <i>Streptococcus</i> spp. <i>N. meningitidis</i>	<i>S. pneumoniae</i> <i>Streptococcus</i> spp. <i>N. meningitidis</i>		
Supplemental QC Strains^h					
<i>E. faecalis</i> ATCC® 29212			Ceftaroline MIC testing		
<i>E. faecalis</i> ATCC® 33186					Alternative to <i>E. faecalis</i> ATCC® 29212 to assess suitability of medium for sulfonamide or trimethoprim MIC tests^e
<i>H. influenzae</i> ATCC® 10211					Assess each batch/lot for growth capabilities of HTM.
<i>K. pneumoniae</i> ATCC® BAA-1705	- KPC-producing strain^c - MHT positive	Phenotypic confirmatory test for carbapenemase production (MHT)			
<i>K. pneumoniae</i> ATCC® BAA-1706	- Resistant to carbapenems by mechanisms other than carbapenemase - MHT negative	Phenotypic confirmatory test for carbapenemase production (MHT)			
<i>S. aureus</i> ATCC® BAA-976	Contains <i>msrA</i>-mediated macrolide-only resistance	Assess disk approximation tests with erythromycin and clindamycin (D-zone test negative).	QC – See M100¹ Tables 2C Supplemental Tables 2 and 3, and Tables 3A and 4A		
<i>S. aureus</i> ATCC® BAA-977	Contains inducible <i>ermA</i>-mediated resistance	Assess disk approximation tests with erythromycin and clindamycin (D-zone test positive).	Routine QC for inducible clindamycin test by MIC method—See M100¹ Tables 2C Supplemental Tables 2 and 3, and Tables 3A and 4A		

Abbreviations: BLNAR, β -lactamase negative, ampicillin-resistant; CMRNG, chromosomally mediated penicillin resistant *Neisseria gonorrhoeae*; ESBL, extended-spectrum β -lactamase; HLAR, high-level aminoglycoside resistance; HTM, *Haemophilus* Test Medium; KPC, *Klebsiella pneumoniae* carbapenemase; MHT, modified Hodge test; MIC, minimal inhibitory concentration; QC, quality control; TEM, Temoneira (first patient from whom a TEM β -lactamase–producing strain was reported).

Appendix D. (Continued)

Footnotes

^a ATCC® is a registered trademark of the American Type Culture Collection.

^b *E. coli* ATCC® 35218 is recommended only as a control organism for β -lactamase inhibitor combinations, such as those containing clavulanic acid, sulbactam, or tazobactam. This strain contains a plasmid-encoded β -lactamase (non-ESBL); subsequently, the organism is resistant to many penicillinase-labile drugs but susceptible to β -lactam/ β -lactamase inhibitor combinations. The plasmid must be present in the QC strain for the QC test to be valid; however, the plasmid may be lost during storage at refrigerator or freezer temperatures. To ensure the plasmid is present, test the strain with a β -lactam agent alone (either ampicillin, amoxicillin, piperacillin, or ticarcillin) in addition to a β -lactam/ β -lactamase inhibitor agent (eg, amoxicillin-clavulanic acid). If the strain loses the plasmid, it will be susceptible to the β -lactam agent when tested alone, indicating that the QC test is invalid and a new culture of *E. coli* ATCC® 35218 must be used.

^c Careful attention to organism maintenance (eg, minimal subcultures) and storage (eg, -60°C or below) is especially important for QC strains *E. coli* ATCC® 35218, *K. pneumoniae* ATCC® 700603, and *K. pneumoniae* ATCC® BAA-1705 because spontaneous loss of the plasmid encoding the β -lactamase or carbapenemase has been documented. Plasmid loss leads to QC results outside the acceptable limit, such as decreased MICs for *E. coli* ATCC® 35218 with enzyme-labile penicillins (eg, ampicillin, piperacillin, ticarcillin), decreased MICs for *K. pneumoniae* ATCC® 700603 with cephalosporins and aztreonam, and false-negative MHT with *K. pneumoniae* ATCC® BAA-1705.

^d Develops resistance to β -lactam antimicrobial agents after repeated transfers onto laboratory media. Minimize by removing new culture from storage at least monthly or whenever the strain begins to show resistance.

^e End points should be easy to read (as 80% or greater reduction in growth as compared to the control) if media have acceptable levels of thymidine.

^f Rasheed JK, Anderson GJ, Yigit H, et al. Characterization of the extended-spectrum β -lactamase reference strain, *Klebsiella pneumoniae* K6 (ATCC® 700603), which produces the novel enzyme SHV-18. *Antimicrob Agents Chemother*. 2000;44(9):2382-2388.

^g Queenan AM, Foleno B, Gownley C, Wira E, Bush K. Effects of inoculum and β -lactamase activity in AmpC- and extended-spectrum β -lactamase (ESBL)-producing *Escherichia coli* and *Klebsiella pneumoniae* clinical isolates tested by using NCCLS ESBL methodology. *J Clin Microbiol*. 2004;42(1):269-275.

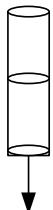
^h See Section 16.3.

Reference for Appendix D

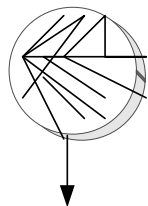
- ¹ CLSI. *Performance Standards for Antimicrobial Susceptibility Testing; Twenty-Second Informational Supplement*. CLSI document M100-S22. Wayne, PA: Clinical and Laboratory Standards Institute; 2012.

Appendix E. Quality Control Strain Maintenance (also refer to Section 16.4)

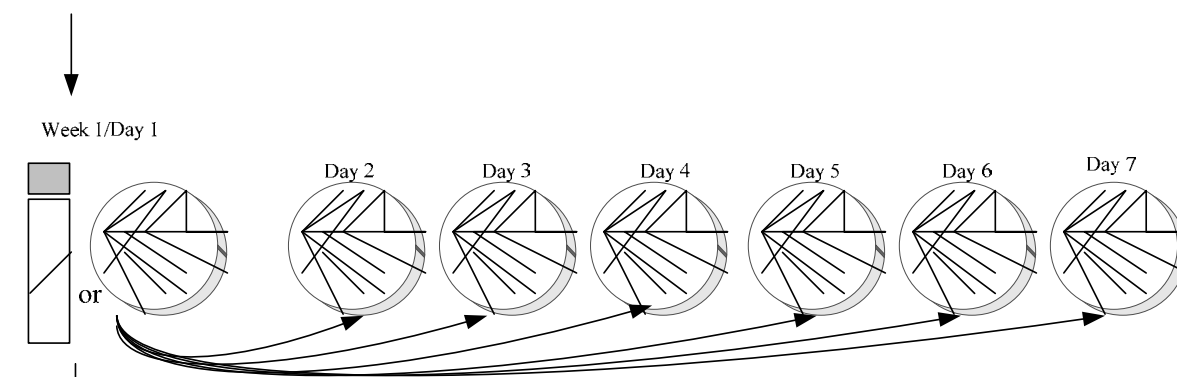
1. Rehydrate new stock culture or obtain strain from frozen stock.



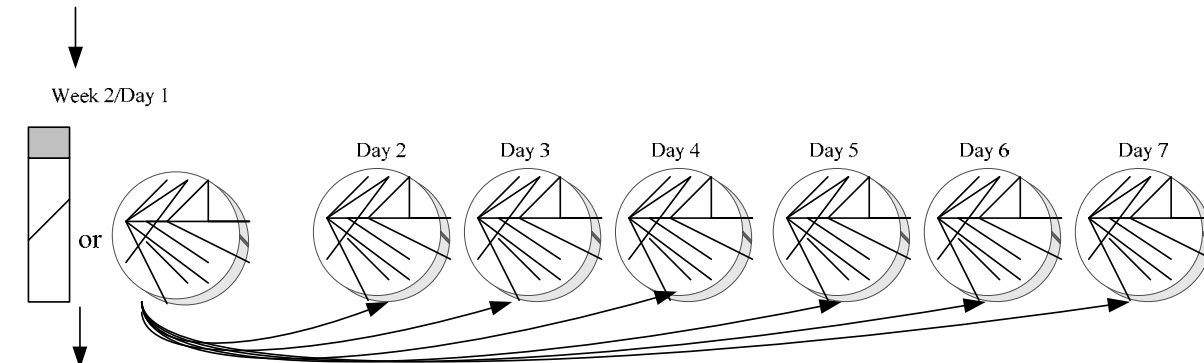
2. Subculture to appropriate media and incubate (primary subculture).



3. Subculture, incubate, and store as appropriate for the organism type. Use isolated colonies from Days 1 to 7 as working cultures for testing.



4. Prepare new subculture every seven days (from slant or Day 1 working culture). Store at appropriate temperature for organism type. Use fresh working cultures each test day.



5. Repeat for Week 3 and Week 4. After four weeks, discard subculture and pull strain from freezer stock or rehydrate new stock culture.

NOTE 1: Subculture frozen or lyophilized cultures twice before use.

NOTE 2: For quality control testing, select isolated colonies from working cultures.

NOTE 3: If contaminated or questionable performance, prepare new primary subculture, working culture, or obtain new stock culture.

NOTE 4: It may be necessary to prepare new subcultures or Week 1/Day 1 working cultures every two weeks for some organisms (eg, *Pseudomonas aeruginosa* ATCC^{®a} 27853, *Enterococcus faecalis* ATCC[®] 51299, *Streptococcus pneumoniae* ATCC[®] 49619).

^aATCC[®] is a registered trademark of the American Type Culture Collection.

The Quality Management System Approach

Clinical and Laboratory Standards Institute (CLSI) subscribes to a quality management system approach in the development of standards and guidelines, which facilitates project management; defines a document structure via a template; and provides a process to identify needed documents. The quality management system approach applies a core set of “quality system essentials” (QSEs), basic to any organization, to all operations in any health care service’s path of workflow (ie, operational aspects that define how a particular product or service is provided). The QSEs provide the framework for delivery of any type of product or service, serving as a manager’s guide. The QSEs are as follows:

Organization	Personnel	Process Management	Nonconforming Event Management
Customer Focus	Purchasing and Inventory	Documents and Records	Assessments
Facilities and Safety	Equipment	Information Management	Continual Improvement

M07-A9 addresses the QSEs indicated by an “X.” For a description of the other documents listed in the grid, please refer to the Related CLSI Reference Materials section on the following page.

Organization	Customer Focus	Facilities and Safety	Personnel	Purchasing and Inventory	Equipment	Process Management	Documents and Records	Information Management	Nonconforming Event Management	Assessments	Continual Improvement
		M29				X M02 M06 M11 M23 M45	X				

Path of Workflow

A path of workflow is the description of the necessary processes to deliver the particular product or service that the organization or entity provides. A laboratory path of workflow consists of the sequential processes: preexamination, examination, and postexamination and their respective sequential subprocesses. All laboratories follow these processes to deliver the laboratory’s services, namely quality laboratory information.

M07-A9 addresses the clinical laboratory path of workflow steps indicated by an “X.” For a description of the other documents listed in the grid, please refer to the Related CLSI Reference Materials section on the following page.

Preexamination				Examination			Postexamination	
Examination ordering	Sample collection	Sample transport	Sample receipt/processing	Examination	Results review and follow-up	Interpretation	Results reporting and archiving	Sample management
				X M02	X M02 M11 M100	X M02 M11 M100	X M02 M11 M100	

Related CLSI Reference Materials*

- M02-A11** **Performance Standards for Antimicrobial Disk Susceptibility Tests; Approved Standard—Eleventh Edition (2012).** This document contains the current Clinical and Laboratory Standards Institute–recommended methods for disk susceptibility testing, criteria for quality control testing, and updated tables for interpretive zone diameters.
- M06-A2** **Protocols for Evaluating Dehydrated Mueller-Hinton Agar; Approved Standard—Second Edition (2006).** This document provides procedures for evaluating production lots of dehydrated Mueller-Hinton agar, and for developing and applying reference media.
- M11-A7** **Methods for Antimicrobial Susceptibility Testing of Anaerobic Bacteria; Approved Standard—Seventh Edition (2007).** This document provides reference methods for the determination of minimal inhibitory concentrations (MICs) of anaerobic bacteria by agar dilution and broth microdilution.
- M23-A3** **Development of *In Vitro* Susceptibility Testing Criteria and Quality Control Parameters; Approved Guideline—Third Edition (2008).** This document addresses the required and recommended data needed for the selection of appropriate interpretive criteria and quality control ranges for antimicrobial agents.
- M29-A3** **Protection of Laboratory Workers From Occupationally Acquired Infections; Approved Guideline—Third Edition (2005).** Based on US regulations, this document provides guidance on the risk of transmission of infectious agents by aerosols, droplets, blood, and body substances in a laboratory setting; specific precautions for preventing the laboratory transmission of microbial infection from laboratory instruments and materials; and recommendations for the management of exposure to infectious agents.
- M45-A2** **Methods for Antimicrobial Dilution and Disk Susceptibility Testing of Infrequently Isolated or Fastidious Bacteria; Approved Guideline—Second Edition (2010).** This document provides guidance to clinical microbiology laboratories for standardized susceptibility testing of infrequently isolated or fastidious bacteria that are not presently included in CLSI documents M02 or M07. The tabular information in this document presents the most current information for drug selection, interpretation, and quality control for the infrequently isolated or fastidious bacterial pathogens included in this guideline.
- M100-S22** **Performance Standards for Antimicrobial Susceptibility Testing; Twenty-Second Informational Supplement (2012).** This document provides updated tables for the Clinical and Laboratory Standards Institute antimicrobial susceptibility testing standards M02-A11 and M07-A9.

* CLSI documents are continually reviewed and revised through the CLSI consensus process; therefore, readers should refer to the most current editions.

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R-Biopharm AG
Radiometer America, Inc.
Roche Diagnostics GmbH
Roche Diagnostics, Inc.
Roche Molecular Systems
RPL Laboratory Solutions, Inc.
DBA RPL Compliance Solutions
Sanofi Pasteur
Sarstedt, Inc.
Sekisui Diagnostics
Seventh Sense Biosystems
Siemens Healthcare Diagnostics Inc.
Siemens Healthcare Diagnostics Products GmbH
Soloy Laboratory Consulting Services, LLC
SomaLogic
Sphere Medical Holding Limited
Streck Laboratories, Inc.
Super Religare Laboratories Ltd
Sysmex America, Inc.
Sysmex Corporation - Japan
Tandem Diagnostics
Tetraphase Pharmaceuticals

The Clinical Microbiology Institute
The Medicines Company
TheraDoc
Theranos
Theravance Inc.
Thermo Fisher Scientific
Thermo Fisher Scientific, Oxoid Products
Thermo Fisher Scientific, Remel
Transasia Bio-Medicals Limited
Tulip Group
Ventana Medical Systems Inc.
VeracYTE, Inc.
Vivacta
Watson Pharmaceuticals
Wellstat Diagnostics, LLC
XDX, Inc.

Associate Active Members

31st Medical Group SGSL (AE)
3rd Medical Group (AK)
48th Medical Group/MDSS RAF Lakenheath (AE)
55th Medical Group/SGSAL (NE)
59th MDW/859th MDTs/MTL Wilford Hall Medical Center (TX)
82 MDG/SGSCL Sheppard AFB (TX)
Academisch Ziekenhuis-VUB UZ Brussel (Belgium)
ACL Laboratories (IL)
ACL Laboratories (WI)
Adams County Hospital (OH)
Adena Regional Medical Center Hospital (OH)
Affiliated Laboratory, Inc. (ME)
Akron Children's Hospital (OH)
Al Ain Hospital (Abu Dhabi, United Arab Emirates)
Al Hada Armed Forces Hospital/TAIF/KSA (Saudi Arabia)
Al Noor Hospital (United Arab Emirates)
Al Rabha Hospital (United Arab Emirates)
Alameda County Medical Center (CA)
Albany Medical Center Hospital (NY)
Albemarle Hospital (NC)
Alberta Health Services (AB, Canada)
Alexandra Hospital (Singapore)
All Children's Hospital (FL)
Allegiance Health (MI)
Alpena Regional Medical Center (MI)
Alta Bates Summit Medical Center (CA)
Alverno Clinical Laboratories, Inc. (IN)
American Esoteric Laboratories (AEL) (TN)
American University of Beirut Medical Center (NJ)
Anand Diagnostic Laboratory (India)
Anne Arundel Medical Center (MD)
Antech Diagnostics (CA)
Antelope Valley Hospital District (CA)
Appalachian Regional Healthcare System (NC)
Arkansas Children's Hospital (AR)
Arkansas Dept of Health Public Health Laboratory (AR)
Arkansas Methodist Medical Center (AR)
ARUP Laboratories (UT)
Asan Medical Center (Korea, Republic Of)
Asante Health System (OR)
Ashley County Medical Center (AR)
Asiri Group of Hospitals Ltd. (Sri Lanka)
Aspen Valley Hospital (CO)
ASPETAR (Qatar Orthopedic and Sports Medicine Hospital) (Qatar)
Aspirus Wausau Hospital (WI)
Auburn Regional Medical Center (WA)
Augusta Health (VA)
Aultman Hospital (OH)
Avera McKennan Hospital & University Health Center (SD)
AZ Sint-Jan (Belgium)
Azienda Ospedale Di Lecco (Italy)
Baptist Hospital of Miami (FL)
Baptist Memorial Health Care Corporation - Hospital Laboratories Works (TN)
Barnes-Jewish Hospital (MO)
Bassett Healthcare (NY)
Baton Rouge General (LA)
Baxter Regional Medical Center (AR)
BayCare Health System (FL)
Baylor Health Care System (TX)
Bayou Pathology, APMC (LA)
Baystate Medical Center (MA)
BC Biomedical Laboratories (BC, Canada)
Beloit Memorial Hospital (WI)
Beth Israel Medical Center (NY)
Bio-Reference Laboratories (NJ)
Blanchard Valley Hospital (OH)
Bon Secours Health Partners (VA)

Bonnyville Health Center (AB, Canada)
 Boston Medical Center (MA)
 Boulder Community Hospital (CO)
 Boyce & Bynum Pathology Labs (MO)
 Brant Community Healthcare
 System/Brant General Hospital
 (Ontario, Canada)
 Bremerton Naval Hospital (WA)
 Brian All Good Community Hospital/121
 Combat (AP)
 Bridgeport Hospital (CT)
 Brooke Army Medical Center (TX)
 Broward General Medical Center (FL)
 Cadham Provincial Laboratory-MB
 Health (MB, Canada)
 Calgary Laboratory Services (AB,
 Canada)
 California Pacific Medical Center (CA)
 Cambridge Health Alliance (MA)
 Cape Fear Valley Medical Center
 Laboratory (NC)
 Capital Coast Health (New Zealand)
 Capital Health System Mercer Campus
 (NJ)
 Caritas Norwood Hospital (MA)
 Carl R. Darnall Army Medical Center
 Department of Pathology (TX)
 Carolina Medical Laboratory (NC)
 Carolinas Healthcare System (NC)
 Carpermor S.A. de C.V. (D.F., Mexico)
 Catholic Health Initiatives (KY)
 Cedars-Sinai Medical Center (CA)
 Cenetron Diagnostics (TX)
 Central Baptist Hospital (KY)
 Central Kansas Medical Center (KS)
 Centre Hospitalier Anna-Laberge
 (Quebec, Canada)
 Centre Hospitalier Regional De Trois
 Rivières (PQ, Canada)
 Centro Médico Imbanaco (Colombia)
 Chaleur Regional Hospital (NB, Canada)
 Chang Gung Memorial Hospital
 (Taiwan)
 Changhua Christian Hospital (Taiwan)
 Changi General Hospital (Singapore)
 Chatham - Kent Health Alliance (ON,
 Canada)
 Chesapeake General Hospital (VA)
 Chester County Hospital (PA)
 Children's Healthcare of Atlanta (GA)
 Childrens Hosp.- Kings Daughters (VA)
 Children's Hospital & Research Center
 At Oakland (CA)
 Childrens Hospital Los Angeles (CA)
 Children's Hospital Medical Center (OH)
 Children's Hospital of Central California
 (CA)
 Children's Hospital of Orange County
 (CA)
 Children's Hospital of Philadelphia (PA)
 Childrens Hospital of Wisconsin (WI)
 Children's Hospitals and Clinics (MN)
 Children's Medical Center (OH)
 Children's Medical Center (TX)
 Christiana Care Health Services (DE)
 CHU - Saint Pierre (Belgium)
 CHU Sainte-Justine (Quebec, Canada)
 CHUM Hospital Saint-Luc (Quebec,
 Canada)
 CHW-St. Mary's Medical Center (CA)
 City of Hope National Medical Center
 (CA)
 Clearstone Central Laboratories (ON,
 Canada)
 Cleveland Clinic (OH)
 Cleveland Heartlab, LLC (OH)
 Clinica Alemana De Santiago (Chile)
 Clinical and Laboratory Standards
 Institute (PA)
 Clinical Labs of Hawaii (HI)
 Clinical Trials Service (Belgium)
 College of Physicians and Surgeons of
 Alberta (AB, Canada)
 Collingwood General & Marine Hospital
 (ON, Canada)
 Commonwealth of Kentucky (KY)
 Commonwealth of Virginia (DCLS)
 (VA)
 Community Hospital (IN)
 Community Hospital of the Monterey
 Peninsula (CA)
 Community Medical Center (NJ)
 Community Memorial Hospital (WI)
 Complexe Hospitalier de la Sagamie
 (Quebec, Canada)
 CompuNet Clinical Laboratories Quest
 Diagnostics JV (OH)
 Concord Hospital (NH)
 Consultants Laboratory of WI LLC (WI)
 Contra Costa Regional Medical Center
 (CA)
 Cook Children's Medical Center (TX)
 Cookeville Regional Medical Center
 (TN)
 Cornwall Community Hospital (ON,
 Canada)
 Covance CLS (IN)
 Covenant Medical Center (TX)

Creighton Medical Lab (NE)
 Crozer-Chester Medical Center (PA)
 Cumberland Medical Center (TN)
 Darwin Library NT Territory Health
 Services (NT, Australia)
 David Grant Medical Center (CA)
 Daviess Community Hospital (IN)
 Deaconess Hospital Laboratory (IN)
 Dean Medical Center (WI)
 DHHS NC State Lab of Public Health
 (NC)
 DiagnoSearch Life Sciences Inc.
 (Maharashtra, India)
 Diagnostic Laboratory Services, Inc. (HI)
 Diagnostic Services of Manitoba (MB,
 Canada)
 DMC University Laboratories (MI)
 Drake Center (OH)
 Driscoll Children's Hospital (TX)
 DUHS Clinical Laboratories Franklin
 Site (NC)
 Dynacare Laboratory (WI)
 Dynacare NW, Inc - Seattle (WA)
 DynaLIFE (AB, Canada)
 E. A. Conway Medical Center (LA)
 East Georgia Regional Medical Center
 (GA)
 East Kootenay Regional Hospital
 Laboratory-Interior Health (BC,
 Canada)
 East Texas Medical Center-Pittsburg
 (TX)
 Eastern Health - Health Sciences Centre
 (NL, Canada)
 Eastern Health Pathology (Victoria,
 Australia)
 Easton Hospital (PA)
 Edward Hospital (IL)
 Effingham Hospital (GA)
 Elmhurst Hospital Center (NY)
 Emory University Hospital (GA)
 Evangelical Community Hospital (PA)
 Evans Army Community Hospital (CO)
 Exeter Hospital (NH)
 Exosome Diagnostics, Inc. (MN)
 Federal Medical Center (MN)
 First Health of the Carolinas Moore
 Regional Hospital (NC)
 Fletcher Allen Health Care (VT)
 Fleury S.A. (Brazil)
 Florida Hospital (FL)
 Fox Chase Cancer Center (PA)
 Fraser Health Authority Royal
 Columbian Hospital Site (BC, Canada)
 Fundação Faculdade de Medicina (SP,
 Brazil)
 Gamma-Dynacare Laboratories (ON,
 Canada)
 Garden City Hospital (MI)
 Garfield Medical Center (CA)
 Gaston Memorial Hospital (NC)
 Geisinger Medical Center (PA)
 Genesis Healthcare System (OH)
 George Washington University Hospital
 (DC)
 Gestión de Calidad (Argentina)
 Gettysburg Hospital (PA)
 Ghent University Hospital (Belgium)
 Good Shepherd Medical Center (TX)
 Grana S.A. (TX)
 Grand Marquis Co., LTD (Taiwan)
 Grand River Hospital (ON, Canada)
 Grand Strand Regional Medical Center
 (SC)
 Grey Bruce Regional Health Center (ON,
 Canada)
 Group Health Cooperative (WA)
 Gundersen Lutheran Medical Center
 (WI)
 Guthrie Clinic Laboratories (PA)
 Hôtel-Dieu de Lévis (PQ, Canada)
 Halton Healthcare Services (ON, Canada)
 Hamad Medical Corporation (Qatar)
 Hamilton Regional Laboratory Medicine
 Program - St. Joseph's (ON, Canada)
 Hanover General Hospital (PA)
 Harford Memorial Hospital (MD)
 Harris Methodist Fort Worth (TX)
 Harris Methodist Hospital Southwest
 (TX)
 Hartford Hospital (CT)
 Health Network Lab (PA)
 Health Sciences Research Institute
 (Japan)
 Health Waikato (New Zealand)
 Heartland Health (MO)
 Heidelberg Army Hospital (AE)
 Helen Hayes Hospital (NY)
 Helix (Russian Federation)
 Henry Ford Hospital (MI)
 Henry M. Jackson Foundation for the
 Advancement of Military Medicine-
 MD (MD)
 Hi-Desert Medical Center (CA)
 Highlands Medical Center (AL)
 HJF Naval Infectious Diseases
 Diagnostic Laboratory (MD)

Hoag Memorial Hospital Presbyterian
 (CA)
 Hoboken University Medical Center (NJ)
 Holy Cross Hospital (MD)
 Holy Name Hospital (NJ)
 Holy Spirit Hospital (PA)
 Hôpital de la Cité-de-La-Santé De Laval
 (Quebec, Canada)
 Hôpital du Haut-Richelieu (PQ, Canada)
 Hôpital Maisonneuve-Rosemont (PQ,
 Canada)
 Hôpital Santa Cabrini Ospedale (PQ,
 Canada)
 Horizon Health Network (N.B., Canada)
 Hospital Albert Einstein (SP, Brazil)
 Hospital Sacre-Coeur de Montreal
 (Quebec, Canada)
 Hôtel-Dieu Grace Hospital Library (ON,
 Canada)
 Hunter Area Pathology Service
 (Australia)
 Hunter Labs (CA)
 Huntington Memorial Hospital (CA)
 Imelda Hospital (Belgium)
 Indian River Memorial Hospital (FL)
 Indiana University Health Bloomington
 Hospital (IN)
 Indiana University Health Care-
 Pathology Laboratory (IN)
 Inova Central Laboratory (VA)
 Institut fur Stand. und Dok. im Med. Lab.
 (Germany)
 Institut National de Santé Publique Du
 Quebec Centre de Doc. - INSPQ (PQ,
 Canada)
 Institute Health Laboratories (PR)
 Institute of Clinical Pathology and
 Medical Research (Australia)
 Institute of Laboratory Medicine
 Landspítali Univ. Hospital (Iceland)
 Institute of Medical & Veterinary
 Science (SA, Australia)
 Intermountain Health Care Lab Services
 (UT)
 International Health Management
 Associates, Inc. (IL)
 Irwin Army Community Hospital (KS)
 Jackson County Memorial Hospital (OK)
 Jackson Memorial Hospital (FL)
 Jackson Purchase Medical Center (KY)
 Jessa Ziekenhuis VZW (Belgium)
 John C. Lincoln Hospital - N.MT. (AZ)
 John F. Kennedy Medical Center (NJ)
 John H. Stroger, Jr. Hospital of Cook
 County (IL)
 John Muir Health (CA)
 John T. Mather Memorial Hospital (NY)
 Johns Hopkins Medical Institutions (MD)
 Johns Hopkins University (MD)
 Johnson City Medical Center Hospital
 (TN)
 JPS Health Network (TX)
 Kallos Genetics (AL)
 Kaiser Permanente (OH)
 Kaiser Permanente (MD)
 Kaiser Permanente Medical Care (CA)
 Kaohsiun Chang Gung Memorial
 Hospital (Taiwan)
 Kenora-Rainy River Reg. Lab. Program
 (ON, Canada)
 King Abdulaziz Hospital, Al Ahsa Dept.
 of Pathology & Laboratory Medicine
 (Al-hasa, Saudi Arabia)
 King Fahad National Guard Hospital
 KAMC - NGH (Saudi Arabia)
 King Fahad Specialist Hospital-
 Dammam, K.S.A. (Eastern Region,
 Saudi Arabia)
 King Faisal Specialist Hospital &
 Research Center (Saudi Arabia)
 King Hussein Cancer Center (Jordan)
 Kingston General Hospital (ON, Canada)
 Laboratória Médico Santa Luzia LTDA
 (Brazil)
 Laboratory Alliance of Central New
 York (NY)
 Laboratory Corporation of America (NJ)
 Laboratory Medicin Dalarna (Dalarna,
 Sweden)
 LabPlus Auckland District Health Board
 (New Zealand)
 LAC/USC Medical Center (CA)
 Lafayette General Medical Center (LA)
 Lakeland Regional Medical Center (FL)
 Lancaster General Hospital (PA)
 Landstuhl Regional Medical Center
 (Germany)
 Langley Air Force Base (VA)
 LeBonheur Children's Hospital (TN)
 Legacy Laboratory Services (OR)
 Letherbridge Regional Hospital (AB,
 Canada)
 Lewis-Gale Medical Center (VA)
 Lexington Medical Center (SC)
 L'Hotel-Dieu de Québec (PQ, Canada)
 Licking Memorial Hospital (OH)
 LifeBridge Health Sinai Hospital (MD)

LifeLabs Medical Laboratory Services
 (BC, Canada)
 Lifeline Hospital (United Arab Emirates)
 Loma Linda University Medical Center
 (LLUMC) (CA)
 Long Beach Memorial Medical Center-
 LBMMC (CA)
 Long Island Jewish Medical Center (NY)
 Louisiana Office of Public Health
 Laboratory (LA)
 Louisiana State University Medical Ctr.
 (LA)
 Lower Columbia Pathologists, P.S. (WA)
 Lower Mainland Laboratories (BC,
 Canada)
 Lyndon B. Johnson General Hospital
 (TX)
 Maccabi Medical Care and Health Fund
 (Israel)
 Madigan Army Medical Center (WA)
 Mafraq Hospital (United Arab Emirates)
 Magnolia Regional Health Center (MS)
 Main Line Clinical Laboratories, Inc.
 Lankenau Hospital (PA)
 Makerere University Walter Reed Project
 Makerere University Medical School
 (Uganda)
 Marquette General Hospital (MI)
 Marshfield Clinic (WI)
 Martha Jefferson Hospital (VA)
 Martin Luther King, Jr./Drew Medical
 Center (CA)
 Martin Memorial Health Systems (FL)
 Mary Hitchcock Memorial Hospital (NH)
 Mary Washington Hospital (VA)
 Mater Health Services - Pathology
 (Australia)
 Maxwell Air Force Base (AL)
 Mayo Clinic (MN)
 MCG Health (GA)
 Meadows Regional Medical Center (GA)
 Medical Center Hospital (TX)
 Medical Center of Louisiana At NO-
 Charity (LA)
 Medical Centre Ljubljana (Slovenia)
 Medical College of Virginia Hospital
 (VA)
 Medical University Hospital Authority
 (SC)
 Memorial Hermann Healthcare System
 (TX)
 Memorial Medical Center (PA)
 Memorial Medical Center (IL)
 Memorial Regional Hospital (FL)
 Mercy Franciscan Mt. Airy (OH)
 Mercy Hospital & Medical Center (IL)
 Methodist Dallas Medical Center (TX)
 Methodist Hospital (TX)
 Methodist Hospital (TX)
 Methodist Hospital Park Nicollet Health
 Services (MN)
 Methodist Hospital Pathology (NE)
 MetroHealth Medical Center (OH)
 Metropolitan Hospital Center (NY)
 Metropolitan Medical Laboratory, PLC
 (IA)
 Miami Children's Hospital (FL)
 Mid Michigan Medical Center - Midland
 (MI)
 Middelheim General Hospital (Belgium)
 Middlesex Hospital (CT)
 Mike O'Callaghan Federal Hospital (NV)
 Minneapolis Medical Research
 Foundation (MN)
 Mississippi Baptist Medical Center (MS)
 Mississippi Public Health Lab (MS)
 Monongalia General Hospital (WV)
 Montreal General Hospital (Quebec,
 Canada)
 Morehead Memorial Hospital (NC)
 Mouwasat Hospital (GA, Saudi Arabia)
 Mt. Carmel Health System (OH)
 Mt. Sinai Hospital (ON, Canada)
 Mt. Sinai Hospital - New York (NY)
 Naples Community Hospital (FL)
 Nassau County Medical Center (NY)
 National B Virus Resource Laboratory
 (GA)
 National Cancer Center (Korea, Republic
 Of)
 National Institutes of Health, Clinical
 Center (MD)
 National Naval Medical Center (MD)
 National University Hospital Department
 of Laboratory Medicine (Singapore)
 National University of Ireland, Galway
 (NUIG) (Ireland)
 Nationwide Children's Hospital (OH)
 Naval Hospital Oak Harbor (WA)
 Naval Medical Center Portsmouth (VA)
 Naval Medical Clinic Hawaii (HI)
 NB Department of Health (NB, Canada)
 New England Baptist Hospital (MA)
 New England Fertility Institute (CT)
 New England Sinai Hospital (MA)
 New Lexington Clinic (KY)
 New York City Department of Health
 and Mental Hygiene (NY)

New York Presbyterian Hospital (NY)	Riyadh Armed Forces Hospital, Sulaymania (Saudi Arabia)	Sunnybrook Health Sciences Centre (ON, Canada)	University of Colorado Health Sciences Center (CO)
New York University Medical Center (NY)	Rockford Memorial Hospital (IL)	Sunrise Hospital and Medical Center (NV)	University of Colorado Hospital (CO)
Newark Beth Israel Medical Center (NJ)	Royal Victoria Hospital (ON, Canada)	Swedish Edmonds Hospital (WA)	University of Illinois Medical Center (IL)
Newfoundland Public Health Laboratory (NL, Canada)	SAAD Specialist Hospital (Saudi Arabia)	Swedish Medical Center (CO)	University of Iowa Hospitals and Clinics (IA)
North Carolina Baptist Hospital (NC)	Sacred Heart Hospital (WI)	Sydney South West Pathology Service	University of Kentucky Medical Center (KY)
North District Hospital (China)	Sacred Heart Hospital (FL)	Liverpool Hospital (NSW, Australia)	University of Maryland Medical System (MD)
North Mississippi Medical Center (MS)	Sahlgrenska Universitetssjukhuset (Sweden)	T.J. Samson Community Hospital (KY)	University of Minnesota Medical Center- Fairview (MN)
North Shore Hospital Laboratory (New Zealand)	Saint Francis Hospital & Medical Center (CT)	Taichung Veterans General Hospital (Taiwan)	University of Missouri Hospital (MO)
North Shore-Long Island Jewish Health System Laboratories (NY)	Saint Mary's Regional Medical Center (NV)	Taiwan Society of Laboratory Medicine (Taiwan)	University of Pennsylvania Health System (PA)
Northridge Hospital Medical Center (CA)	Saints Memorial Medical Center (MA)	Tallaght Hospital (Ireland)	University of Pittsburgh Medical Center (PA)
Northside Hospital (GA)	Salem Memorial District Hospital (MO)	Tartu University Clinics (Estonia)	University of Texas Health Center (TX)
Northside Medical Center (OH)	Sampson Regional Medical Center (NC)	Temple Univ. Hospital - Parkinson Pav. (PA)	University of the Ryukyus (Japan)
Northwest Texas Hospital (TX)	Samsung Medical Center (Korea, Republic Of)	Tenet Healthcare (PA)	University of Virginia Medical Center (VA)
Northwestern Memorial Hospital (IL)	San Francisco General Hospital- University of California San Francisco (CA)	Texas Children's Hospital (TX)	UPMC Bedford Memorial (PA)
Norton Healthcare (KY)	Sanford USD Medical Center (SD)	Texas Department of State Health Services (TX)	US Naval Hospital Naples
Ochsner Clinic Foundation (LA)	Santa Clara Valley Medical Center (CA)	Texas Health Presbyterian Hospital Dallas (TX)	UZ-KUL Medical Center (Belgium)
Ohio State University Hospitals (OH)	SARL Laboratoire Caron (France)	The Brooklyn Hospital Center (NY)	VA (Asheville) Medical Center (NC)
Ohio Valley Medical Center (WV)	Scott & White Memorial Hospital (TX)	The Charlotte Hungerford Hospital (CT)	VA (Bay Pines) Medical Center (FL)
Onze Lieve Vrouweziekenhuis (Belgium)	Seattle Children's Hospital/Children's Hospital and Regional Medical Center (WA)	The Children's Mercy Hospital (MO)	VA (Central Texas) Veterans Health Care System (TX)
Ordre Professionnel Des Technologistes Médicaux Du Quebec (Quebec, Canada)	Seoul National University Hospital (Korea, Republic Of)	The Cooley Dickinson Hospital, Inc. (MA)	VA (Chillicothe) Medical Center (OH)
Orebro University Hospital (Sweden)	Seoul St. Mary's Hospital (Korea, Republic Of)	The Credit Valley Hospital (ON, Canada)	VA (Cincinnati) Medical Center (OH)
Orlando Health (FL)	Seton Healthcare Network (TX)	The Hospital for Sick Children (ON, Canada)	VA (Dallas) Medical Center (TX)
Ospedale Casa Sollievo Della Sofferenza - IRCCS (Italy)	Seton Medical Center (CA)	The Medical Center of Aurora (CO)	VA (Dayton) Medical Center (OH)
Our Lady's Hospital For Sick Children (Ireland)	Sharp Health Care Laboratory Services (CA)	The Michener Inst. for Applied Health Sciences (ON, Canada)	VA (Hines) Medical Center (IL)
Palmetto Baptist Medical Center (SC)	Sheik Kalifa Medical City (United Arab Emirates)	The Naval Hospital of Jacksonville (FL)	VA (Indianapolis) Medical Center (IN)
Pamela Youde Nethersole Eastern Hospital (Hong Kong East Cluster) (Hong Kong)	Shore Memorial Hospital (NJ)	The Nebraska Medical Center (NE)	VA (Iowa City) Medical Center (IA)
Pathgroup (TN)	Singapore General Hospital (Singapore)	The Ottawa Hospital (ON, Canada)	VA (Miami) Medical Center (FL)
Pathlab (IA)	Slotervaart Ziekenhuis (Netherlands)	The Permanente Medical Group (CA)	VA (San Diego) Medical Center (CA)
Pathology and Cytology Laboratories, Inc. (KY)	South Bend Medical Foundation (IN)	The Toledo Hospital (OH)	VA (Tampa) Hospital (FL)
Pathology Associates Medical Lab. (WA)	South Eastern Area Laboratory Services (NSW, Australia)	The University of Texas Medical Branch (TX)	VA (Wilmington) Medical Center (DE)
Pathology Inc. (CA)	South Miami Hospital (FL)	Thomas Jefferson University Hospital, Inc. (PA)	Vancouver Island Health Authority (SI) (BC, Canada)
Penn State Hershey Medical Center (PA)	Southern Community Laboratories (Canterbury, New Zealand)	Timmins and District Hospital (ON, Canada)	Vanderbilt University Medical Center (TN)
Pennsylvania Hospital (PA)	Southern Health Care Network (Australia)	Tokyo Metro. Res. Lab of Public Health (Japan)	Verinata Health, Inc. (CA)
Peterborough Regional Health Centre (ON, Canada)	Southwest Healthcare System (CA)	Touro Infirmary (LA)	Via Christi Regional Medical Center (KS)
PHS Indian Hospital - Pine Ridge (SD)	Southwestern Medical Center (OK)	TriCore Reference Laboratories (NM)	Viracor-IBT Reference Laboratory (MO)
Piedmont Hospital (GA)	Spectra East (NJ)	Trident Medical Center (SC)	Virginia Regional Medical Center (MN)
Pitt County Memorial Hospital (NC)	Spectra Laboratories (CA)	Trinity Medical Center (AL)	Virtua - West Jersey Hospital (NJ)
Potomac Hospital (VA)	St. Agnes Healthcare (MD)	Tripler Army Medical Center (HI)	WakeMed (NC)
Prairie Lakes Hospital (SD)	St. Anthony Hospital (OK)	Tuen Mun Hospital, Hospital Authority (China)	Walter Reed Army Medical Center (DC)
Presbyterian Hospital - Laboratory (NC)	St. Barnabas Medical Center (NJ)	Tufts Medical Center Hospital (MA)	Warren Hospital (NJ)
Presbyterian/St. Luke's Medical Center (CO)	St. Elizabeth Community Hospital (CA)	Tulane Medical Center Hospital & Clinic (LA)	Washington Hospital Center (DC)
Prince of Wales Hospital (Hong Kong)	St. Eustache Hospital (Quebec, Canada)	Turku University Central Hospital (Finland)	Washington Hospital Healthcare System (CA)
Princess Margaret Hospital (Hong Kong, China)	St. Francis Hospital (SC)	Twin Lakes Regional Medical Center (KY)	Waterbury Hospital (CT)
Providence Alaska Medical Center (AK)	St. Francis Memorial Hospital (CA)	UCI Medical Center (CA)	Waterford Regional Hospital (Ireland)
Providence Health Services, Regional Laboratory (OR)	St. John Hospital and Medical Center (MI)	UCLA Medical Center Clinical Laboratories (CA)	Wayne Memorial Hospital (NC)
Provincial Laboratory for Public Health (AB, Canada)	St. John's Episcopal Hospital (NY)	UCSD Medical Center (CA)	Weirton Medical Center (WV)
Queen Elizabeth Hospital (P.E.I, Canada)	St. John's Hospital & Health Center (CA)	UCSF Medical Center China Basin (CA)	West Jefferson Medical Center (LA)
Queen Elizabeth Hospital (China)	St. John's Mercy Medical Center (MO)	UMC of El Paso- Laboratory (TX)	West Penn Allegheny Health System- Allegheny General Hospital (PA)
Queensland Health Pathology Services (Australia)	St. John's Regional Health Center (MO)	UMC of Southern Nevada (NV)	West Shore Medical Center (MI)
Queensway Carleton Hospital (ON, Canada)	St. Jude Children's Research Hospital (TN)	UNC Hospitals (NC)	West Valley Medical Center Laboratory (ID)
Quest Diagnostics, Incorporated (CA)	St. Luke's Hospital (IA)	Unidad De Patología Clínica (Mexico)	Westchester Medical Center (NY)
Quintiles Laboratories, Ltd. (GA)	St. Luke's Hospital (PA)	Union Clinical Laboratory (Taiwan)	Western Baptist Hospital (KY)
Rady Children's Hospital San Diego (CA)	St. Mary Medical Center (CA)	United Christian Hospital (Kowloon, Hong Kong)	Western Healthcare Corporation (NL, Canada)
Ramathibodi Hospital (Thailand)	St. Mary's Hospital (WI)	United States Air Force School of Aerospace Medicine / PHE (TX)	Wheaton Franciscan Laboratories (WI)
Redington-Fairview General Hospital (ME)	St. Michael's Medical Center, Inc. (NJ)	Unity HealthCare (IA)	Wheeling Hospital (WV)
Regions Hospital (MN)	St. Tammany Parish Hospital (LA)	Universitair Ziekenhuis Antwerpen (Belgium)	White Memorial Medical Center (CA)
Reid Hospital & Health Care Services (IN)	Stanford Hospital and Clinics (CA)	University College Hospital (Ireland)	Whitehorse General Hospital (YT, Canada)
Reinier De Graaf Groep (Netherlands)	Stanton Territorial Health Authority (NT, Australia)	University Hospital (GA)	William Beaumont Army Medical Center (TX)
Renown Regional Medical Center (NV)	State of Connecticut Department of Public Health (CT)	University Hospital Center Sherbrooke (CHUS) (Quebec, Canada)	William Beaumont Hospital (MI)
Research Medical Center (MO)	State of Ohio/Corrections Medical Center Laboratory (OH)	University Medical Center at Princeton (NJ)	William Osler Health Centre (ON, Canada)
Response Genetics, Inc. (CA)	State of Washington Public Health Labs (WA)	University of Alabama Hospital Lab (AL)	Winchester Hospital (MA)
RIPAS Hospital (Brunei-Maura, Brunei Darussalam)	Stillwater Medical Center (OK)	University of Chicago Hospitals Laboratories (IL)	Winn Army Community Hospital (GA)
Riverside County Regional Medical Center (CA)	Stony Brook University Hospital (NY)		Wishard Health Sciences (IN)
Riverside Health System (VA)	Stormont-Vail Regional Medical Ctr. (KS)		Womack Army Medical Center Department of Pathology (NC)
Riverside Methodist Hospital (OH)	Strong Memorial Hospital (NY)		York Hospital (PA)
	Sudbury Regional Hospital (ON, Canada)		Yukon-Kuskokwim Delta Regional Hospital (AK)

NOTES

NOTES

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